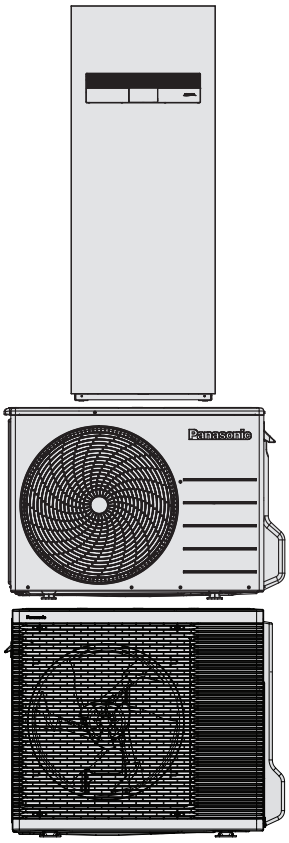
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Service Manual

Air-to-Water Hydromodule + Tank



Indoor Unit
WH-ADC0309K0E5A

Outdoor Unit
WH-UDZ03KE5
WH-UDZ05KE5
WH-UDZ07KE5
WH-UDZ09KE5

Destination
Europe
UK

⚠ WARNING

This service information is designed for experienced repair technicians only and is not designed for use by the general public.
It does not contain warnings or cautions to advise non-technical individuals of potential dangers in attempting to service a product.
Products powered by electricity should be serviced or repaired only by experienced professional technicians. Any attempt to service or repair the products dealt with in this service information by anyone else could result in serious injury or death.

IMPORTANT SAFETY NOTICE

There are special components used in this equipment which are important for safety. These parts are marked by ⚠ in the Schematic Diagrams, Circuit Board Diagrams, Exploded Views and Replacement Parts List. It is essential that these critical parts should be replaced with manufacturer's specified parts to prevent shock, fire or other hazards. Do not modify the original design without permission of manufacturer.

⚠ PRECAUTION OF LOW TEMPERATURE

In order to avoid frostbite, be assured of no refrigerant leakage during the installation or repairing of refrigerant circuit.

⚠ CAUTION

R32 REFRIGERANT – This Air-to-Water Hydromodule + Tank contains and operates with refrigerant R32.
THIS PRODUCT MUST ONLY BE INSTALLED OR SERVICED BY QUALIFIED PERSONNEL.
Refer to National, State, Territory and local legislation, regulations, codes, installation & operation manuals, before the installation, maintenance and/or service of this product.

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• Specifications, designs and contents in this Service Manual are subject to change without notice.

1. Safety Precautions

- Read the following “SAFETY PRECAUTIONS” carefully before installation of Air-To-Water Hydromodule + Tank (here after referred to as “Tank Unit”).
- Electrical works and water installation works must be done by licensed electrician and licensed water system installer respectively. Be sure to use the correct rating and main circuit for the model to be installed.
- The caution items stated here must be followed because these important contents are related to safety. The meaning of each indication used is as below. Incorrect installation due to ignorance or negligence of the instructions will cause harm or damage, and the seriousness is classified by the following indications.
- Please leave this installation manual with the unit after installation.

⚠ **WARNING**	This indication shows the possibility of causing death or serious injury.
⚠ **CAUTION**	This indication shows the possibility of causing injury or damage to properties only.

The items to be followed are classified by the symbols:

⃠	Symbol with white background denotes item that is PROHIBITED.
❗ ⏚	Symbol with dark background denotes item that must be carried out.

- Carry out test run to confirm that no abnormality occurs after the installation. Then, explain to user the operation, care and maintenance as stated in instructions. Please remind the customer to keep the operating instructions for future reference.
- If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.

⚠ **WARNING**	
1. Do not use means to accelerate the defrosting process or to clean, other than those recommended by the manufacturer. Any unfit method or using incompatible material may cause product damage, burst and serious injury.	⃠
2. Do not install outdoor unit near handrail of veranda. When installing outdoor unit at veranda of high rise building, child may climb up to outdoor unit and cross over the handrail and causing accident.	⃠
3. Do not use unspecified cord, modified cord, joint cord or extension cord for power supply cord. Do not share the single outlet with other electrical appliances. Poor contact, poor insulation or over current will cause electrical shock or fire.	⃠
4. Do not tie up the power supply cord into a bundle by band. Abnormal temperature rise on power supply cord may happen.	⃠
5. Do not insert your fingers or other objects into the unit, high speed rotating fan may cause injury.	⃠
6. Do not sit or step on the unit, you may fall down accidentally.	⃠
7. Keep plastic bag (packaging material) away from small children, it may cling to nose and mouth and prevent breathing.	⃠
8. When install or relocate outdoor unit, do not let any substance other than the specified refrigerant, e.g. air etc. mix into refrigerant cycle (piping). Mixing of air etc. will cause abnormal high pressure in refrigeration cycle and result in explosion, injury etc.	⃠
9. Do not use pipe wrench to install refrigerant piping. It might deform the piping and cause the unit to malfunction.	⃠
10. Do not purchase unauthorized electrical parts for installation, service, maintenance and etc.. They might cause electrical shock or fire.	⃠
11. Do not modify the wiring of outdoor unit for installation of other components (i.e. heater, etc). Overloaded wiring or wire connection points may cause electrical shock or fire.	⃠
12. Do not pierce or burn as the appliance is pressurized. Do not expose the appliance to heat, flame, sparks, or other sources of ignition. Else, it may explode and cause injury or death.	⃠
13. Do not add or replace refrigerant other than specified type. It may cause product damage, burst and injury etc.	⃠
14. Do not place containers with liquids on top of the Tank Unit. It may cause Tank Unit damage and/or fire could occurs if they leak or spill onto the Tank Unit.	⃠
15. Do not use joint cable for Tank Unit / Outdoor Unit connection cable. Use specified Tank Unit / Outdoor Unit connection cable, refer to instruction **12.2.4 CONNECT THE CABLE TO THE TANK UNIT** and connect tightly for Tank Unit / Outdoor Unit connection. Clamp the cable so that no external force will be acted on the terminal. If connection or fixing is not perfect, it will cause heat up or fire at the connection.	⃠
16. Do not use joint cable for outdoor connection cable. Use specified outdoor connection cable, refer to instruction **12.3.6 CONNECT THE CABLE TO THE OUTDOOR UNIT** and connect tightly for outdoor connection. Clamp the cable so that no external force will be acted on the terminal. If connection or fixing is not perfect, it will cause heat up or fire at the connection.	❗
17. Wire routing must be properly arranged so that control board cover is fixed properly. If control board cover is not fixed perfectly, it will cause fire or electrical shock.	❗

⚠ WARNING	
18. For electrical work, follow the national regulation, legislation and this installation instructions. An independent circuit and single outlet must be used. If electrical circuit capacity is not enough or defect found in the electrical work, it will cause electrical shock or fire.	
19. For electrical work, follow local wiring standard, regulation and this installation instruction. An independent circuit and single outlet must be used. If electrical circuit capacity is not enough or defect found in electrical work, it will cause electrical shock or fire.	
20. For water circuit installation work, follow to relevant European and national regulations (including EN61770) and local plumbing and building regulation codes.	
21. Engage authorized dealer or specialist for installation. If installation done by the user is incorrect, it will cause water leakage, electrical shock or fire.	
22. Engage dealer or specialist for installation. If installation done by the user is defective, it will cause water leakage, electrical shock or fire.	
23. • This is a R32 model, use piping, flare nut and tools which is specified for R32 refrigerant. Using of existing (R22) piping, flare nut and tools may cause abnormally high pressure in the refrigerant cycle (piping), and possibly result in explosion and injury. • Thickness for copper pipes used with R32 must be more than 0.8 mm. Never use copper pipes thinner than 0.8 mm. • It is desirable that the amount of residual oil is less than 40 mg/10 m.	
24. When installing or relocating Tank Unit, do not let any substance other than the specified refrigerant, eg. air etc. mix into refrigeration cycle (piping). Mixing of air etc. will cause abnormal high pressure in refrigeration cycle and result in explosion, injury etc.	
25. For refrigeration system work, install according to this installation instructions strictly. If installation is defective, it will cause water leakage, electrical shock or fire.	
26. Install at a strong and firm location which is able to withstand weight of the set. If the strength is not enough or installation is not properly done, the set will drop and cause injury.	
27. This equipment is strongly recommended to be installed with Residual Current Device (RCD) on-site according to the respective national wiring rules or country-specific safety measures in terms of residual current.	
28. During installation, install the refrigerant piping properly before running the compressor. Operation of compressor without fixing refrigeration piping and valves at opened position will cause suck-in of air, abnormal high pressure in refrigeration cycle and result in explosion, injury etc.	
29. During pump down operation, stop the compressor before removing the refrigeration piping. Removal of refrigeration piping while compressor is operating and valves are opened will cause suck-in of air, abnormal high pressure in refrigeration cycle and result in explosion, injury etc.	
30. Tighten the flare nut with torque wrench according to specified method. If the flare nut is over tightened, after a long period, the flare may break and cause refrigerant gas leakage.	
31. After completion of installation, confirm there is no leakage of refrigerant gas. It may generate toxic gas when the refrigerant contacts with fire.	
32. Ventilate if there is refrigerant gas leakage during operation. It may cause toxic gas when the refrigerant contacts with fire.	
33. Use the attached accessories parts and specified parts for installation. Otherwise, it will cause the set to fall, water leakage, fire or electrical shock.	
34. Only use the supplied or specified installation parts. Else, it may causes unit vibrate, fall, water leakage, electrical shock or fire.	
35. If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.	
36. Select a location where in case of water leakage, the leakage will not cause damage to other properties.	
37. When installing electrical equipment at wooden building of metal lath or wire lath, in accordance with electrical facility standard, no electrical contact between equipment and building is allowed. Insulator must be installed in between.	
38. Any work carried out on the Tank Unit after removing any panels which is secured by screws, must be carried out under the supervision of authorized dealer and licensed installation contractor.	
39. This system is multi supply appliance. All circuits must be disconnected before accessing the unit terminals.	
40. For cold water supply has a backflow regulator, check valve or water meter with check valve, provisions for thermal expansion of water in the hot water system must be provided. Otherwise it will cause water leakage.	
41. The piping installation work must be flushed before Tank Unit is connected to remove contaminants. Contaminants may damage the Tank Unit components.	
42. This installation may be subjected to building regulation approval applicable to respective country that may require to notify the local authority before installation.	
43. The Tank Unit must be shipped and stored in upright condition and dry environment. It may laid on its back when being moved into the building.	
44. Work done to the Tank Unit after remove the front plate cover that secured by screws, must be carried out under the supervision of authorized dealer, licensed installation contractor, skilled person and instructed person.	
45. Be aware that refrigerants may not contain an odour.	
46. This equipment must be properly earthed. Earth line must not be connected to gas pipe, water pipe, earth of lightning rod and telephone. Otherwise, it may cause electrical shock in case of equipment breakdown or insulation breakdown.	
47. This unit must be properly earthed. The electrical earth must not be connected to a gas pipe, water pipe, the earth of lightening rod or a telephone. Otherwise there is a danger of electrical shock in the event of an insulation breakdown or electrical earth fault in the outdoor unit.	

⚠ CAUTION	
1. Do not install the Tank Unit at place where leakage of flammable gas may occur. In case gas leaks and accumulates at surrounding of the unit, it may cause fire.	⊘
2. Prevent liquid or vapor from entering sumps or sewers since vapor is heavier than air and may form suffocating atmospheres.	⊘
3. Do not release refrigerant during piping work for installation, re-installation and during repairing a refrigeration parts. Take care of the liquid refrigerant, it may cause frostbite.	⊘
4. Do not install this appliance in a laundry room or other high humidity location. This condition will cause rust and damage to the unit.	⊘
5. Make sure the insulation of power supply cord does not contact hot part (i.e. refrigerant piping, water piping) to prevent from insulation failure (melt).	⊘
6. Do not touch the sharp aluminium fin, sharp parts may cause injury. 	⊘
7. Do not apply excessive force to water pipes that may damage the pipes. If water leakage occurs, it will cause flooding and damage to other properties.	⊘
8. Do not transport the Tank Unit with water inside the unit. It may cause damage to the unit.	⊘
9. Carry out drainage piping as mentioned in installation instructions. If drainage is not perfect, water may enter the room and damage the furniture.	❗
10. Select an installation location which is easy for maintenance. Incorrect installation, service or repair of this Tank Unit may increase the risk of rupture and this may result in loss damage or injury and/or property.	❗
11. Power supply connection to Tank Unit. • Power supply point should be in easily accessible place for power disconnection in case of emergency. • Must follow local national wiring standard, regulation and this installation instruction. • Strongly recommended to make permanent connection to a circuit breaker. ■ For Tank Unit WH-ADC0309K0E5A: – Power Supply 1: For WH-UDZ03KE5* and WH-UDZ05KE5*, use approved 15/16A 2-poles circuit breaker with a minimum contact gap of 3.0mm. For WH-UDZ07KE5* and WH-UDZ09KE5*, use approved 25A 2-poles circuit breaker with a minimum contact gap of 3.0mm. – Power Supply 2 (Optional): Use approved 16A 2-poles circuit breaker with a minimum contact gap of 3.0mm (if Optional Back-up Heater CZ-NB3 installed).	❗
12. Ensure the correct polarity is maintained throughout all wiring. Otherwise, it will cause electrical shock or fire.	❗
13. After installation, check the water leakage condition in connection area during test run. If leakage occurs, it will cause damage to other properties.	❗
14. If the Tank Unit not operates for long time, the water inside the Tank Unit should be drained.	❗
15. Installation work. It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.	❗
16. Keep any required ventilation openings clear of obstruction.	❗

2. Precaution For Using R32 Refrigerant

- The basic installation work procedures are the same as conventional refrigerant (R410A, R22) models. However, pay careful attention to the following points:

	⚠ WARNING	
1.	Since the working pressure is higher than that of refrigerant R22 models, some of the piping and installation and service tools are special. Especially, when replacing a refrigerant R22 model with a new refrigerant R32 model, always replace the conventional piping and flare nuts with the R32 and R410A piping and flare nuts on the outdoor unit side. For R32 and R410A, the same flare nut on the outdoor unit side and pipe can be used.	❗
2.	The mixing of different refrigerants within a system is prohibited. Models that use refrigerant R32 and R410A have a different charging port thread diameter to prevent erroneous charging with refrigerant R22 and for safety. Therefore, check beforehand. [The charging port thread diameter for R32 and R410A is 12.7 mm (1/2 inch).]	❗
3.	Ensure that foreign matter (oil, water, etc.) does not enter the piping. Also, when storing the piping, securely seal the opening by pinching, taping, etc. (Handling of R32 is similar to R410A.)	❗
4.	Operation, maintenance, repairing and refrigerant recovery should be carried out by trained and certified personnel in the use of flammable refrigerants and as recommended by the manufacturer. Any personnel conducting an operation, servicing or maintenance on a system or associated parts of the equipment should be trained and certified.	❗
5.	Any part of refrigerating circuit (evaporators, air coolers, AHU, condensers or liquid receivers) or piping should not be located in the proximity of heat sources, open flames, operating gas appliance or an operating electric heater.	❗
6.	The user/owner or their authorized representative shall regularly check the alarms, mechanical ventilation and detectors, at least once a year, where as required by national regulations, to ensure their correct functioning.	❗
7.	A logbook shall be maintained. The results of these checks shall be recorded in the logbook.	❗
8.	In case of ventilations in occupied spaces shall be checked to confirm no obstruction.	❗
9.	Before a new refrigerating system is put into service, the person responsible for placing the system in operation should ensure that trained and certified operating personnel are instructed on the basis of the instruction manual about the construction, supervision, operation and maintenance of the refrigerating system, as well as the safety measures to be observed, and the properties and handling of the refrigerant used.	❗
10.	The general requirement of trained and certified personnel are indicated as below: a) Knowledge of legislation, regulations and standards relating to flammable refrigerants; and, b) Detailed knowledge of and skills in handling flammable refrigerants, personal protective equipment, refrigerant leakage prevention, handling of cylinders, charging, leak detection, recovery and disposal; and, c) Able to understand and to apply in practice the requirements in the national legislation, regulations and Standards; and, d) Continuously undergo regular and further training to maintain this expertise.	❗
11.	Air-to-Water Heatpump piping in the occupied space shall be installed in such a way to protect against accidental damage in operation and service.	❗
12.	Precautions shall be taken to avoid excessive vibration or pulsation to refrigerating piping.	❗
13.	Ensure protection devices, refrigerating piping and fittings are well protected against adverse environmental effects (such as the danger of water collecting and freezing in relief pipes or the accumulation of dirt and debris).	❗
14.	Expansion and contraction of long runs piping in refrigerating systems shall be designed and installed securely (mounted and guarded) to minimize the likelihood hydraulic shock damaging the system.	❗
15.	Protect the refrigerating system from accidental rupture due to moving furniture or reconstruction activities.	❗
16.	To ensure no leaking, field-made refrigerant joints indoors shall be tightness tested. The test method shall have a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>1.04MPa, max 4.15MPa). No leak shall be detected.	❗

⚠ CAUTION

1.

Installation (Space)

- Must ensure the installation of pipe-work shall be kept to a minimum. Avoid use dented pipe and do not allow acute bending.
- Must ensure that pipe-work shall be protected from physical damage.
- Must comply with national gas regulations, state municipal rules and legislation. Notify relevant authorities in accordance with all applicable regulations.
- Must ensure mechanical connections be accessible for maintenance purposes.
- In cases that require mechanical ventilation, ventilation openings shall be kept clear of obstruction.
- When disposal of the product, do follow to the precautions in #12 and comply with national regulations.
- In case of field charge, the effect on refrigerant charge caused by the different pipe length has to be quantified, measured and labelled.
- Always contact to local municipal offices for proper handling.

2.

Servicing

2-1. Service personnel

- Any qualified person who is involved with working on or breaking into a refrigerant circuit should hold a current valid certificate from an industry-accredited assessment authority, which authorizes their competence to handle refrigerants safely in accordance with an industry recognized assessment specification.
- Servicing shall only be performed as recommended by the equipment manufacturer. Maintenance and repair requiring the assistance of other skilled personnel shall be carried out under the supervision of the person competent in the use of flammable refrigerants.
- Servicing shall be performed only as recommended by the manufacturer.
- The system is inspected, regularly supervised and maintained by a trained and certified service personnel who is employed by the person user or party responsible.
- Ensure the actual refrigerant charge is in accordance with the room size within which the refrigerant containing parts are installed.
- Ensure refrigerant charge not to leak.

2-2. Work

- Prior to beginning work on systems containing flammable refrigerants, safety checks are necessary to ensure that the risk of ignition is minimised.
  For repair to the refrigerating system, the precautions in #2-2 to #2-8 must be followed before conducting work on the system.
- Work shall be undertaken under a controlled procedure so as to minimize the risk of a flammable gas or vapour being present while the work is being performed.
- All maintenance staff and others working in the local area shall be instructed and supervised on the nature of work being carried out.
- Avoid working in confined spaces. Always ensure away from source, at least 2 meter of safety distance, or zoning of free space area of at least 2 meter in radius.
- Wear appropriate protective equipment, including respiratory protection, as conditions warrant.
- Keep all sources of ignition and hot metal surfaces away.

2-3. Checking for presence of refrigerant

- The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially flammable atmospheres.
- Ensure that the leak detection equipment being used is suitable for use with flammable refrigerants, i.e. non sparking, adequately sealed or intrinsically safe.
- In case of leakage/spillage happened, immediately ventilate area and stay upwind and away from spill/release.
- In case of leakage/spillage happened, do notify persons down wind of the leaking/spill, isolate immediate hazard area and keep unauthorized personnel out.

2-4. Presence of fire extinguisher

- If any hot work is to be conducted on the refrigerating equipment or any associated parts, appropriate fire extinguishing equipment shall be available at hand.
- Have a dry powder or CO_2 fire extinguisher adjacent to the charging area.

2-5. No ignition sources

- No person carrying out work in relation to a refrigerating system which involves exposing any pipe work that contains or has contained flammable refrigerant shall use any sources of ignition in such a manner that it may lead to the risk of fire or explosion. He/She must not be smoking when carrying out such work.
- All possible ignition sources, including cigarette smoking, should be kept sufficiently far away from the site of installation, repairing, removing and disposal, during which flammable refrigerant can possibly be released to the surrounding space.
- Prior to work taking place, the area around the equipment is to be surveyed to make sure that there are no flammable hazards or ignition risks.
- “No Smoking” signs shall be displayed.

2-6. Ventilated area

- Ensure that the area is in the open or that it is adequately ventilated before breaking into the system or conducting any hot work.
- A degree of ventilation shall continue during the period that the work is carried out.
- The ventilation should safely disperse any released refrigerant and preferably expel it externally into the atmosphere.

	⚠ **CAUTION**	
2.	2-7. Checks to the refrigerating equipment • Where electrical components are being changed, they shall be fit for the purpose and to the correct specification. • At all times the manufacturer's maintenance and service guidelines shall be followed. • If in doubt consult the manufacturer's technical department for assistance. • The following checks shall be applied to installations using flammable refrigerants. - The actual refrigerant charge is in accordance with the room size within which the refrigerant containing parts are installed. - The ventilation machinery and outlets are operating adequately and are not obstructed. - If an indirect refrigerating circuit is being used, the secondary circuit shall be checked for the presence of refrigerant. - Marking to the equipment continues to be visible and legible. Markings and signs that are illegible shall be corrected. - Refrigerating pipe or components are installed in a position where they are unlikely to be exposed to any substance which may corrode refrigerant containing components, unless the components are constructed of materials which are inherently resistant to being corroded or are properly protected against being so corroded. 2-8. Checks to electrical devices • Repair and maintenance to electrical components shall include initial safety checks and component inspection procedures. • Initial safety checks shall include but not limit to:- - That capacitors are discharged: this shall be done in a safe manner to avoid possibility of sparking. - That there is no live electrical components and wiring are exposed while charging, recovering or purging the system. - That there is continuity of earth bonding. • At all times the manufacturer's maintenance and service guidelines shall be followed. • If in doubt consult the manufacturer's technical department for assistance. • If a fault exists that could compromise safety, then no electrical supply shall be connected to the circuit until it is satisfactorily dealt with. • If the fault cannot be corrected immediately but it is necessary to continue operation, an adequate temporary solution shall be used. • The owner of the equipment must be informed or reported so all parties are advised thereinafter.	❗
3.	Repairs to sealed components • During repairs to sealed components, all electrical supplies shall be disconnected from the equipment being worked upon prior to any removal of sealed covers, etc. • If it is absolutely necessary to have an electrical supply to equipment during servicing, then a permanently operating form of leak detection shall be located at the most critical point to warn of a potentially hazardous situation. • Particular attention shall be paid to the following to ensure that by working on electrical components, the casing is not altered in such a way that the level of protection is affected. This shall include damage to cables, excessive number of connections, terminals not made to original specification, damage to seals, incorrect fitting of glands, etc. • Ensure that apparatus is mounted securely. • Ensure that seals or sealing materials have not degraded such that they no longer serve the purpose of preventing the ingress of flammable atmospheres. • Replacement parts shall be in accordance with the manufacturer's specifications. NOTE: The use of silicon sealant may inhibit the effectiveness of some types of leak detection equipment. Intrinsically safe components do not have to be isolated prior to working on them.	❗
4.	Repair to intrinsically safe components • Do not apply any permanent inductive or capacitance loads to the circuit without ensuring that this will not exceed the permissible voltage and current permitted for the equipment in use. • Intrinsically safe components are the only types that can be worked on while live in the presence of a flammable atmosphere. • The test apparatus shall be at the correct rating. • Replace components only with parts specified by the manufacturer. Unspecified parts by manufacturer may result ignition of refrigerant in the atmosphere from a leak.	❗
5.	Cabling • Check that cabling will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects. • The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.	❗
6.	Detection of flammable refrigerants • Under no circumstances shall potential sources of ignition be used in the searching or detection of refrigerant leaks. • A halide torch (or any other detector using a naked flame) shall not be used.	❗

⚠ CAUTION

7.

The following leak detection methods are deemed acceptable for all refrigerant systems.

- No leaks shall be detected when using detection equipment with a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>1.04MPa, max 4.15MPa). For example, a universal sniffer.
- Electronic leak detectors may be used to detect flammable refrigerants, but the sensitivity may not be adequate, or may need re-calibration.
  (Detection equipment shall be calibrated in a refrigerant-free area.)
- Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used.
- Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed and the appropriate percentage of gas (25 % maximum) is confirmed.
- Leak detection fluids are also suitable for use with most refrigerants, for example, bubble method and fluorescent method agents. The use of detergents containing chlorine shall be avoided as the chlorine may react with the refrigerant and corrode the copper pipe-work.
- If a leak is suspected, all naked flames shall be removed/extinguished.
- If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak.
  The precautions in #8 must be followed to remove the refrigerant.

8.

Removal and evacuation

- When breaking into the refrigerant circuit to make repairs – or for any other purpose – conventional procedures shall be used.
  However, it is important that best practice is followed since flammability is a consideration.
  The following procedure shall be adhered to:

> • remove refrigerant -> • purge the circuit with inert gas -> • evacuate -> • purge with inert gas ->
> • open the circuit by cutting or brazing

- The refrigerant charge shall be recovered into the correct recovery cylinders.
- The system shall be purged with OFN to render the appliance safe. (remark: OFN = oxygen free nitrogen, type of inert gas)
- This process may need to be repeated several times.
- Compressed air or oxygen shall not be used for this task.
- Purging shall be achieved by breaking the vacuum in the system with OFN and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum.
- This process shall be repeated until no refrigerant is within the system.
- When the final OFN charge is used, the system shall be vented down to atmospheric pressure to enable work to take place.
- This operation is absolutely vital if brazing operations on the pipe work are to take place.
- Ensure that the outlet for the vacuum pump is not close to any potential ignition sources and there is ventilation available.

9.

Charging procedures

- In addition to conventional charging procedures, the following requirements shall be followed.
  - Ensure that contamination of different refrigerants does not occur when using charging equipment.
  - Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them.
  - Cylinders shall be kept in an appropriate position according to the instructions.
  - Ensure that the refrigerating system is earthed prior to charging the system with refrigerant.
  - Label the system when charging is complete (if not already).
  - Extreme care shall be taken not to over fill the refrigerating system.
- Prior to recharging the system it shall be pressure tested with OFN (refer to #7).
- The system shall be leak tested on completion of charging but prior to commissioning.
- A follow up leak test shall be carried out prior to leaving the site.
- Electrostatic charge may accumulate and create a hazardous condition when charging and discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

⚠ CAUTION

10.

Decommissioning

- Before carrying out this procedure, it is essential that the technician is completely familiar with the equipment and all its details.
- It is recommended good practice that all refrigerants are recovered safely.
- Prior to the task being carried out, an oil and refrigerant sample shall be taken in case analysis is required prior to re-use of recovered refrigerant.
- It is essential that electrical power is available before the task is commenced.
  a) Become familiar with the equipment and its operation.
  b) Isolate system electrically.
  c) Before attempting the procedure ensure that:

> • mechanical handling equipment is available, if required, for handling refrigerant cylinders;
> • all personal protective equipment is available and being used correctly;
> • the recovery process is supervised at all times by a competent person;
> • recovery equipment and cylinders conform to the appropriate standards.

  d) Pump down refrigerant system, if possible.
  e) If a vacuum is not possible, make a manifold so that refrigerant can be removed from various parts of the system.
  f) Make sure that cylinder is situated on the scales before recovery takes place.
  g) Start the recovery machine and operate in accordance with instructions.
  h) Do not over fill cylinders. (No more than 80 % volume liquid charge).
  i) Do not exceed the maximum working pressure of the cylinder, even temporarily.
  j) When the cylinders have been filled correctly and the process completed, make sure that the cylinders and the equipment are removed from site promptly and all isolation valves on the equipment are closed off.
  k) Recovered refrigerant shall not be charged into another refrigerating system unless it has been cleaned and checked.
- Electrostatic charge may accumulate and create a hazardous condition when charging or discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

11.

Labelling

- Equipment shall be labelled stating that it has been de-commissioned and emptied of refrigerant.
- The label shall be dated and signed.
- Ensure that there are labels on the equipment stating the equipment contains flammable refrigerant.

12.

Recovery

- When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.
- When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed.
- Ensure that the correct number of cylinders for holding the total system charge are available.
- All cylinders to be used are designated for the recovered refrigerant and labelled for that refrigerant (i.e. special cylinders for the recovery of refrigerant).
- Cylinders shall be complete with pressure relief valve and associated shut-off valves in good working order.
- Recovery cylinders are evacuated and, if possible, cooled before recovery occurs.
- The recovery equipment shall be in good working order with a set of instructions concerning the equipment that is at hand and shall be suitable for the recovery of flammable refrigerants.
- In addition, a set of calibrated weighing scales shall be available and in good working order.
- Hoses shall be complete with leak-free disconnect couplings and in good condition.
- Before using the recovery machine, check that it is in satisfactory working order, has been properly maintained and that any associated electrical components are sealed to prevent ignition in the event of a refrigerant release. Consult manufacturer if in doubt.
- The recovered refrigerant shall be returned to the refrigerant supplier in the correct recovery cylinder, and the relevant Waste Transfer Note arranged.
- Do not mix refrigerants in recovery units and especially not in cylinders.
- If compressors or compressor oils are to be removed, ensure that they have been evacuated to an acceptable level to make certain that flammable refrigerant does not remain within the lubricant.
- The evacuation process shall be carried out prior to returning the compressor to the suppliers.
- Only electric heating to the compressor body shall be employed to accelerate this process.
- When oil is drained from a system, it shall be carried out safely.

3. Specifications

3.1 WH-ADC0309K0E5A WH-UDZ03KE5

Item		Unit	Outdoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	3.20		
		BTU/h	10900		
Cooling EER		W/W	3.52		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	3.20	3.20	
		BTU/h	10900	10900	
Heating COP		W/W	5.33	3.64	
Heating Erp	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	4.0	4.0	3.0
	Tbivalent/TOL	°C	2 / 2	-10 / -10	-20 / -22
	SCOP/ns	(W/W)/%	6.20 / 245	4.90 / 193	4.00 / 157
	Annual Consumption	kWh	862	1685	1848
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	4.0	3.0	2.0
	Tbivalent/TOL	°C	2 / 2	-10 / -10	-20 / -22
	SCOP/ns	(W/W)/%	4.20 / 165	3.30 / 129	2.83 / 110
	Annual Consumption	kWh	1274	1878	1740
	Class		A+++	A++	A+
	DHW		Warmer	Average	Colder
	Application	Climate			
	COP/nwh	(W/W)/%	3.86 / 154	3.10 / 124	2.48 / 99
	AEC	kWh	654	816	1023
	Class		A++	A+	A
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 45***	Heating: 44***	
		Power Level dB	Cooling: 61***	Heating: 60*** Heating: 55****	
Air Flow		m^3/min (ft^3/min)	Cooling: 33.9 (1200) Heating: 28.9 (1020)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (450)		
Refrigerant		kg (oz)	R32, 0.90 (31.8) (Pre-charged) R32, 1.20 (42.4) (Maximum)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged/Maximum)		0.608 / 0.810		

Item		Unit	Outdoor Unit		
Dimension	Height	mm (inch)	622 (24-1/2)		
	Width	mm (inch)	824 (32-15/32)		
	Depth	mm (inch)	298 (11-24/32)		
Net Weight		kg (lbs)	37 (82)		
Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	12.70 (1/2)		
Standard Length		m (ft)	7 (23.0)		
Pipe Length Range		m (ft)	3 (9.8) ~ 25 (82.0)		
I/D & O/D Height Difference		m (ft)	20 (65.6)		
Additional Gas Amount		g/m (oz/ft)	20 (0.2)		
Refrigeration Charge Less		m (ft)	10 (32.8)		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (6-poles)		
	Rated Output	kW	0.90		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	W	-		
	Output Power	W	40		
	Fan Speed	rpm	Cooling: 840 Heating: 720		
Heat Exchanger	Fin material		Aluminium (Pre Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 28 × 19		
	Size (W × H × L)	mm	36.4 × 588 × 827.7:856.3		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 0.91	Heating: 0.60	Heating: 0.88
Maximum Input Power For Heatpump System		kW	2.59		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 12.0 / 2.59k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	2.9		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 4.3	Heating: 2.9	Heating: 4.2
Maximum Current For Heatpump System		A	12.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 92	Heating: 90	Heating: 91
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min./max.)	Cooling: 10 / 43 Heating (Tank): -20 / 35 Heating (Circuit): -20 / 35		
	Water Outlet	°C (min./max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -15 °C) *4 Heating (Circuit): 20 / 60 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 5.0 Heating: 5.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 28***	Heating: 28***	
		Power Level dB	Cooling: 41***	Heating: 41***	
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1642 (64-41/64)		
Net Weight		kg (lbs)	96 (212)		
Refrigerant Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	12.70 (1/2)		
Water Pipe Diameter	Room	mm (inch)	1		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		36		
	Size (W × H × L)	mm	120 × 65 × 376		
	Water Flow Rate	l/min (m^3/h)	Cooling: 9.2 (0.6) Heating: 9.2 (0.6)		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer Pressure range		Bar	0 ~ 4		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW/°C	- / -		
Tank Volume (Spec / Nett)		L	200 / 185		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat/Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN-1.4521
	Volume	L	185
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.8
	Surface Area	m^2	1.8
	Total Length	m	25
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60°C	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65°C	µS/cm	< 1200
	Saturation Index (LSI) @ 20°C		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in accordance with EU directive 2003/32/EC.
- *3 Above 55°C, only possible with backup heater operation.
- *4 Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- It is recommended to follow DHW tank water quality limit for Panasonic Air to Water All in One according to Drinking Water Directive 98/83 EC

3.2 WH-ADC0309K0E5A WH-UDZ05KE5

Item		Unit	Outdoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	5.00		
		BTU/h	17100		
Cooling EER		W/W	3.05		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	5.00	5.00	
		BTU/h	17100	17100	
Heating COP		W/W	5.10	3.57	
Heating Erp	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	5.0	5.0	6.0
	Tbivalent/TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP/ns	(W/W)/%	6.00 / 237	4.60 / 181	4.08 / 160
	Annual Consumption	kWh	1113	2245	3625
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	4.0	5.0	4.0
	Tbivalent/TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP/ns	(W/W)/%	4.20 / 165	3.40 / 133	2.95 / 115
	Annual Consumption	kWh	1274	3035	3338
	Class		A+++	A++	A+
	DHW		Warmer	Average	Colder
	Application	Climate			
	COP/nwh	(W/W)/%	4.00 / 160	3.34 / 134	2.80 / 112
	AEC	kWh	630	750	900
	Class		A++	A+	A
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 49***	Heating: 48***	
		Power Level dB	Cooling: 62***	Heating: 60*** Heating: 55****	
Air Flow		m^3/min (ft^3/min)	Cooling: 55.0 (1942) Heating: 35.3 (1246)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (600)		
Refrigerant		kg (oz)	R32, 1.30 (45.9) (Pre-charged) R32, 2.30 (81.2) (Maximum)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged/Maximum)		0.878 / 1.553		
Dimension	Height	mm (inch)	795 (31-19/64)		
	Width	mm (inch)	875 (34-29/64)		
	Depth	mm (inch)	380 (14-31/32)		
Net Weight		kg (lbs)	55 (121)		

Item		Unit	Outdoor Unit		
Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	15.88 (5/8)		
Standard Length		m (ft)	7 (23.0)		
Pipe Length Range		m (ft)	3 (9.8) ~ 40 (131.2) / 3 (9.8) ~ 50 (164.0)		
I/D & O/D Height Difference		m (ft)	30 (98.4)		
Additional Gas Amount		g/m (oz/ft)	25 (0.3)		
Refrigeration Charge Less		m (ft)	10 (32.8)		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (6-poles)		
	Rated Output	kW	1.50		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	W	-		
	Output Power	W	60		
	Fan Speed	rpm	Cooling: 640 Heating: 520		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 36 × 19		
	Size (W × H × L)	mm	36.38 × 756.0 × 868.8:897		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 1.64	Heating: 0.98	Heating: 1.40
Maximum Input Power For Heatpump System		kW	3.36		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 14.9 / 3.36k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	4.4		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 7.3	Heating: 4.4	Heating: 6.3
Maximum Current For Heatpump System		A	14.9		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		%	A35W7 Cooling: 98	A7W35 Heating: 97	A2W35 Heating: 97
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min./max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min./max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -15 °C) *4 Heating (Circuit): 20 / 60 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 11.0 Heating: 11.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 28***	Heating: 28***	
		Power Level dB	Cooling: 41***	Heating: 41***	
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1642 (64-41/64)		
Net Weight		kg (lbs)	96 (212)		
Refrigerant Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	15.88 (5/8)		
Water Pipe Diameter	Room	mm (inch)	1		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		DC Motor		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		36		
	Size (W × H × L)	mm	120 × 65 × 376		
	Water Flow Rate	l/min (m^3/h)	Cooling: 14.3 (0.9) Heating: 14.3 (0.9)		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer Pressure range		Bar	0 ~ 4		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW/°C	- / -		
Tank Volume (Spec / Nett)		L	200 / 185		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat/Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		En-1.4521
	Volume	L	185
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.8
	Surface Area	m^2	1.8
	Total Length	m	25
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60°C	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65°C	µS/cm	< 1200
	Saturation Index (LSI) @ 20°C		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in accordance with EU directive 2003/32/EC.
- *3 Above 55°C, only possible with backup heater operation.
- *4 Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- It is recommended to follow DHW tank water quality limit for Panasonic Air to Water All in One according to Drinking Water Directive 98/83 EC

3.3 WH-ADC0309K0E5A WH-UDZ07KE5

Item		Unit	Outdoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	6.70		
		BTU/h	22800		
Cooling EER		W/W	3.03		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	7.00	6.85	
		BTU/h	23900	23400	
Heating COP		W/W	4.86	3.43	
Heating Erp	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	7.0	7.0	7.0
	Tbivalent/TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP/ns	(W/W)/%	5.75 / 227	4.65 / 183	4.18 / 164
	Annual Consumption	kWh	1627	3111	4132
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	6.0	7.0	6.0
	Tbivalent/TOL	°C	2 / 2	-7 / -10	-15 / -22
	SCOP/ns	(W/W)/%	4.07 / 160	3.43 / 134	2.98 / 116
	Annual Consumption	kWh	1971	4216	4967
	Class		A+++	A++	A+
	DHW		Warmer	Average	Colder
	Application	Climate			
	COP/nwh	(W/W)/%	4.00 / 160	3.34 / 134	2.80 / 112
	AEC	kWh	630	750	900
	Class		A++	A++	A
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 50***	Heating: 50***	
		Power Level dB	Cooling: 64***	Heating: 62*** Heating: 56****	
Air Flow		m^3/min (ft^3/min)	Cooling: 55.0 (1942) Heating: 43.2 (1525)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (600)		
Refrigerant		kg (oz)	R32, 1.30 (45.9) (Pre-charged) R32, 2.30 (81.2) (Maximum)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged/Maximum)		0.878 / 1.553		

Item		Unit	Outdoor Unit		
Dimension	Height	mm (inch)	795 (31-5/16)		
	Width	mm (inch)	875 (34-15/32)		
	Depth	mm (inch)	380 (14-31/32)		
Net Weight		kg (lbs)	55 (121)		
Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	15.88 (5/8)		
Standard Length		m (ft)	7 (23.0)		
Pipe Length Range		m (ft)	3 (9.8) ~ 40 (131.2) / 3 (9.8) ~ 50 (164.0)		
I/D & O/D Height Difference		m (ft)	30 (98.4)		
Additional Gas Amount		g/m (oz/ft)	25 (0.3)		
Refrigeration Charge Less		m (ft)	10 (32.8)		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (4-poles)		
	Rated Output	kW	1.50		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	W	-		
	Output Power	W	60		
	Fan Speed	rpm	Cooling: 670 Heating: 520		
Heat Exchanger	Fin material		Aluminium (Pre Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 36 × 19		
	Size (W × H × L)	mm	36.38 × 756.0 × 868.8:897.0		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.21	Heating: 1.44	Heating: 2.00
Maximum Input Power For Heatpump System		kW	3.58		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 15.9 / 3.58k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	6.5		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 9.8	Heating: 6.5	Heating: 8.9
Maximum Current For Heatpump System		A	15.9		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 96	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min./max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min./max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -15 °C) *4 Heating (Circuit): 20 / 60 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 18.0 Heating: 19.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 28***	Heating: 28***	
		Power Level dB	Cooling: 41***	Heating: 41***	
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1642 (64-41/64)		
Net Weight		kg (lbs)	96 (212)		
Refrigerant Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	15.88 (5/8)		
Water Pipe Diameter	Room	mm (inch)	1		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		DC Motor		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		36		
	Size (W × H × L)	mm	120 × 65 × 376		
	Water Flow Rate	l/min (m^3/h)	Cooling: 19.2 (1.2) Heating: 20.1 (1.2)		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer Pressure range		Bar	0 ~ 4		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW/°C	- / -		
Tank Volume (Spec / Nett)		L	200 / 185		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat/Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		En-1.4521
	Volume	L	185
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.8
	Surface Area	m^2	1.8
	Total Length	m	25
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60°C	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65°C	µS/cm	< 1200
	Saturation Index (LSI) @ 20°C		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in accordance with EU directive 2003/32/EC.
- *3 Above 55°C, only possible with backup heater operation.
- *4 Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- It is recommended to follow DHW tank water quality limit for Panasonic Air to Water All in One according to Drinking Water Directive 98/83 EC

3.4 WH-ADC0309K0E5A WH-UDZ09KE5

Item		Unit	Outdoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	8.20		
		BTU/h	28000		
Cooling EER		W/W	2.72		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	9.00	7.00	
		BTU/h	30700	23900	
Heating COP		W/W	4.55	3.40	
Heating Erp	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	7.0	8.0	7.0
	Tbivalent/TOL	°C	2 / 2	-7 / -10	-15 / -22
	SCOP/ns	(W/W)/%	5.75 / 227	4.44 / 175	4.18 / 164
	Annual Consumption	kWh	1627	3720	4132
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	6.0	8.0	6.0
	Tbivalent/TOL	°C	2 / 2	-7 / -10	-15 / -22
	SCOP/ns	(W/W)/%	4.07 / 160	3.41 / 133	2.98 / 116
	Annual Consumption	kWh	1971	4851	4967
	Class		A+++	A++	A+
	DHW		Warmer	Average	Colder
	Application	Climate			
	COP/nwh	(W/W)/%	4.00 / 160	3.34 / 134	2.80 / 112
	AEC	kWh	630	750	900
	Class		A++	A+	A
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 51***	Heating: 51***	
		Power Level dB	Cooling: 65***	Heating: 63*** Heating: 56****	
Air Flow		m^3/min (ft^3/min)	Cooling: 55.0 (1942) Heating: 52.7 (1860)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (600)		
Refrigerant		kg (oz)	R32, 1.30 (45.9) (Pre-charged) R32, 2.30 (81.2) (Maximum)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged/Maximum)		0.878 / 1.553		
Dimension	Height	mm (inch)	795 (31-5/16)		
	Width	mm (inch)	875 (34-15/32)		
	Depth	mm (inch)	380 (14-31/32)		
Net Weight		kg (lbs)	55 (121)		

Item		Unit	Outdoor Unit		
Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	15.88 (5/8)		
Standard Length		m (ft)	7 (23.0)		
Pipe Length Range		m (ft)	3 (9.8) ~ 40 (131.2) / 3 (9.8) ~ 50 (164.0)		
I/D & O/D Height Difference		m (ft)	30 (98.4)		
Additional Gas Amount		g/m (oz/ft)	25 (0.3)		
Refrigeration Charge Less		m (ft)	10 (32.8)		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (4-poles)		
	Rated Output	kW	1.50		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	W	-		
	Output Power	W	60		
	Fan Speed	rpm	Cooling: 700 Heating: 550		
Heat Exchanger	Fin material		Aluminium (Pre Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 36 × 19		
	Size (W × H × L)	mm	36.38 × 756.0 × 868.0:897.0		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 3.02	Heating: 1.98	Heating: 2.06
Maximum Input Power For Heatpump System		kW	3.58		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 15.9 / 3.58k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	8.9		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 13.4	Heating: 8.9	Heating: 9.1
Maximum Current For Heatpump System		A	15.9		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 97	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min./max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min./max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -15 °C) *4 Heating (Circuit): 20 / 60 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 24.0 Heating: 27.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 28***	Heating: 28***	
		Power Level dB	Cooling: 41***	Heating: 41***	
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1642 (64-41/64)		
Net Weight		kg (lbs)	96 (212)		
Refrigerant Pipe Diameter	Liquid	mm (inch)	6.35 (1/4)		
	Gas	mm (inch)	15.88 (5/8)		
Water Pipe Diameter	Room	mm (inch)	1		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		DC Motor		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		36		
	Size (W × H × L)	mm	120 × 65 × 376		
	Water Flow Rate	l/min (m^3/h)	Cooling: 23.5 (1.4) Heating: 25.8 (1.5)		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer Pressure range		Bar	0 ~ 4		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW/°C	- / -		
Tank Volume (Spec / Nett)		L	200 / 185		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat/Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		En-1.4521
	Volume	L	185
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.8
	Surface Area	m^2	1.8
	Total Length	m	25
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60°C	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65°C	µS/cm	< 1200
	Saturation Index (LSI) @ 20°C		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in accordance with EU directive 2003/32/EC.
- *3 Above 55°C, only possible with backup heater operation.
- *4 Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- It is recommended to follow DHW tank water quality limit for Panasonic Air to Water All in One according to Drinking Water Directive 98/83 EC

4. Features

- **Inverter Technology**
  - Energy saving

- **High Efficiency**

- **Environment Protection**
  - Non-ozone depletion substances refrigerant (R32)

- **Long Installation Piping**
  - Long piping up to 25 meter (UDZ03KE5), 50 meter (UDZ05/07/09KE5) with height difference 20 meter (UDZ03KE5), 30 meter (UDZ05/07/09KE5)

- **Easy to use control panel**
  - Auto mode
  - Holiday mode
  - Dry concrete function
  - Weekly timer setting

- **A-class energy efficiency pump**
  - Water pump speed can be set by selection at control panel

- **Improved deice cycle**

- **Protection Feature**
  - Random auto restart after power failure for safety restart operation
  - Gas leakage protection
  - Prevent compressor reverse cycle
  - Inner protector to protect compressor

- **Serviceability Feature**
  - Breakdown Self Diagnosis function
  - System Status Check Buttons for servicing purpose
  - System Pumpdown Button for servicing purpose

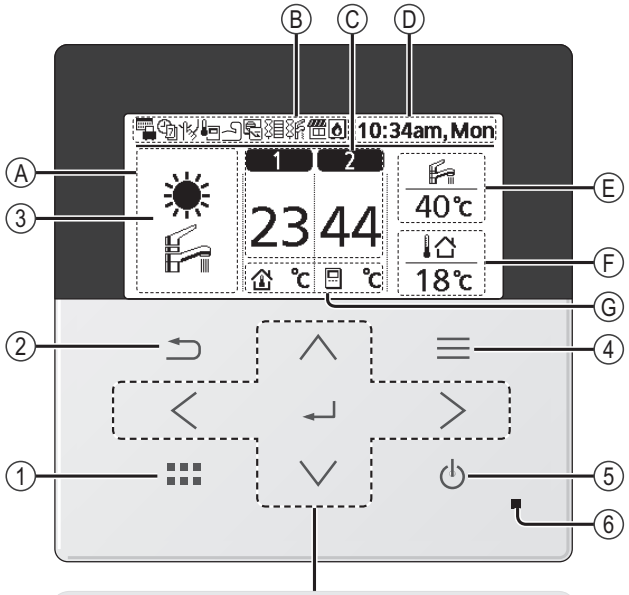
5. Location of Controls and Components

5.1 Indoor Unit

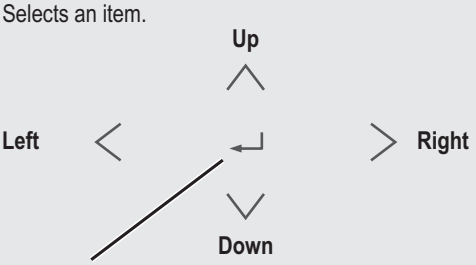
5.1.1 Remote Controller buttons and display

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.

Buttons / Indicator	
①	**Quick Menu button**
②	**Back button** Returns to the previous screen
③	**LCD Display** (Actual - Dark background with white icons)
④	**Main Menu button** For function setup
⑤	**ON/OFF button** Starts/Stops operation
⑥	**Operation indicator** Illuminates during operation, blinks during alarm.

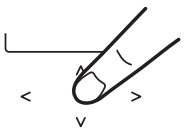


Cross key buttons
Selects an item.

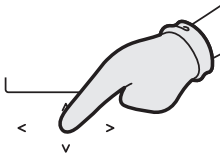


Enter button
Fixes the selected content.

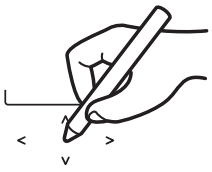
Press centre



No glove

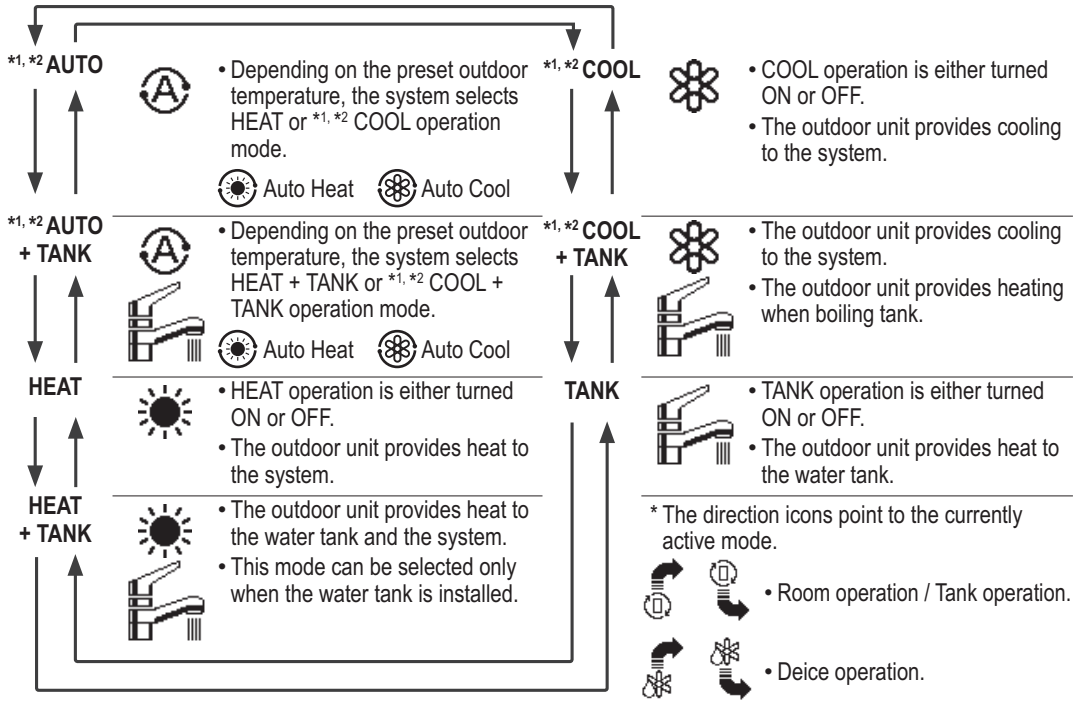


No pen



Display

Ⓐ Mode selection



Ⓑ Operation icons

The status of operation is displayed.
Icon will not display (under operation OFF screen) whenever operation is OFF except weekly timer.

- Holiday operation status
- Weekly Timer operation status
- Quiet operation status
- Zone:Room Thermostat →Internal sensor status
- Powerful operation status
- Demand Control or SG ready or SHP status
- Room Heater status
- Tank Heater status
- Solar status
- Bivalent status (Boiler)

Ⓒ Temperature of each zone

Ⓓ Time and day

Ⓔ Water Tank temperature

Ⓕ Outdoor temperature

Ⓖ Sensor type/Set temperature type icons

- Water Temperature →Compensation curve
- Water Temperature →Direct
- Pool only
- Room Thermostat →External
- Room Thermostat →Internal
- Room Thermistor

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

5.1.2 Initialization

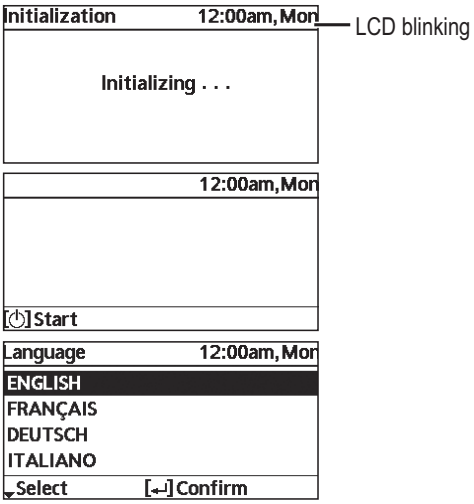
Before starting to install the various menu settings, please initiate the Remote Controller by selecting the language of operation and installing the date and time correctly.
When power is turned on for the first time, it becomes the setting screen automatically. It can also be set from personal setting of the menu.

Selecting the language

Wait while the display is initializing.
When initializing screen ends, it turns to normal screen.
When any button is pressed, language setting screen appears.

① **Scroll with ∨ and ∧ to select the language.**

② **Press ↵ to confirm the selection.**



Setting the clock

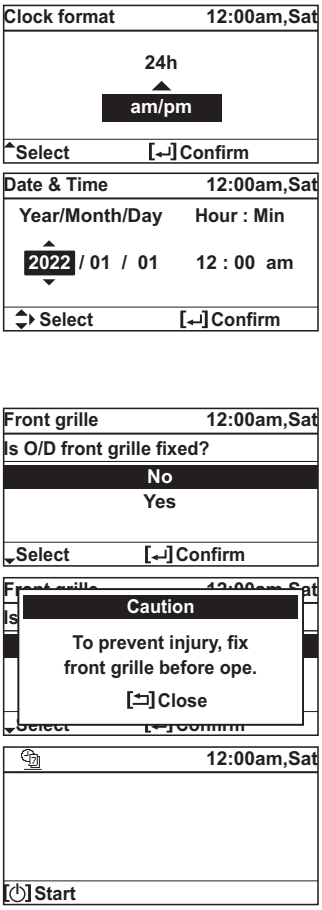
① **Select with ∨ or ∧ how to display the time, either 24h or am/pm format (for example, 15:00 or 3:00 pm).**

② **Press ↵ to confirm the selection.**

③ **Use ∨ and ∧ to select year, month, day, hour and minutes. (Select and move with ∨ and press ↵ to confirm.)**

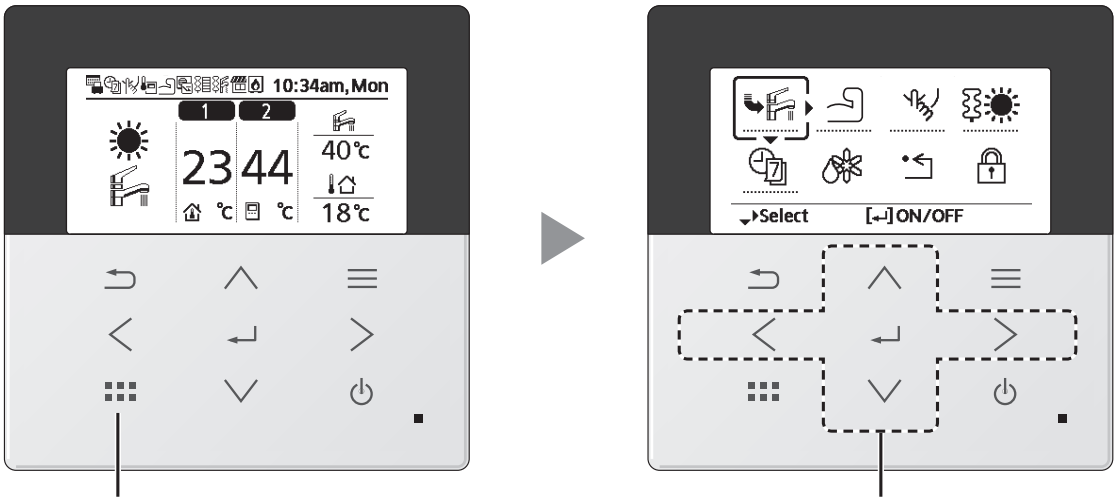
④ **Once the time is set, time and day will appear on the display even if the Remote Controller is turned OFF.**

⑤ **Final precaution step to check and confirm whether outdoor front grille is fixed before operating the unit for safety purpose.**
Select Yes if outdoor front grille is already fixed. Then it will proceed to main screen.
Select No if outdoor front grille is not yet fixed. A caution message will pop up to remind on the installation.



5.1.3 Quick Menu

After the initial settings have been completed, you can select a quick menu from the following options and edit the setting.



① Press ⋮⋮⋮ to display the quick menu.

② Use ∧ ∨ < > to select menu.

③ Press ↵ to turn on/off the select menu.

Quick Menu

Force DHW	Powerful	Quiet	Force Heater
Weekly Timer	Force Defrost	Error Reset	R/C Lock

▾Select	[↵] ON/OFF

Select each setting and confirm the setting according to the instructions displayed at the bottom of the screen. (The icons refer to each selection key.)

To return to the Main Screen,

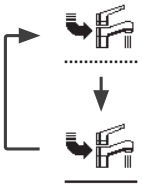
Press ⋮⋮⋮ or ⮌ .

5.1.4 How to use the Quick Menu

Force DHW

Select this icon to turn the Tank DHW on or off.

Press ↵ to confirm your selection.



• Force DHW is turned off.

• Force DHW is turned on.

Note:

• Force DHW is disabled when Force Heater is turned on.

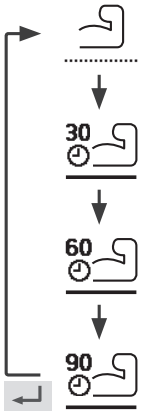
• When Force DHW is turned off, operation & mode should change back to the previous memorized status.

Powerful

Select this icon to operate the heating/cooling system powerfully.

Press ↵ to confirm your selection.

(The powerful operation starts approximately 1 minute after ↵ is pressed.)



• Powerful mode is set off.

• The Powerful mode operates for 30 minutes.

• The Powerful mode operates for 60 minutes.

• The Powerful mode operates for 90 minutes.

Note:

• Powerful is disabled when operation is turned OFF.

Quiet

Select this icon to operate quietly.

Press ↵ to confirm your selection.

(The quiet operation starts approximately 1 minute after ↵ is pressed.)

- • Quiet mode is set off.
- ↵
- • The Quiet mode operates at level 1.
- ↵
- • The Quiet mode operates at level 2.
- ↵
- • The Quiet mode operates at level 3.
  Level 3 is the most quiet mode.
- ↵
- • Select this to set a timer for the Quiet mode operation.
- After 1 sec delay.



Select "Yes".

• Select "Yes" using < > buttons.

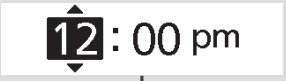


Select pattern "1" ~ "6".



Select "Edit".

• If you select "Delete", the timer setting of the selected pattern will be deleted.



Set the hour and minutes.



Select the level of Quiet.



Note:

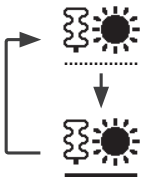
• If the time overlaps with another pattern, "Set time is overlapped!" will appear on the screen.

Force Heater

Select to force the Heater on.

Press ↵ to confirm your selection.

(The Force Heater mode starts approximately 1 minute after ↵ is pressed.)

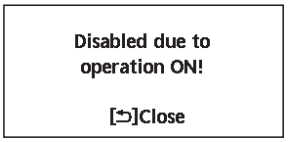


• Force Heater is turned off.

• Force Heater is turned on.

Note:

• Force Heater is disabled whenever operation is already on and “Disabled due to operation ON!” will be displayed.



Weekly Timer

Select this icon to delete (cancel) or change the pre-set Weekly Timer.

Press ↵ to confirm your selection.

• Timer is not ON.

• Timer is set.

After 1 sec delay.



Select "Yes".

• If you select "No", the screen will return to the Main Screen.



• Timer setup: Select Timer setup to edit the Weekly Timer.

• Timer copy: Select to copy a timer setting.

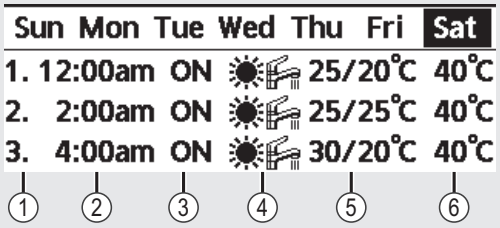


[Example of a Timer setup]

Select the day(s) which you wish to edit using ∧ ∨ buttons.



If all 6 patterns are not preset, this screen will be displayed.



① **Select pattern "1" ~ "6".**

② **Set the hour and minutes of the Timer.**

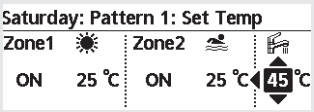
③ **Select ON/OFF of the Timer.**

④ **Select the operation mode.**

Ⓐ / Ⓐ / ☀ / ☀ / / / ❀ / ❀

• Select mode using ∧ ∨ buttons.

⑤ **Set the temperature for both Zone 1 and 2 (if your system has the 2-Zone setting).**



⑥ **Set the Tank temperature.**

Note:

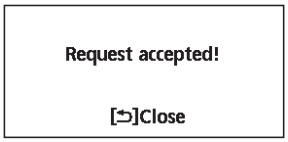
• Timer is disabled when Force Heater is turned on or Heat-Cool SW is enabled.

• If you have preset the Weekly Timer on 2 zones, you must repeat the same procedure with Zone 2.

Force Defrost

Select to defrost the frozen pipes.

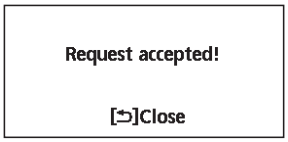
Press ↵ to confirm your selection.
(When the mode is accepted, below screen will be displayed.)



Error Reset

Select to restore the previous settings when error has occurred.

Press ↵ to confirm your selection.
(When the mode has been accepted, below screen will be displayed.)

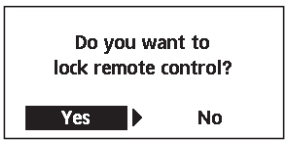


- Make sure all units are turned off before selecting this mode which restores the whole system to the previous settings.

R/C Lock

Select to lock the Remote Controller.

Press ↵ to confirm your selection.
(When the mode has been accepted, below screen will be displayed.)



Select "Yes".
(The Main Screen will be locked.)
- If "No" is selected, the screen will return to the Main Screen.

To unlock the Remote Controller

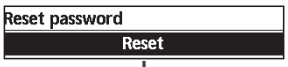
Press any key.
(When the mode has been accepted, below screen will be displayed.)



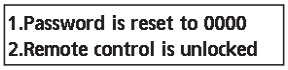
Enter any 4 digits of number (if the number is correct, the screen will be unlocked).

To reset forgotten password (under operation OFF screen)

Press ⮌ , ↵ and > continuously for 5 seconds.
(When the mode has been accepted, below screen will be displayed.)



Select "Reset".



(The screen will be off after 3 seconds.)

5.1.5 Menus (For user)

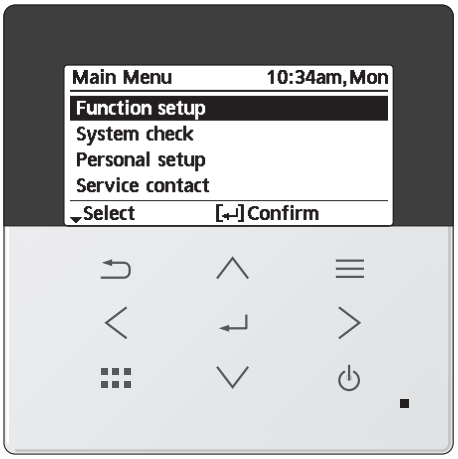
Select menus and determine settings according to the system available in the household. All initial settings must be done by an authorised dealer or a specialist. It is recommended that all alterations of the initial settings are also done by an authorised dealer or a specialist.

- After initial installation, you may manually adjust the settings.
- The initial setting remains active until the user changes it.
- The Remote Controller can be used for multiple installations.
- Ensure the operation indicator is OFF before setting.
- The system may not work properly if set wrongly. Please consult an authorised dealer.

To display <Main Menu>:

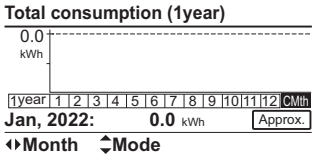
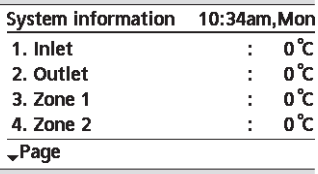
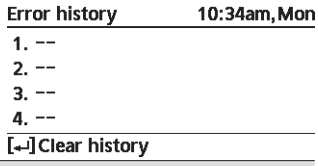
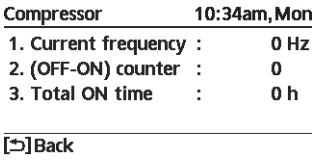
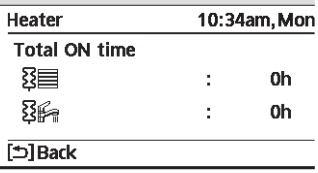
To select menu:

To confirm the selected content:

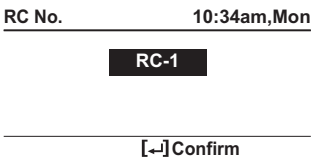
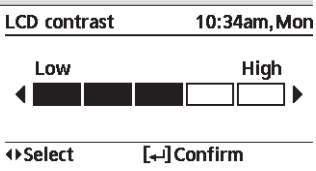
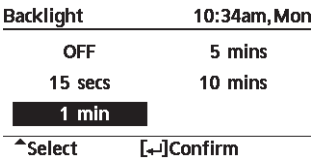
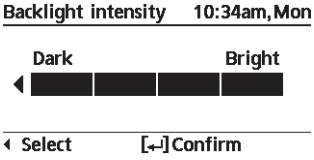
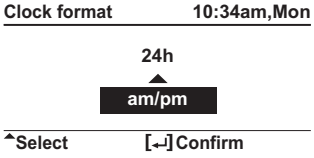
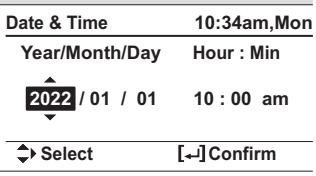


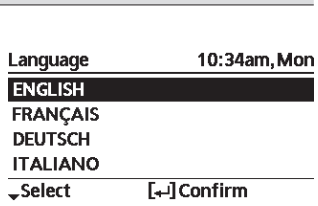
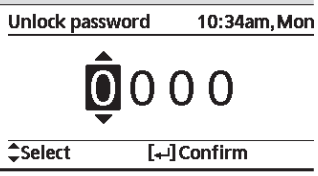
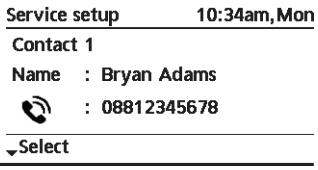
Menu	Default Setting	Setting Options / Display
1 Function setup		
1.1 > Weekly timer		
Once the weekly timer is set up, User can edit from Quick Menu. To set up to 6 patterns of operation on a daily basis. • Disabled if Heat-Cool SW is select "Yes" or if Force Heater is on.	**Timer setup** Select day of the week and set the patterns needed (Time / Operation ON/OFF / Mode)	Weekly timer 10:34am,Mon Sun Mon Tue Wed Thu Fri Sat 1. 8:00am ON 40℃ 2. 12:00pm ON 24/28℃ 40℃ 3. 1:00pm ON 12/10℃ Day Pattern [↵]Edit
	Timer copy Select day of the week	
1.2 > Holiday timer		
To save energy, a holiday period may be set to either turn OFF the system or lower the temperature during the period.	OFF	ON OFF
	> ON	
	Holiday start and end. Date and time	Holiday: End 10:34am,Mon Year/Month/Day Hour : Min 2022 / 01 / 01 10 : 00 am Select [↵]Confirm
	OFF or lowered temperature	
• Weekly timer setting may be temporarily disabled during Holiday timer setting but it will be restored once the Holiday timer is completed.		
1.3 > Quiet timer		
To operate quietly during the preset period. 6 patterns may be set. Level 0 means the mode is off.	Time to start Quiet : Date and time	Quiet 10:34am,Mon Pattern Time Level 1 8:00am 0 2 5:00pm 1 3 11:00pm 3 Select [↵]Edit
	Level of quietness: 0 ~ 3	

Menu	Default Setting	Setting Options / Display
1.4 > Quiet priority		
• To select priority during Quiet mode between Sound and Capacity. • If Sound priority is selected, unit will operate in quiet condition only. • If Capacity priority is selected, unit will operate in quiet condition but it will prioritize on providing required capacity at the same time.	Sound	Sound Capacity
1.5 > Room heater		
To set the room heater ON or OFF.	OFF	ON OFF
1.6 > Tank heater		
To set the tank heater ON or OFF.	OFF	ON OFF
1.7 > Sterilization		
To set the auto sterilization ON or OFF.	ON	ON OFF
• Do not use the system during sterilization in order to prevent scalding with hot water, or overheating of shower. • Ask an authorised dealer to determine the level of sterilization function field settings according to the local laws and regulations.		
1.8 > DHW mode (Domestic Hot Water)		
To set the DHW mode to Standard or Smart. • Standard mode have faster DHW Tank heat up time. Meanwhile Smart mode take longer time to heat up DHW time with lower energy consumption.	Standard	Standard Smart
To set the tank sensor to Top or Center. • Selection of the tank sensor to top slow down the start of boiling up the tank and reduce power consumption. Please change this selection to “Center” when the hot water becomes insufficient.	Top	Top Center

Menu	Default Setting	Setting Options / Display
2 System check		
2.1 > Energy monitor		
Present or historical chart of energy consumption, generation or COP.	**Present** Select and retrieve **Historical chart** Select and retrieve	
• COP= Coefficient of Performance. • For historical chart, the period is selected from 1 day/1 week/1year. • Energy consumption (kWh) of heating, *1, *2 cooling, tank and total may be retrieved. • The total power consumption is an estimated value based on AC 230 V and may differ from value measured by precise equipment.		
2.2 > System information		
Shows all system information in each area.	**Actual system information of 11 items: Inlet / Outlet / Zone 1 / Zone 2 / Tank / Buffer tank / Solar / Pool / COMP frequency / Pump flowrate / Water pressure** Select and retrieve	
2.3 > Error history		
• Refer to Troubleshooting for error codes. • The most recent error code is displayed at the top.	Select and retrieve	
2.4 > Compressor		
Shows the compressor performance.	Select and retrieve	
2.5 > Heater		
Total hours of ON time for Room heater/Tank heater.	Select and retrieve	

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
3 Personal setup		
3.1 > Remote control No.		
• To display remote control number of a particular remote controller so that installer and end user are well informed. • Main remote controller is displayed as RC-1. Second remote controller is displayed as RC-2.	Select and retrieve	
3.2 > Touch sound		
Turns the operation sound ON/ OFF.	ON	
3.3 > LCD contrast		
Sets the screen contrast.	3	
3.4 > Backlight		
Sets the duration of screen backlight.	1 min	
3.5 > Backlight intensity		
Sets screen backlight brightness.	4	
3.6 > Clock format		
Sets the type of clock display.	am/pm	
3.7 > Date & Time		
Sets the present date and time.	Year / Month / Day / Hour / Min	

Menu	Default Setting	Setting Options / Display
3.8 > Language		
Sets the display language for the top screen.	ENGLISH / FRANÇAIS / DEUTSCH / ITALIANO / ESPAÑOL / DANISH / SWEDISH / NORWEGIAN / POLISH / CZECH / NEDERLANDS / TÜRKÇE / SUOMI / MAGYAR / SLOVENŠČINA / HRVATSKI / LIETUVIŲ / PORTUGUÊS / БЪЛГАРСКИ / EESTI / LATVIEŠU / ROMÂNĂ / SHQIP / SLOVENČINA / МАКЕДОНСКИ / УКРАЇНСЬКА / ΕΛΛΗΝΙΚΑ	
3.9 > Unlock password		
4 digit password for all the settings.	0000	
4 Service contact		
4.1 > Contact 1 / Contact 2		
Preset contact number for installer.	Select and retrieve	

5.1.6 Menus (For installer)

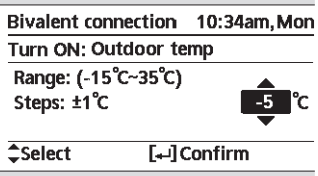
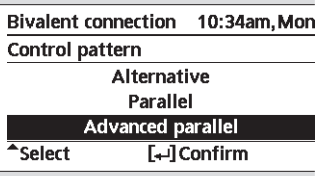
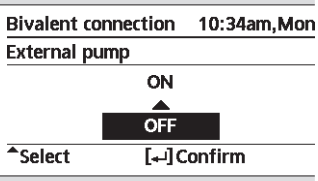
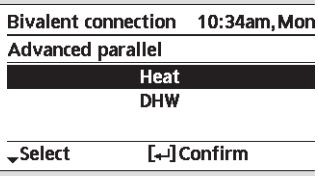
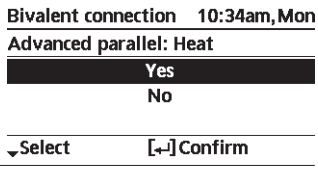
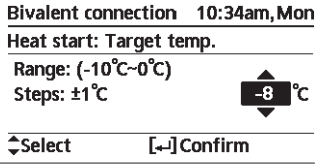
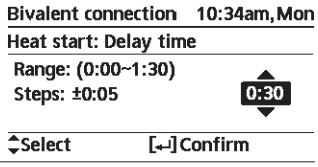
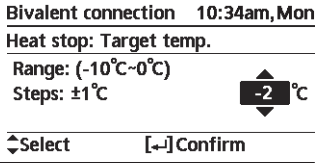
Menu	Default Setting	Setting Options / Display
5 Installer setup > System setup		
5.1 > Optional PCB connectivity		
To connect to the external PCB required for servicing.	No	Yes ▲ No

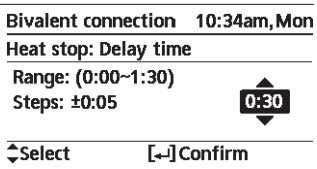
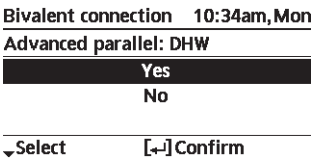
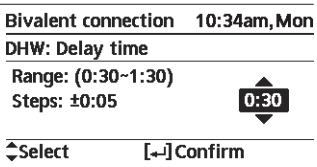
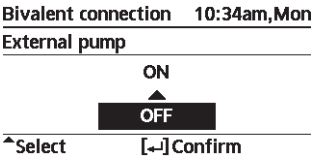
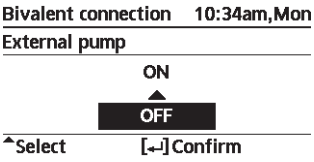
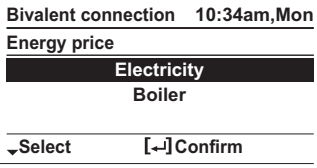
• If the external PCB is connected (optional), the system will have following additional functions:

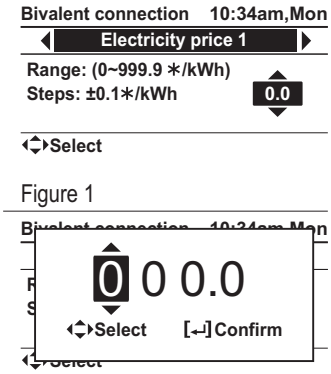
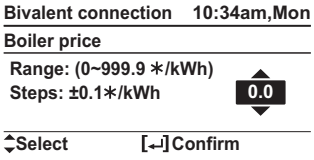
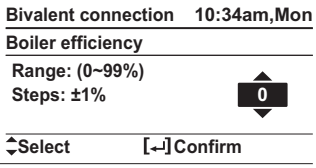
① Control over 2 zones (including the swimming pool and the function to heat water in it).
② Solar function (the solar thermal panels connected to either the DHW (Domestic Hot Water) Tank or the Buffer Tank.
  • DHW is not applicable for WH-ADC *models.
③ External compressor switch.
④ External error signal.
⑤ SG ready control.
⑥ Demand control.
⑦ Heat-Cool SW

5.2 > Zone & Sensor		
To select the sensors and to select either 1 zone or 2 zone system.	**Zone** • After selecting 1 or 2 zone system, proceed to the selection of room or swimming pool. • If the swimming pool is selected, the temperature must be selected for △T temperature between 0°C ~ 10 °C.	Zone & Sensor 10:34am,Mon Zone 1 Zone system 2 Zones system ▼Select [↵]Confirm
	Sensor * For room thermostat, there is a further selection of external or internal. • If select internal, there is a further selection of RC-1 or RC-2 (only available when Zone selection is 1 zone system). Select RC-1 if main remote controller's thermistor is to be used for room temperature control and vice versa.	Zone & Sensor 10:34am,Mon Sensor Water temperature Room thermostat Room thermistor ▼Select [↵]Confirm
5.3 > Heater capacity		
To reduce the heater power if unnecessary.* 3 kW / 6 kW / 9 kW * Options of kW vary depending on the model.		Heater capacity 10:34am,Mon 3 kW [↵]Confirm
5.4 > Anti freezing		
To activate or deactivate the water freeze prevention when the system is OFF	Yes	Yes ▼ No

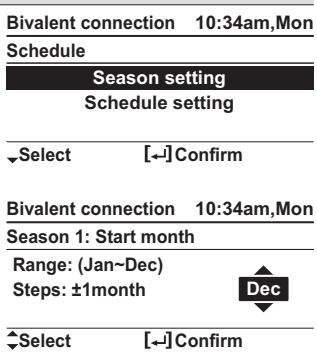
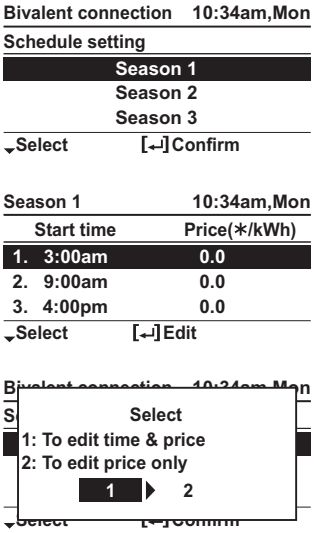
Menu	Default Setting	Setting Options / Display
5.5 > DHW capacity		
To select tank heating capacity to variable or standard. Variable capacity heat up tank with fast mode and keep the tank temperature with efficient mode. While standard capacity heat up tank with rated heating capacity.	Variable	Variable Standard
5.6 > Buffer tank connection		
To connect tank to the system and if selected YES, to set △T temperature.	No	Yes No
	> Yes	
	5 °C	Set △T for Buffer Tank Buffer tank 10:34am,Mon ΔT for Buffer tank Range: (0℃~10℃) Steps: ±1℃ 5 ℃ Select [↵] Confirm
5.7 > Base pan heater		
To select whether or not optional base pan heater is connected. * Type A - The base pan heater activates only during deice operation. * Type B - The base pan heater activates when outdoor ambient temperature is 5 °C or lower.	No	Yes No
	> Yes	
	A	Set base pan heater type*. Base pan heater type 10:34am,Mon A B Select [↵] Confirm
5.8 > Alternative outdoor sensor		
To select an alternative outdoor sensor.	No	Yes No
5.9 > Bivalent connection		
To select to enable or disable bivalent connection.	No	Yes No
> Yes		
To select either auto control pattern or SG ready input control pattern or smart control pattern. * This selection only display to select when optional pcb connection set to Yes.	Auto	Auto SG ready Smart

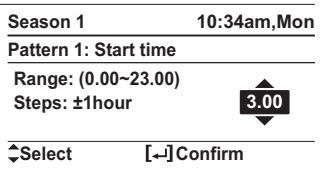
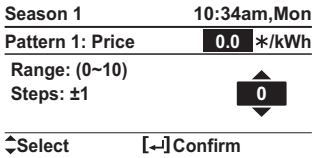
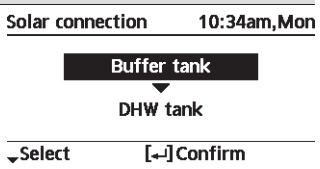
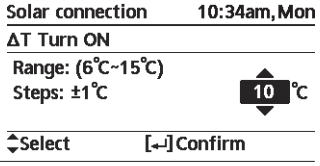
Menu	Default Setting	Setting Options / Display
To select a bivalent connection to allow an additional heat source such as a boiler to heat-up the buffer tank and domestic hot water tank when heatpump capacity is insufficient at low outdoor temperature. The bivalent feature can be set-up either in alternative mode (heatpump and boiler operate alternately), or in parallel mode (both heatpump and boiler operate simultaneously), or in advance parallel mode (heatpump operates and boiler turns on for buffer-tank and/or domestic hot water depending on the control pattern setting options).	**> Yes > Auto**	
	-5 °C	Set outdoor temperature for turn ON Bivalent connection. 
	Yes > After selecting the outdoor temperature	
	Control pattern Alternative / Parallel / Advanced parallel • Select advanced parallel for bivalent use of the tanks.	
	Control pattern > Alternative	
	OFF	Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection. 
	Control pattern > Advanced parallel	
	Heat • "Heat" implies Buffer Tank and "DHW" implies Domestic Hot Water Tank.	Selection of the tank 
	Control pattern > Advanced parallel > Heat > Yes	
	• Buffer Tank is activated only after selecting "Yes".	
	-8 °C	Set the temperature threshold to start the bivalent heat source. 
	0:30	Delay timer to start the bivalent heat source (in hour and minutes). 
	-2 °C	Set the temperature threshold to stop the bivalent heat source. 

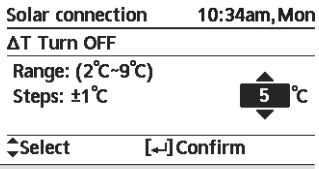
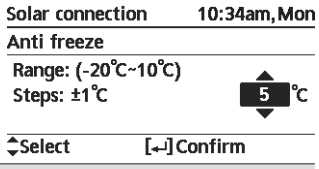
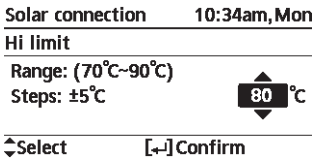
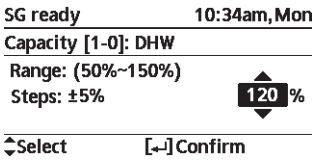
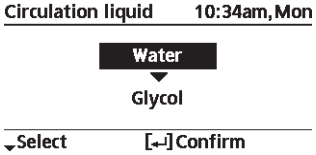
<table>
<tr><th>Menu</th><th>Default Setting</th><th>Setting Options / Display</th></tr>
<tr><td></td><td>0:30</td><td>Delay timer to stop the bivalent heat source
(in hour and minutes).</td><td> 
</td></tr>
<tr><td></td><td colspan="3">Control pattern > Advanced parallel > DHW > Yes</td></tr>
<tr><td></td><td colspan="2">• DHW Tank is activated only after selecting “Yes”.</td><td> 
</td></tr>
<tr><td></td><td>0:30</td><td>Delay timer to start the bivalent heat source
(in hour and minutes).</td><td> 
</td></tr>
<tr><td rowspan="2">SG ready input control for bivalent system follow below input condition.
<table><tr><th colspan="2">SG signal</th><th>Operation pattern</th></tr><tr><th>Vcc-bit1</th><th>Vcc-bit2</th><th></th></tr><tr><td>Open</td><td>Open</td><td>Heat Pump OFF, Boiler OFF</td></tr><tr><td>Short</td><td>Open</td><td>Heat Pump ON, Boiler OFF</td></tr><tr><td>Open</td><td>Short</td><td>Heat Pump OFF, Boiler ON</td></tr><tr><td>Short</td><td>Short</td><td>Heat Pump ON, Boiler ON</td></tr></table></td><td colspan="3">> Yes > SG ready</td></tr>
<tr><td>OFF</td><td>Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.</td><td> 
</td></tr>
<tr><td rowspan="3">To do settings related to electricity and boiler so that unit is able to determine whether to operate heat pump or boiler at a particular period depends on operating cost of both heat sources. These settings are electricity price, boiler price, season, schedule etc.</td><td colspan="3">> Yes > Smart</td></tr>
<tr><td>OFF</td><td>Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.</td><td> 
</td></tr>
<tr><td colspan="3">> Yes > Smart > After selecting for the external pump > Energy price</td></tr>
<tr><td></td><td colspan="2">- Select Electricity to set on electricity price.
- Select Boiler to set on boiler price and its efficiency.</td><td> 
</td></tr>
</table>

Menu	Default Setting	Setting Options / Display
	> Yes > Smart > After selecting for the external pump > Energy price > Electricity	
	0.0 * / kWh - There are total 10 different prices can be set for Electricity: Electricity price 1 ~ Electricity price 10 - Range is 0 ~ 999.9 * / kWh - Press ∧ or ∨ to enter a setting screen as shown in Figure 1. Then start setting the value of electricity price. - After finish setting a particular electricity price (eg. Electricity price 1), press < or > to go and set for other electricity price. * Set the price according to value provided by electrical supply company.	
	> Yes > Smart > After selecting for the external pump > Energy price > Boiler	
	0.0 * / kWh - Refer to method of Electricity price setting above for setting of boiler price. - After finish setting of boiler price, set the boiler efficiency (Range : 0 ~ 99%).	
	0% * Set the price according to value provided by boiler or gas supply company.	

Remark : * implies cents in most currency except Czech crown.

Menu	Default Setting	Setting Options / Display
	> Yes > Smart > After selecting for the external pump > Schedule > Season setting	
	Season 1 : Dec (Refers to Winter season) Season 2 : Mar (Refers to Spring season) Season 3 : Jun (Refers to Summer season) Season 4 : Oct (Refers to Autumn season) - There are total 4 seasons to be set - Set the starting month for each season. (Eg. when Season 1 is set to Dec and Season 2 is set to Mar, month of December to February will be treated as Season 1).	
	> Yes > Smart > After selecting for the external pump > Schedule > Schedule setting	
	Start time (Pattern 1) : 3:00am Start time (Pattern 2) : 9:00am Start time (Pattern 3) : 4:00pm Start time (Pattern 4) : 9:00pm - For each season, there are total 4 patterns can be set. Price (Pattern 1/2/3/4) : 1 - Set the target start time and the appropriate electricity price for each pattern. - Select “1” to edit both start time and electricity price. Select “2” to edit electricity price only.	

Menu	Default Setting	Setting Options / Display
	- Range of start time displayed can be in “24h” or “am/pm” format depend on setting of “Clock format”.	
	- Range of electricity price is 0 ~ 10 which refers back to the 10 different electricity price set previously (under “Energy price > Electricity”: Electricity price 1 ~ Electricity price 10). The price displayed on the upper right corner indicates the previous set value of Electricity price 1 to Electricity price 10. * When the price is set to “0”, the electricity price will be treated as 0.0 * / kWh. It is for the convenience of installer when 0.0 is the desired setting value for a particular time.	
5.10 **> External SW**		
	No	
5.11 **> Solar connection**		
• The optional PCB connectivity must be selected YES to enable the function. • If the optional PCB connectivity is not selected, the function will not appear on the display. • DHW is not applicable for WH-ADC *models.	No	
	> Yes	
	Buffer tank — Selection of the tank	
	> Yes > After selecting the tank	
	10 °C — Set △T ON temperature	

Menu	Default Setting	Setting Options / Display
	> Yes > After selecting the tank > △T ON temperature	
	5 °C	Set △T OFF temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature	
	5 °C	Set Antifreeze temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature > After setting the antifreeze temperature	
	80 °C	Set Hi limit 
5.12	**> External error signal**	
	No	
5.13	**> Demand control**	
	No	
5.14	**> SG ready**	
	No	
	> Yes	
	120 %	Capacity (1) & (2) of DHW (in %), Heat (in %) and Cool (in °C) 
5.15	**> External compressor SW**	
	No	
5.16	**> Circulation liquid**	
	To select whether to circulate water or glycol in the system.	Water 

Menu	Default Setting	Setting Options / Display
5.17 > Heat-Cool SW		
	No	Yes No
5.18 > Force heater		
To turn on Force heater either manually (by default) or automatically.	Manual	Force heater 10:34am,Mon Auto Manual Select [↵] Confirm
5.19 > Force defrost		
If auto selection is set, outdoor unit will start defrost operation if long heating hour operate during low outdoor temperature.	Manual	Auto Manual
5.20 > Defrost signal		
To turn on defrost signal to stop fan coil during defrost operation. (If defrost signal set to yes, bivalent function will not available to use)	No	Yes No
5.21 > Pump flowrate		
To set variable flow pump control or fix pump duty control.	△T	ΔT Max. Duty
5.22 > DHW Defrost		
Allow system to run defrost by using hot water instead of room unit for better room comfort.	Yes	Yes No
5.23 > Heating control		
To select unit operation condition whether to achieve set temperature faster or to save energy.	Comfort	Comfort Efficiency

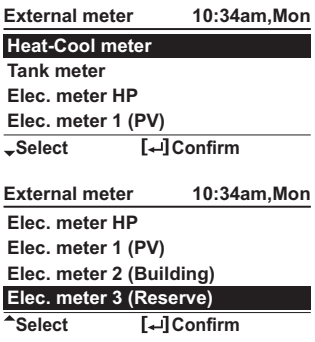
Menu	Default Setting	Setting Options / Display

5.24 › External meter

To set which external meter to be used depends on meter connection.
There are generation meters and various types of electricity meters.
For generation meters, there are two connection systems :-
a) One generation meter system :
Heat-cool meter only
b) Two generation meter system :
Heat-cool meter and Tank meter

Heat-cool meter : No
* Tank meter : No
Elec. meter HP : No
Elec. meter 1 (PV) : No
Elec. meter 2 (Building) : No
Elec. meter 3 (Reserve) : No

* Only available when Heat-cool meter select Yes



› Heat-cool meter

- Set Heat-cool meter to Yes when this generation meter is connected.
- It is to measure energy generation of heat pump unit during heating, cooling and DHW operation (one generation meter system) or during heating and cooling only (two generation meter system).

Yes / No

› Tank meter

- Set Tank meter to Yes when this generation meter is connected.
- It is to measure energy generation of heat pump unit during DHW operation.
  * Only available to select when Heat-cool meter is set to Yes.
    Only set Tank meter to Yes when the connection is two generation meter system.

Yes / No

› Elec. meter HP

- Set Elec. meter HP to Yes when this electricity meter is connected.
- It is to measure energy consumption of heat pump unit.

Yes / No

› Elec. meter 1 (PV)

- Set Elec. meter 1 (PV) to Yes when this electricity meter is connected.
- It is to measure energy generation of solar system.

* This data will be displayed only on Cloud system.

Yes / No

› Elec. meter 2 (Building)

- Set Elec. meter 2 (Building) to Yes when this electricity meter is connected.
- It is to measure energy consumption of the building.

* This data will be displayed only on Cloud system.

Yes / No

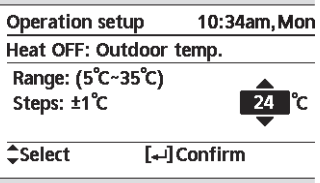
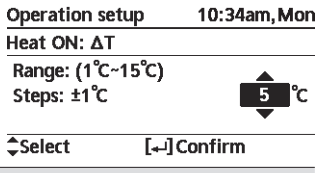
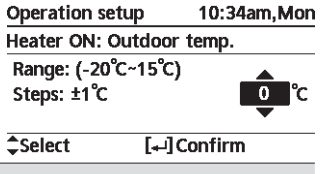
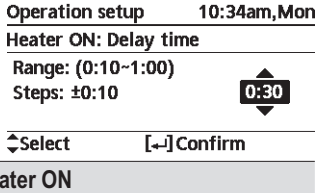
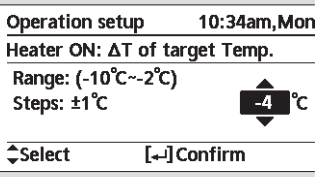
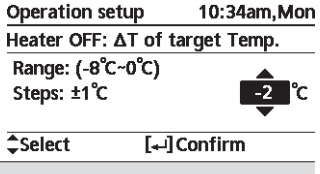
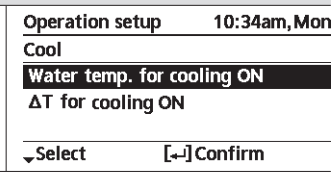
Menu	Default Setting	Setting Options / Display
	> Elec. meter 3 (Reserve)	
	- Set Elec. meter 3 (Reserve) to Yes when this electricity meter is connected. - It is to measure energy consumption * This data will be displayed only on Cloud system.	Yes ▲ No
5.25 > Electrical anode		
To enable or disable operation of electrical anode.	Yes (only for WH-ADC0309K6E5AN model) No (for non -AN models)	Yes ▼ No

(NOTE) : If [Approx.] is shown on Energy Monitor display, data displayed on the remote controller is obtained through heat pump's internal calculation.
If [Approx.] is NOT shown on Energy Monitor display, data** displayed on the remote controller is obtained by External Meters.
Data stored on the Aquarea unit can be mixed between internal calculation and External Meters.
**In order to know the exact consumption or generation, please use as reference always the External Meters' data.

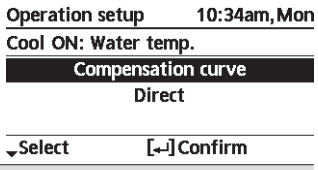
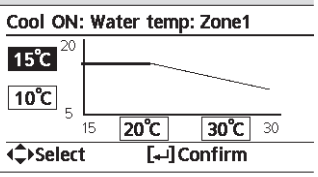
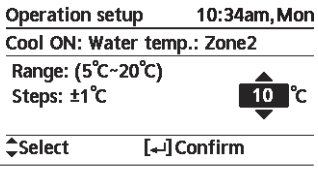
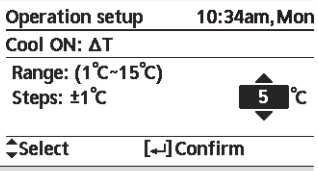
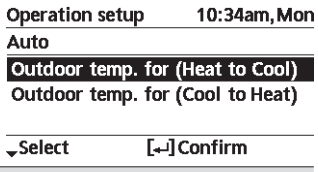
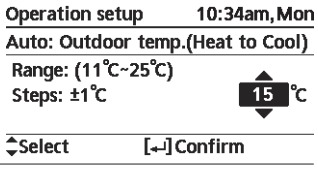
Remark : Elec. stands for "Electricity"
HP stands for "Heat pump"

Menu	Default Setting	Setting Options / Display
6 Installer setup > Operation setup		
To access to the four major functions or modes.	4 main modes Heat / *1, *2 Cool / *1, *2 Auto / Tank	Operation setup 10:34am,Mon Heat Cool Auto Tank Select [↵]Confirm
6.1 > Heat		
To set various water & ambient temperatures for heating.	Water temp. for heating ON / Outdoor temp. for heating OFF / △T for heating ON / Heater ON/OFF	Operation setup 10:34am,Mon Heat Water temp. for heating ON Outdoor temp. for heating OFF ΔT for heating ON Select [↵]Confirm
> Water temp. for heating ON		
Compensation curve	Heating ON temperatures in compensation curve or direct input.	Operation setup 10:34am,Mon Heat ON: Water temp. Compensation curve Direct Select [↵]Confirm
> Water temp. for heating ON > Compensation curve		
X axis: -5 °C, 15 °C Y axis: 55 °C, 35 °C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis).	Heat ON: Water temp.:Zone1 55℃ 35℃ 60 20 -20 -5℃ 15℃ 15 Select [↵]Confirm
• Temperature range: X axis: -20 °C ~ 15 °C, Y axis: See below • Temperature range for the Y axis input: 1. WH-UD model: 20 °C ~ 60 °C 2. WH-UH model & Back up heater is enabled: 25 °C ~ 65 °C 3. WH-UH model & Back up heater is disabled: 35 °C ~ 65 °C 4. WH-UX model: 20 °C ~ 60 °C • If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • “Zone 1” and “Zone 2” will not appear on the display if only 1 zone system.		
> Water temp. for heating ON > Direct		
35 °C	Temperature for heating ON	Operation setup 10:34am,Mon Heat ON: Water temp.:Zone2 Range: (20℃~60℃) Steps: ±1℃ 35 ℃ Select [↵]Confirm
• Min. ~ Max. range is conditional as follows: 1. WH-UD model: 20 °C ~ 60 °C 2. WH-UH model & Back up heater is enabled: 25 °C ~ 65 °C 3. WH-UH model & Back up heater is disabled: 35 °C ~ 65 °C 4. WH-UX model: 20 °C ~ 60 °C • If 2 zone system is selected, temperature set point must input for Zone 2. • “Zone 1” and “Zone 2” will not appear on the display if only 1 zone system.		

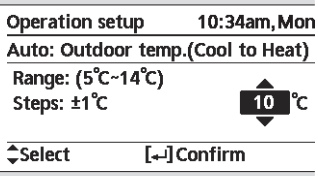
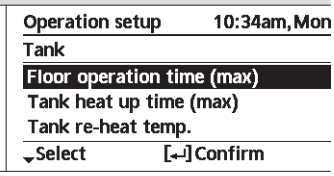
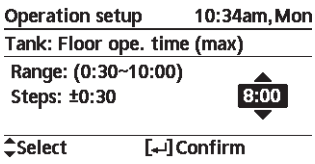
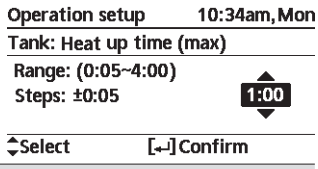
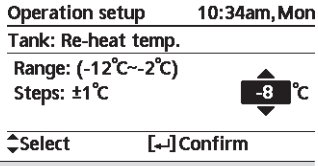
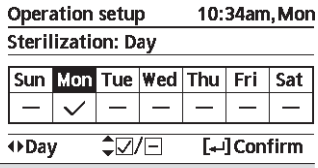
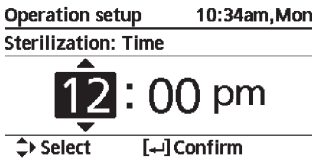
*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Outdoor temp. for heating OFF	
	24 °C	Temperature for heating OFF 
	> △T for heating ON	
	5 °C	Set △T for heating ON. * This setting will not available to set when pump flowrate set to Max. duty. 
	> Heater ON/OFF	
	> Heater ON/OFF > Outdoor temp. for heater ON	
	0 °C	Temperature for heater ON 
	> Heater ON/OFF > Delay time for heater ON	
	0:30 min	Delay time for heater to turn on 
	> Heater ON/OFF > Water temperature for heater ON	
	-4 °C	Setting of water temperature to turn on from water set temperature. 
	> Heater ON/OFF > Water temperature for heater OFF	
	-2 °C	Setting of water temperature to turn off from water set temperature. 
6.2	**> *1, *2 Cool**	
To set various water & ambient temperatures for cooling.	Water temperatures for cooling ON and △T for cooling ON.	

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Water temp. for cooling ON	
	Compensation curve	Cooling ON temperatures in compensation curve or direct input. 
	> Water temp. for cooling ON > Compensation curve	
	X axis: 20 °C, 30 °C Y axis: 15 °C, 10 °C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis) 
	• If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Water temp. for cooling ON > Direct	
	10 °C	Set temperature for Cooling ON 
	• If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> △T for cooling ON	
	5 °C	Set △T for cooling ON * This setting will not available to set when pump flowrate set to Max. duty. 
6.3	**> *1, *2 Auto**	
Automatic switch from Heat to Cool or Cool to Heat.	Outdoor temperatures for switching from Heat to Cool or Cool to Heat. Outdoor temp. for (Heat to Cool) / Outdoor temp. for (Cool to Heat)	
	> Outdoor temp. for (Heat to Cool)	
	15 °C	Set outdoor temperature for switching from Heat to Cool. 

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Outdoor temp. for (Cool to Heat)	
	10 °C	Set outdoor temperature for switching from Cool to Heat. 
6.4 **> Tank**		
Setting functions for the tank.	Floor operation time (max) / Tank heat up time (max) / Tank re-heat temp. / Sterilization	
	• The display will show 3 functions at a time.	
	> Floor operation time (max)	
	8:00	Maximum time for floor operation (in hours and minutes) 
	> Tank heat up time (max)	
	1:00	Maximum time for heating the tank (in hours and minutes) 
	> Tank re-heat temp.	
	-8 °C	Set temperature to perform reboil of tank water. 
	> Sterilization	
	Monday	Sterilization may be set for 1 or more days of the week. Sun / Mon / Tue / Wed / Thu / Fri / Sat 
	> Sterilization: Time	
	12:00	Time of the selected day(s) of the week to sterilize the tank 0:00 ~ 23:59 

Menu	Default Setting	Setting Options / Display
	> Sterilization: Boiling temp.	
	65 °C	Set boiling temperatures for sterilize the tank.
	> Sterilization: Ope. time (max)	
	0:10	Set sterilizing time (in hours and minutes)

7 Installer setup > Service setup

7.1 > Pump maximum speed

To set the maximum speed of the pump.

Setting the flow rate, max. duty and operation ON/OFF of the pump.

Flow rate: XX:X L/min
Max. Duty: 0x40 ~ 0xFE,
Pump: ON/OFF/Air Purge

7.2 > Pump down

To set the pump down operation.

Pump down operation

ON

7.3 > Dry concrete

To dry the concrete (floor, walls, etc.) during construction.

Do not use this menu for any other purposes and in period other than during construction

Edit to set the temperature of dry concrete.

ON / Edit

> Edit

Stages: 1
Temperature: 25 °C

Heating temperature for drying the concrete.
Select the desired stages: 1 ~ 10, range: 1 ~ 99

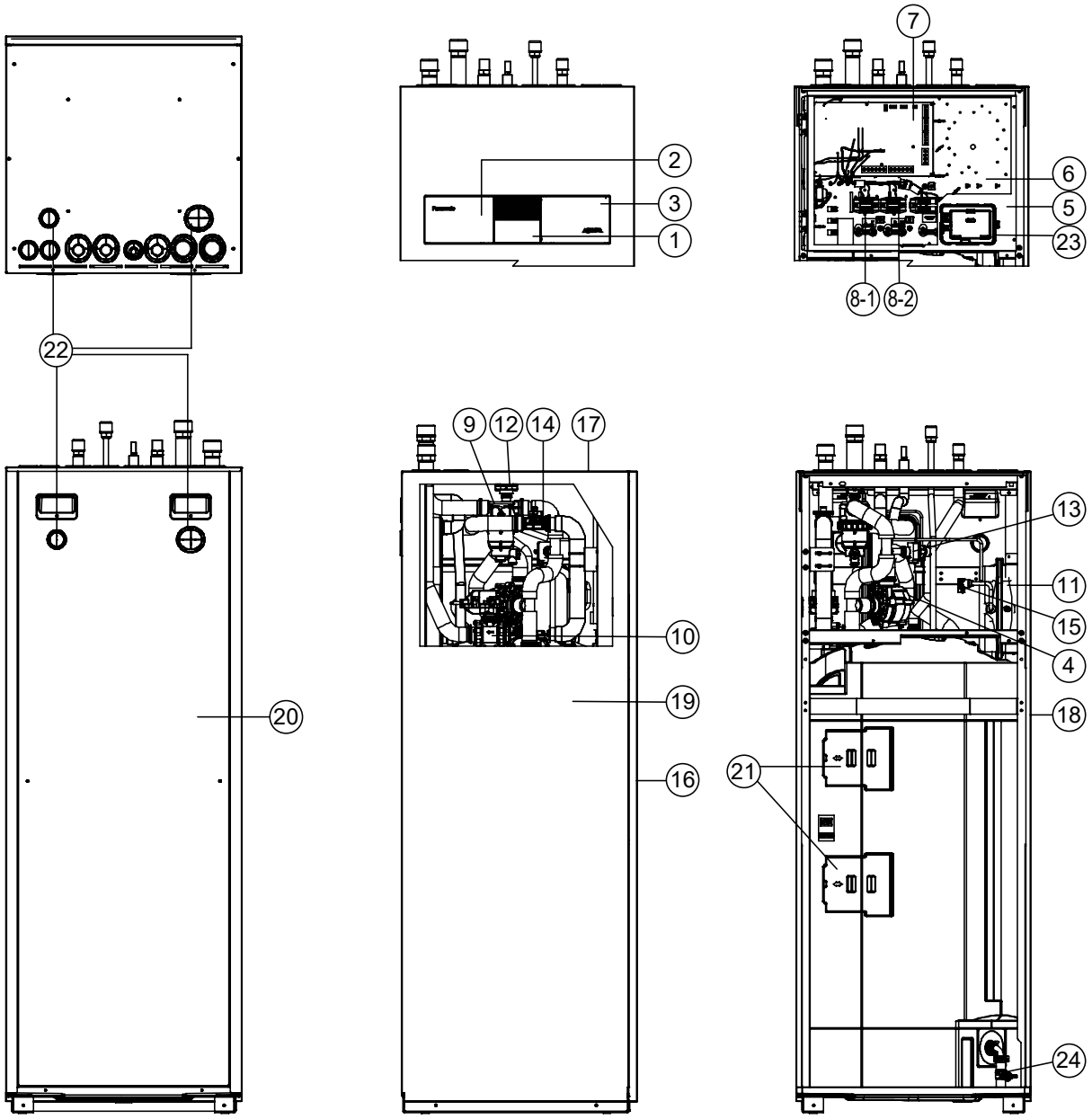
> ON

Confirm the setting temperatures of dry concrete for each stage.

Menu	Default Setting	Setting Options / Display
7.4 > Service contact		
To set up to 2 contact names and numbers for the User.	Service engineer's name and contact number. Contact 1 / Contact 2	Service setup 10:34am,Mon Service contact: Contact 1 Contact 2 Select [↵]Confirm
	> Contact 1 / Contact 2	
	Contact name or number. Name / phone icon	Service contact 10:34am,Mon Contact 1 Name : Bryan Adams : 08812345678 Select [↵]Edit
	Input name and number Contact name: alphabet a ~ z. Contact number: 1 ~ 9	Contact-1 ABC/abc 0-9/Other ABCDEFGHIJKLMNOPQR Space STUVWXYZ abcdefghi BS jklmnopqrstuvwxyz Conf Select [↵]Enter Number: 1 2 3 (4 5 6) 7 8 9 - BS * 0 # _ Conf Select [↵]Enter

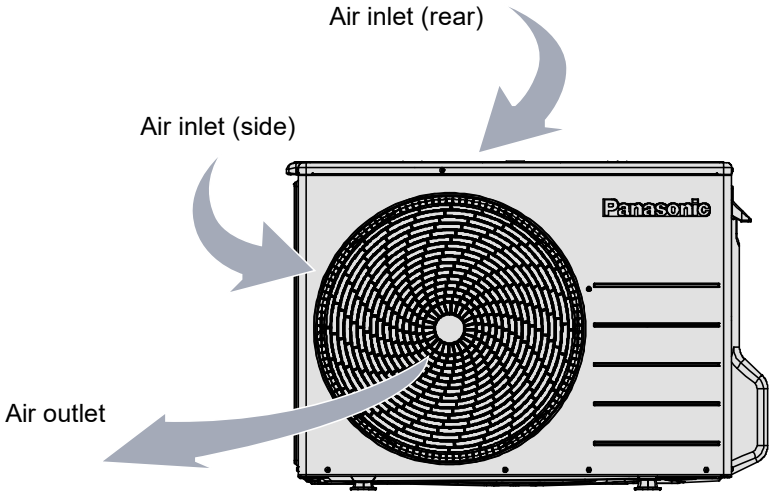
8 Installer setup > Remote control setup			
• To select whether to use one remote controller or two remote controllers. • Select Single when one remote controller is connected. Select Dual when two remote controllers are connected. Second remote controller can be used for zone 2 room temperature control.	Single	Selection of one or two remote controllers.	Single Dual
		When Dual is selected, Main remote controller (RC-1) will start to communicate with second remote controller (RC-2) and display "RC-1 & RC-2 sync. in progress". They are ready to be used after this pop up screen disappears.	RC-1 & RC-2 sync. in progress!
		When both remote controllers have communication failure, it will display "Communication with RC-2 failed".	Communication with RC-2 failed! [⮌]Close

5.1.7 Main Components

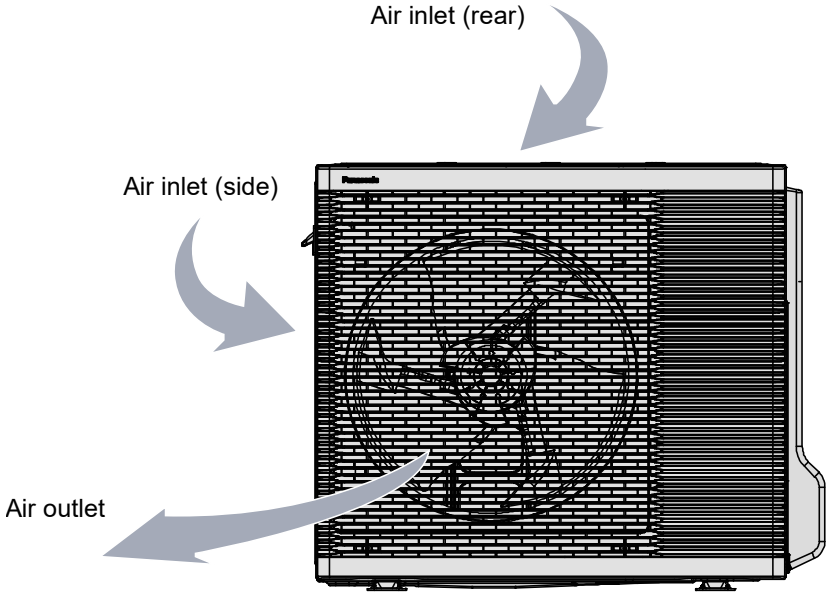


① Remote Controller
② Left Decoration Panel
③ Right Decoration Panel
④ Water Pump
⑤ Control Board Cover
⑥ Control Board
⑦ Main PCB
⑧-1 Terminal Board (Main Power)
⑧-2 Terminal Board (Back-up Heater, if applicable)
⑨ Magnetic Water Filter Set
⑩ 3-Way Valve
⑪ Expansion Vessel
⑫ Air Purge Valve
⑬ Pressure Relief Valve
⑭ Flow Sensor
⑮ Water Pressure Sensor
⑯ Front Plate
⑰ Top Plate
⑱ Right Plate
⑲ Left Plate
⑳ Rear Plate
㉑ Tank Sensor (Not Visible)
㉒ Bushing (4 pieces)
㉓ Network Adaptor Holder
㉔ Drain Tap

5.2 Outdoor Unit



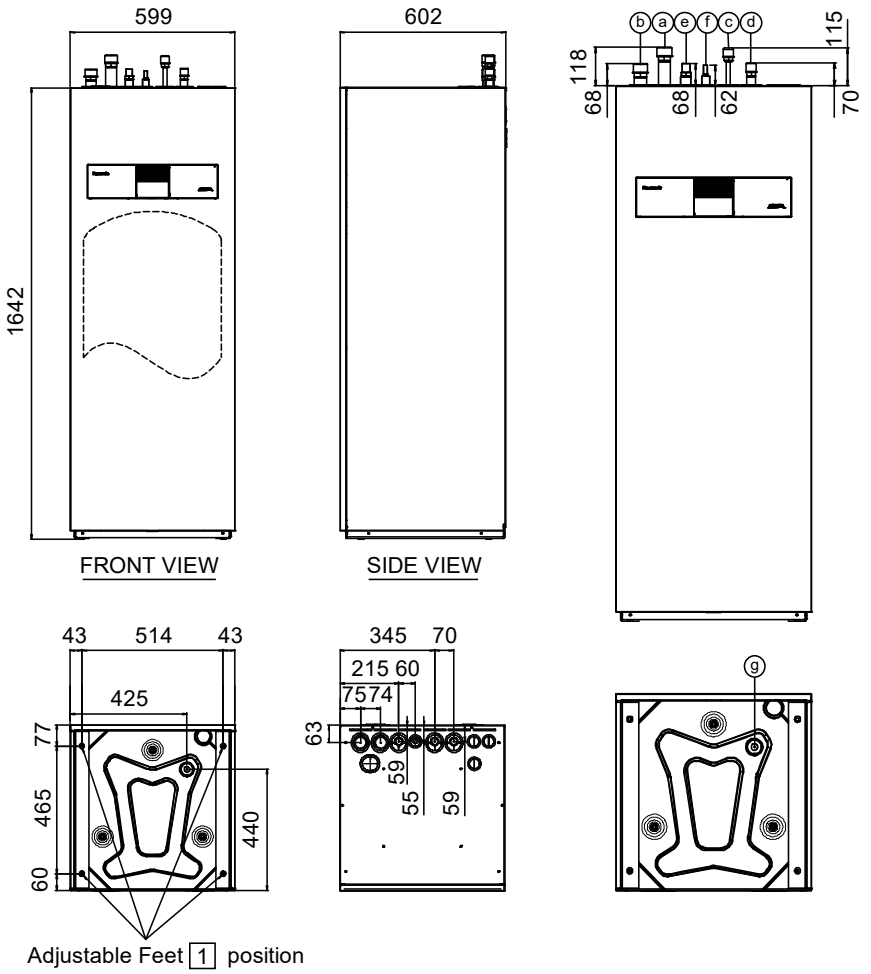
WH-UDZ03KE5



WH-UDZ05_07_09KE5

6. Dimensions

6.1 Indoor Unit

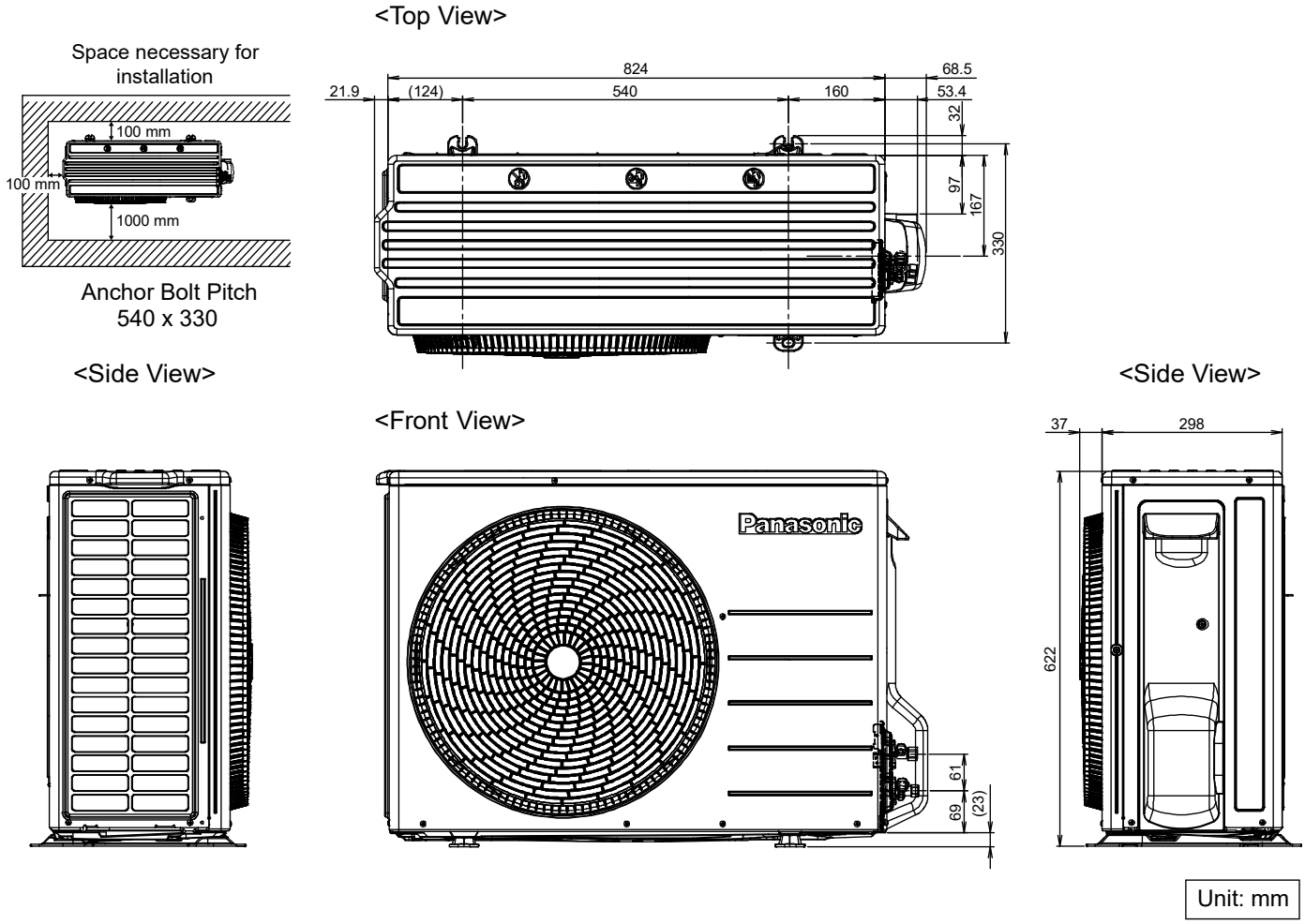


Tube Connector	Function	Connector Size
ⓐ	Water Inlet (From Space Heating/Cooling)	R 1"
ⓑ	Water Outlet (To Space Heating/Cooling)	R 1"
ⓒ	Cold Water Inlet (Domestic Hot Water Tank)	R ¾"
ⓓ	Hot Water Outlet (Domestic Hot Water Tank)	R ¾"
ⓔ	Refrigerant Gas	7/8-14UNF
ⓕ	Refrigerant Liquid	7/16-20UNF
ⓖ	Drain Water Hole	---

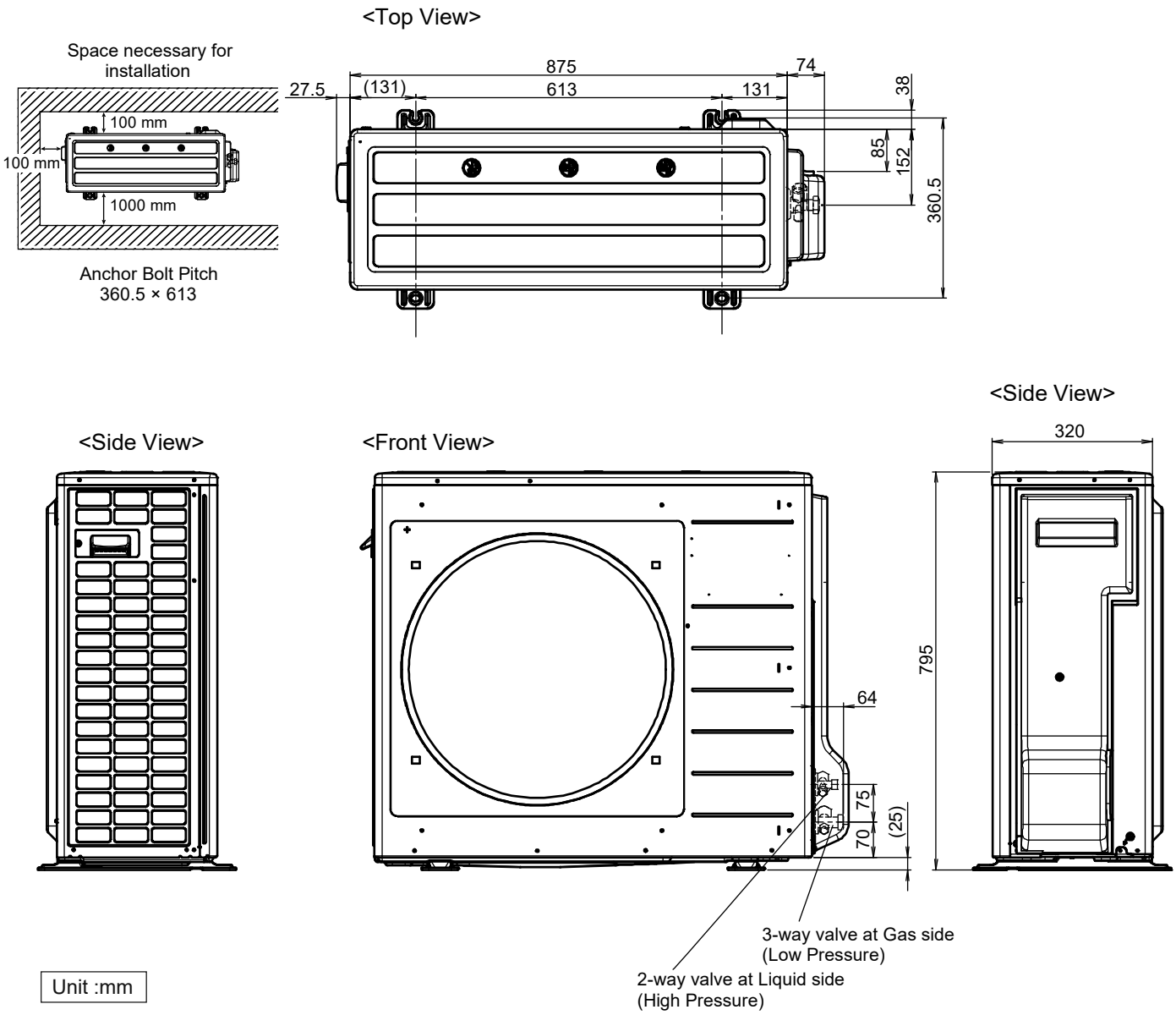
Unit: mm

6.2 Outdoor Unit

6.2.1 WH-UDZ03KE5

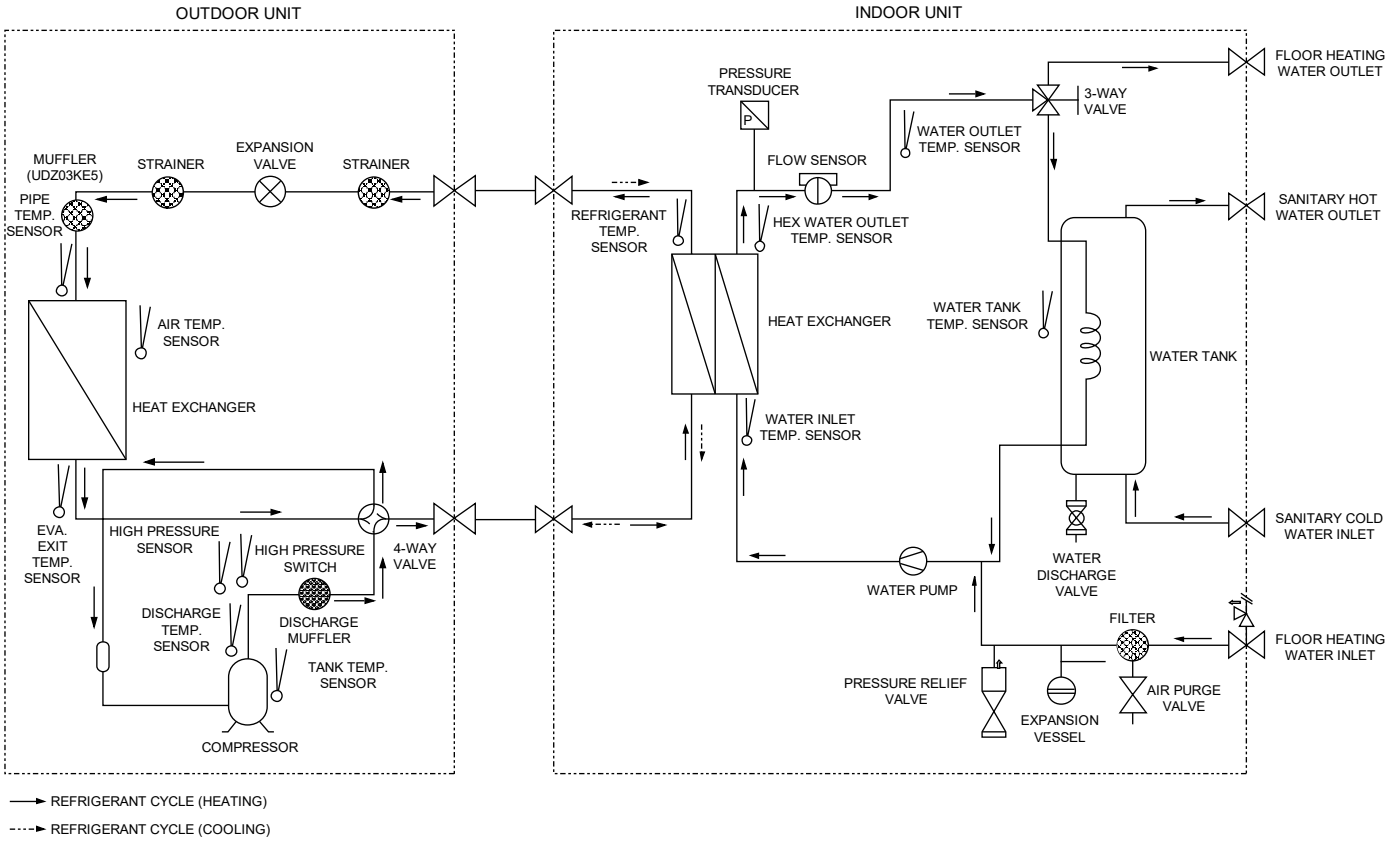


6.2.2 WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



7. Refrigeration and Water Cycle Diagram

1 ZONE

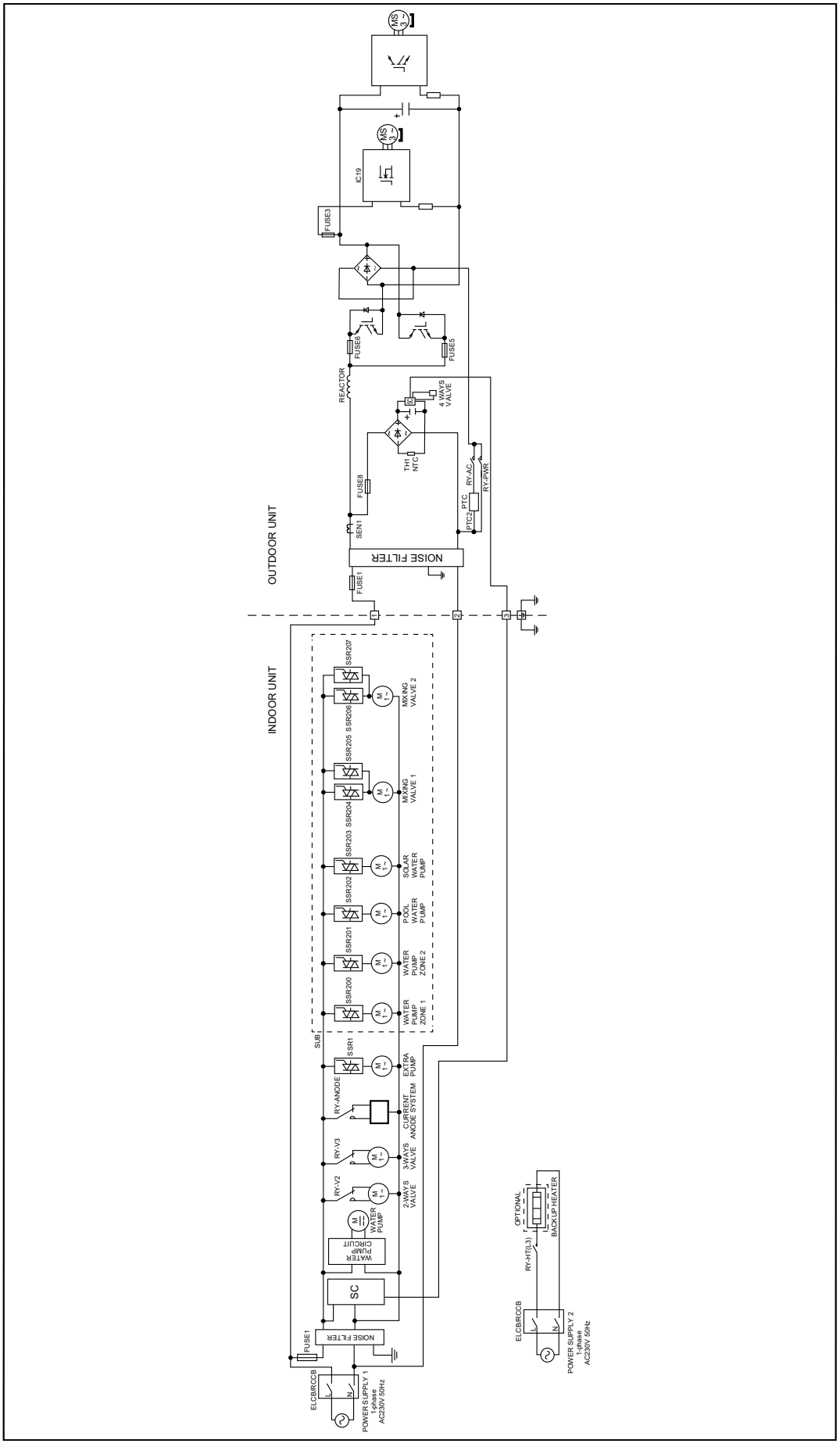


Model	Piping size		Pre-charged Refrigerant (kg)	Rated Length (m)		Max. Elevation (m)	Min. Piping Length (m)	Max. Piping Length (m)	Additional Refrigerant (g/m)
	Gas	Liquid		For Heat Pump Indoor Unit	For Hydromodule + Tank				
WH-UDZ03KE5*	Ø12.7mm (1/2")	Ø6.35mm (1/4")	0.90	7	7	20	3	25	20
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	Ø15.88mm (5/8")	Ø6.35mm (1/4")	1.30	7	7	30	3	50	25

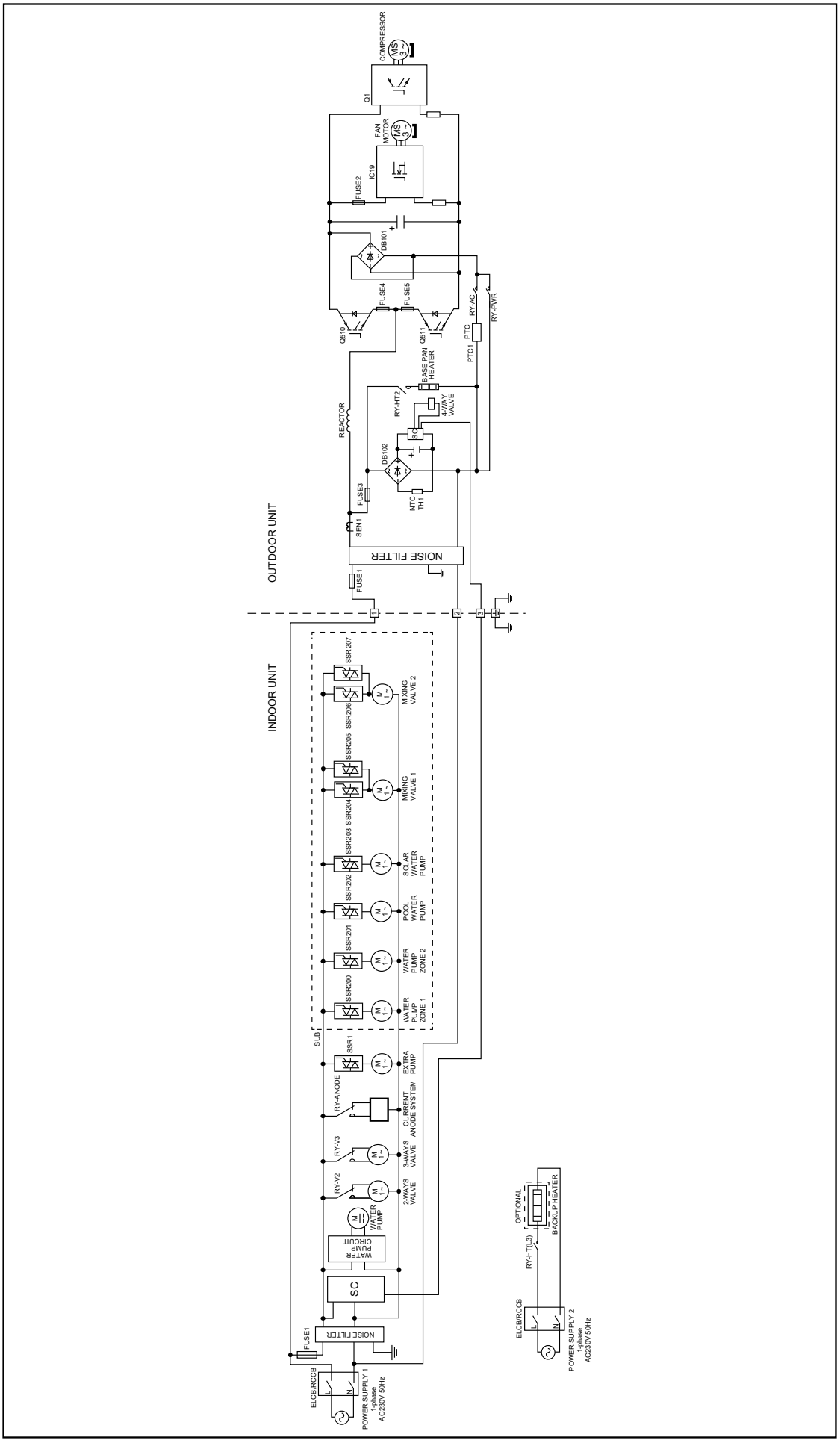
Example: WH-UDZ03KE5*
If piping length is 15m, the quantity of additional refrigerant should be 100g. [(15-10)m × 20 g/m = 100g]

8. Block Diagram

8.1 WH-ADC0309K0E5A WH-UDZ03KE5

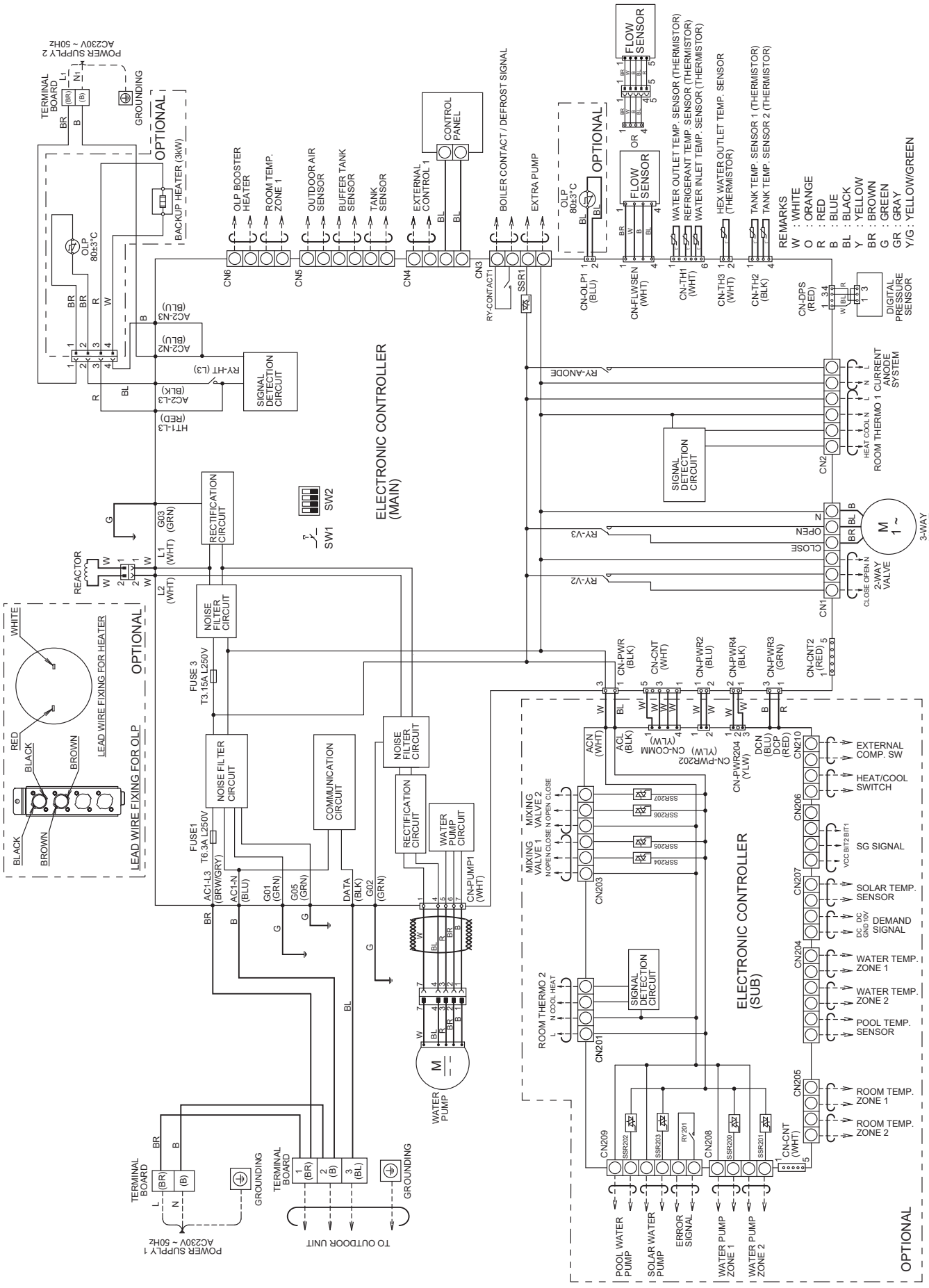


8.2 WH-ADC0309K0E5A WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



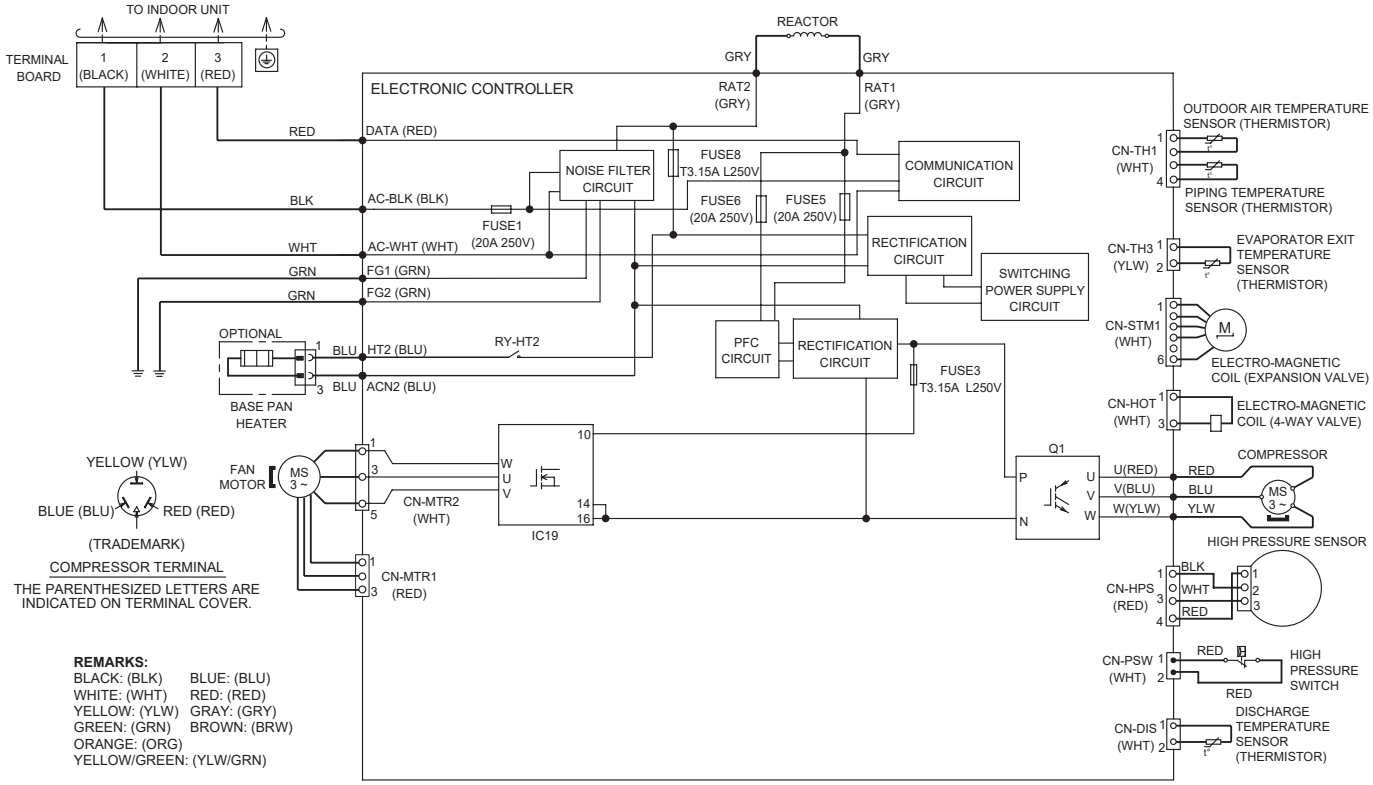
9. Wiring Connection Diagram

9.1 Indoor Unit



9.2 Outdoor Unit

9.2.1 WH-UDZ03KE5

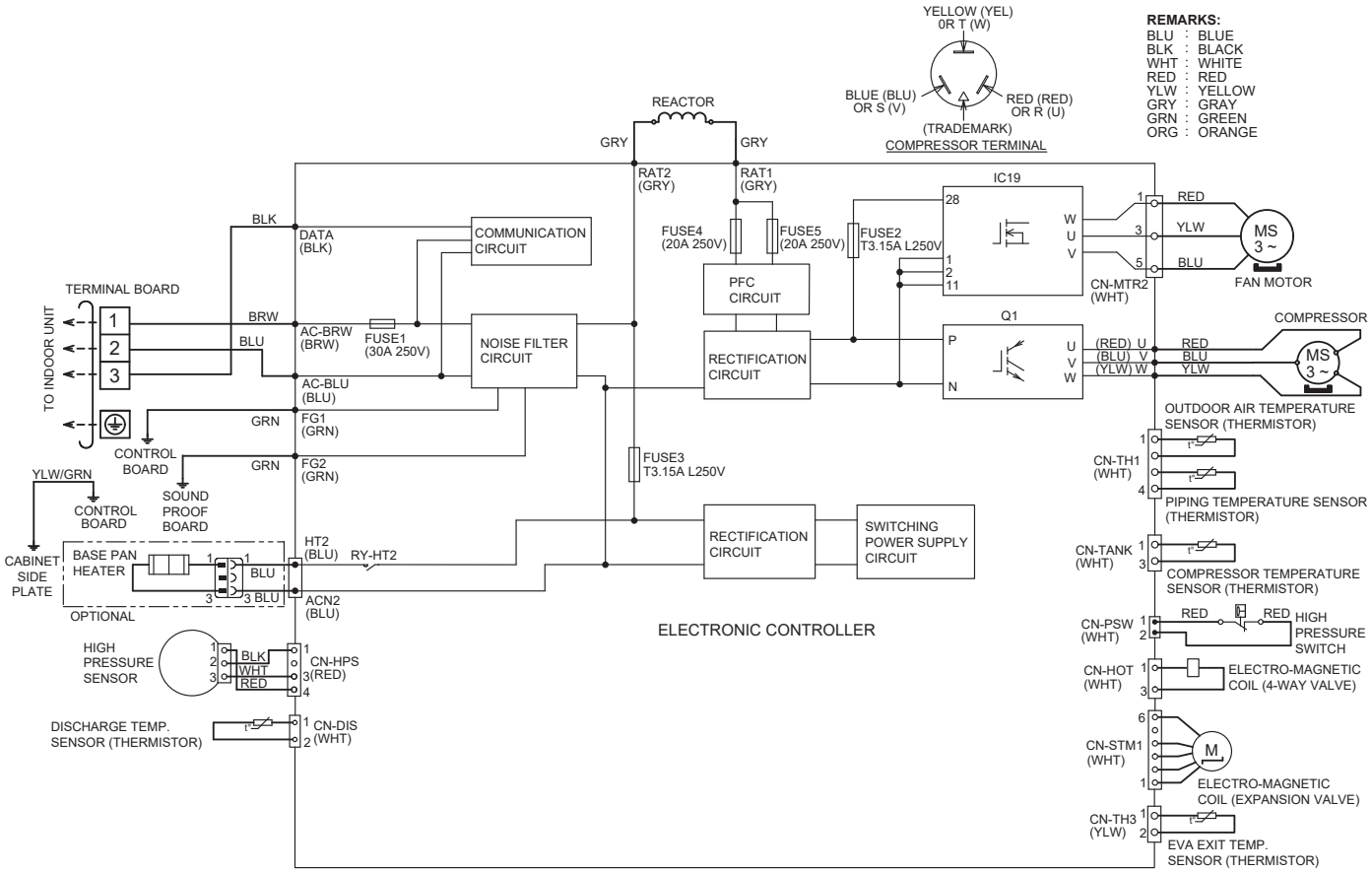


Resistance of Compressor Windings

MODEL	WH-UDZ03KE5
CONNECTION	9RD138ZAB21
U - V	2.215 Ω
V - W	2.194 Ω
U - W	2.208 Ω

Note: Resistance at 20°C of ambient temperature.

9.2.2 WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



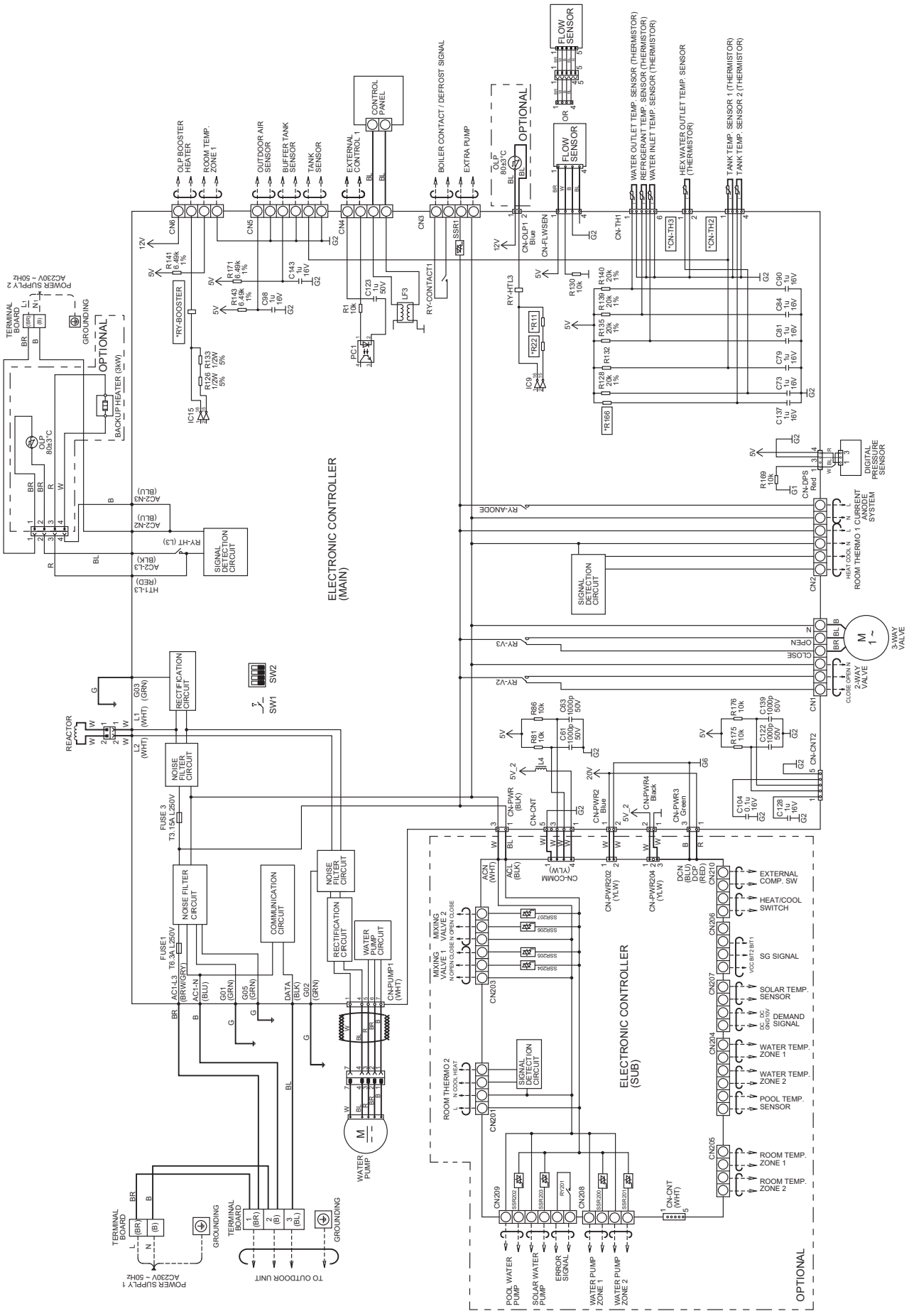
Resistance of Compressor Windings

MODEL	WH-UDZ05/07/09KE5
CONNECTION	9RD220XBC21
U - V	0.998 Ω
U - W	0.998 Ω
V - W	0.998 Ω

Note: Resistance at 20°C of ambient temperature.

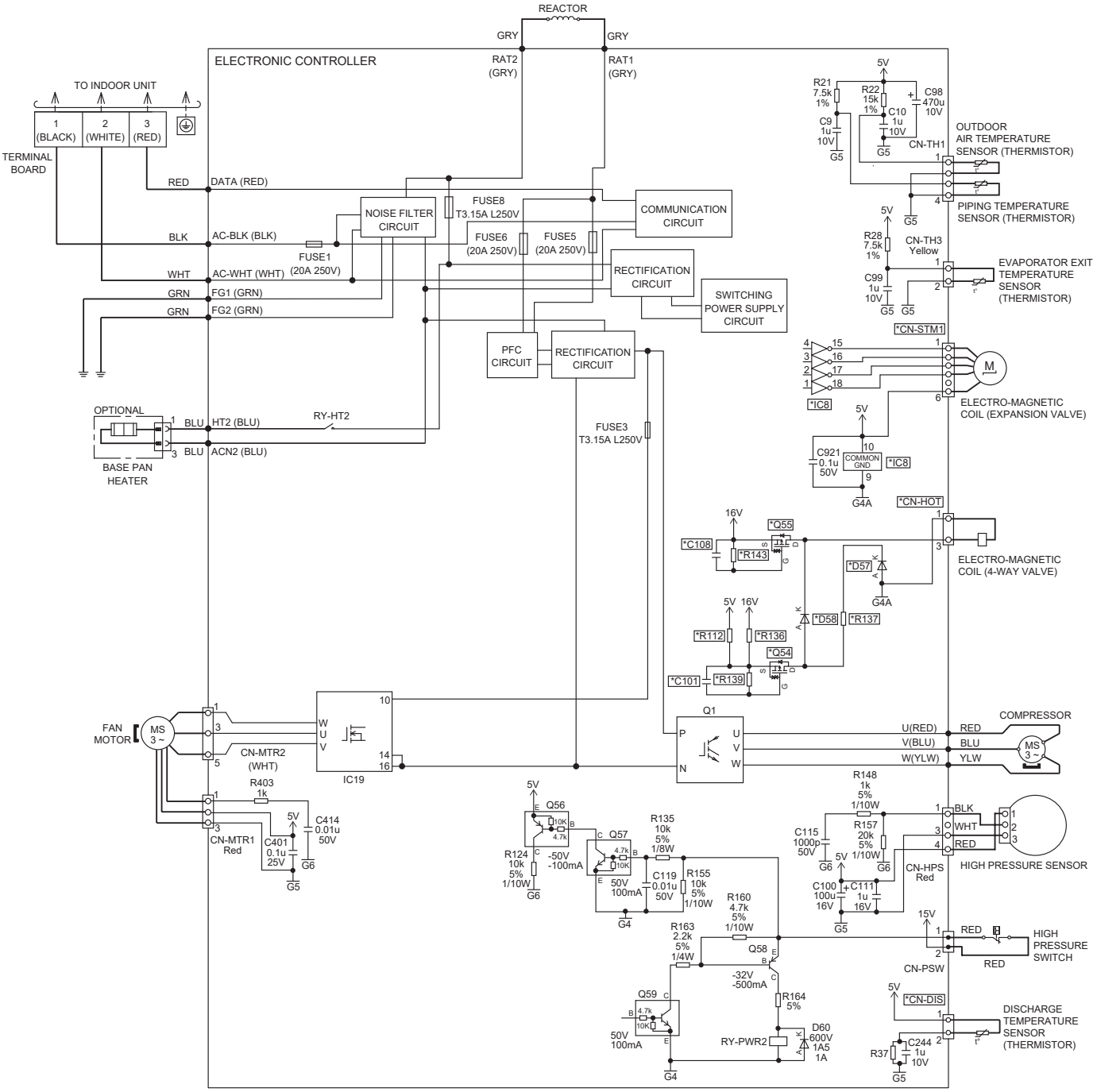
10. Electronic Circuit Diagram

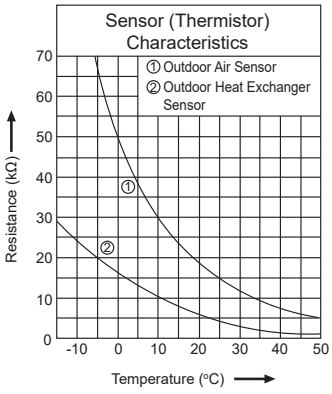
10.1 Indoor Unit

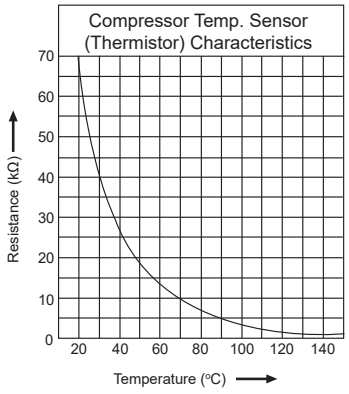


10.2 Outdoor Unit

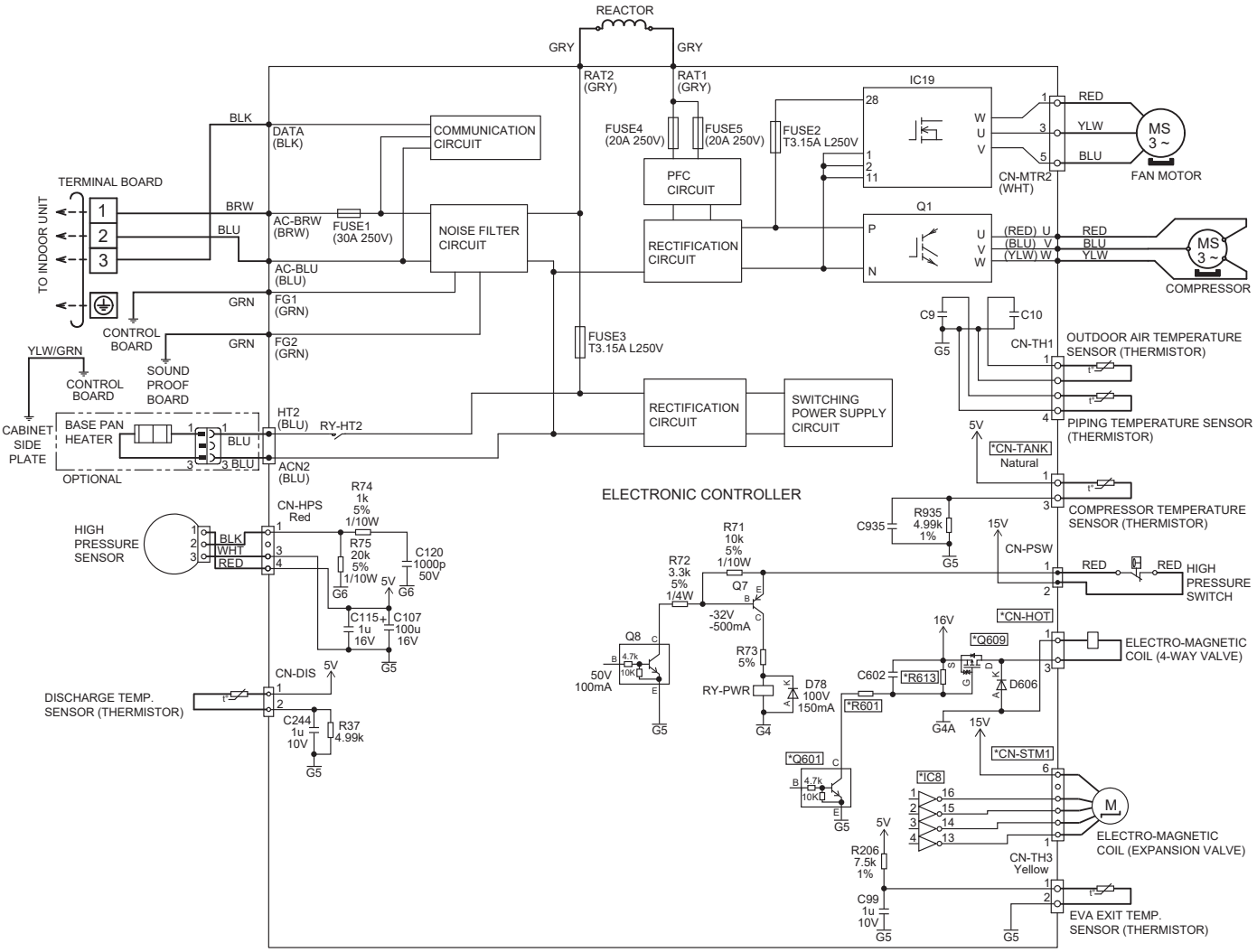
10.2.1 WH-UDZ03KE5

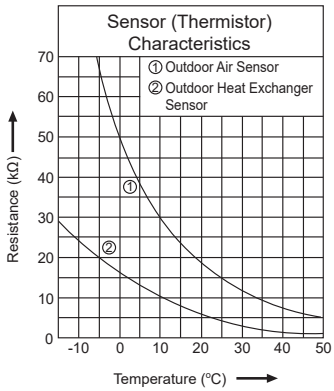


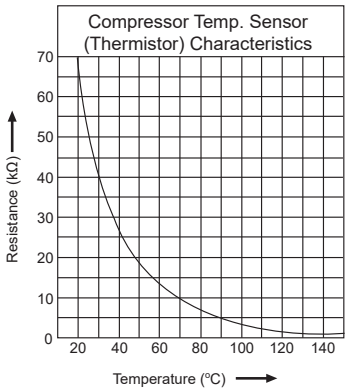




10.2.2 WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



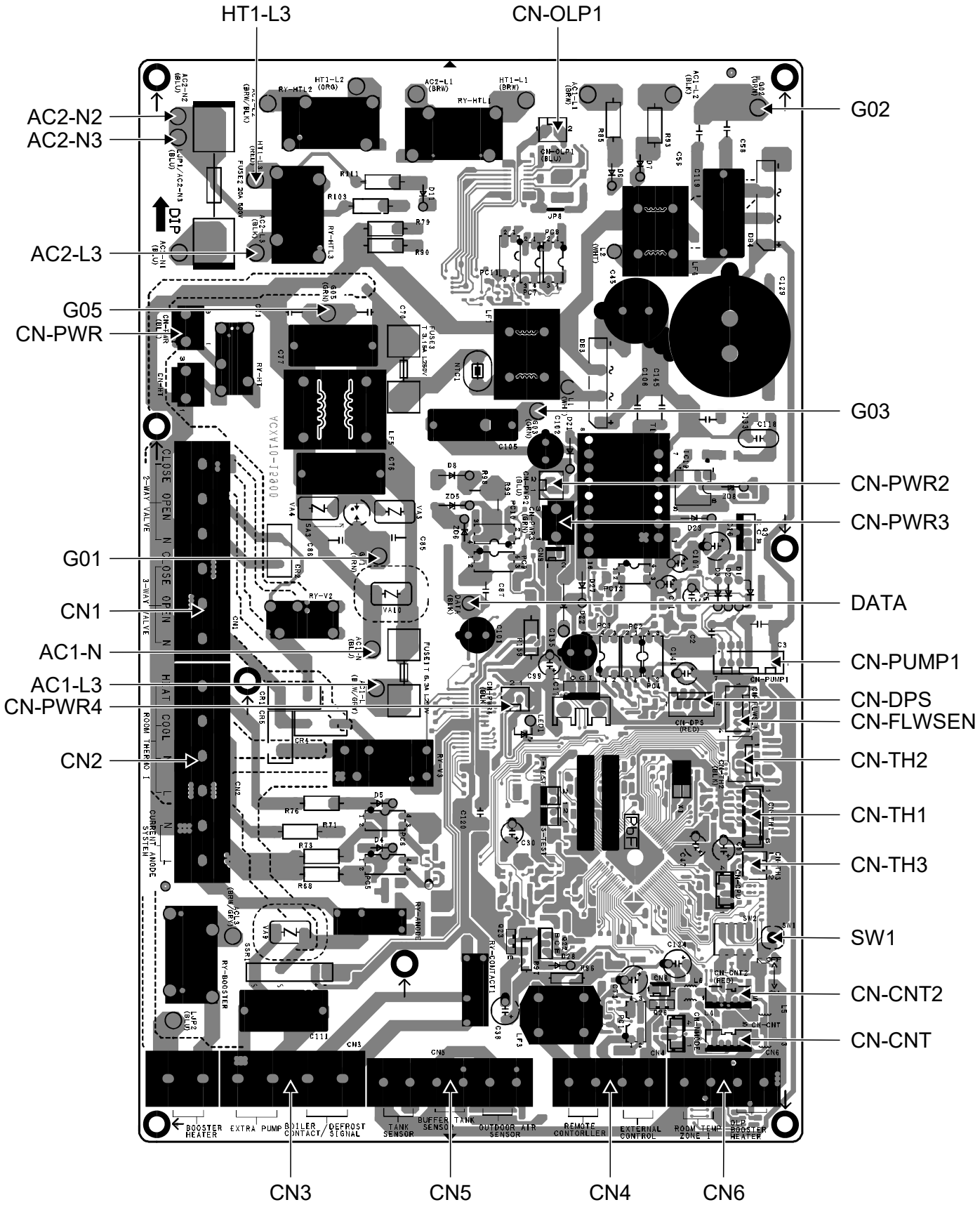




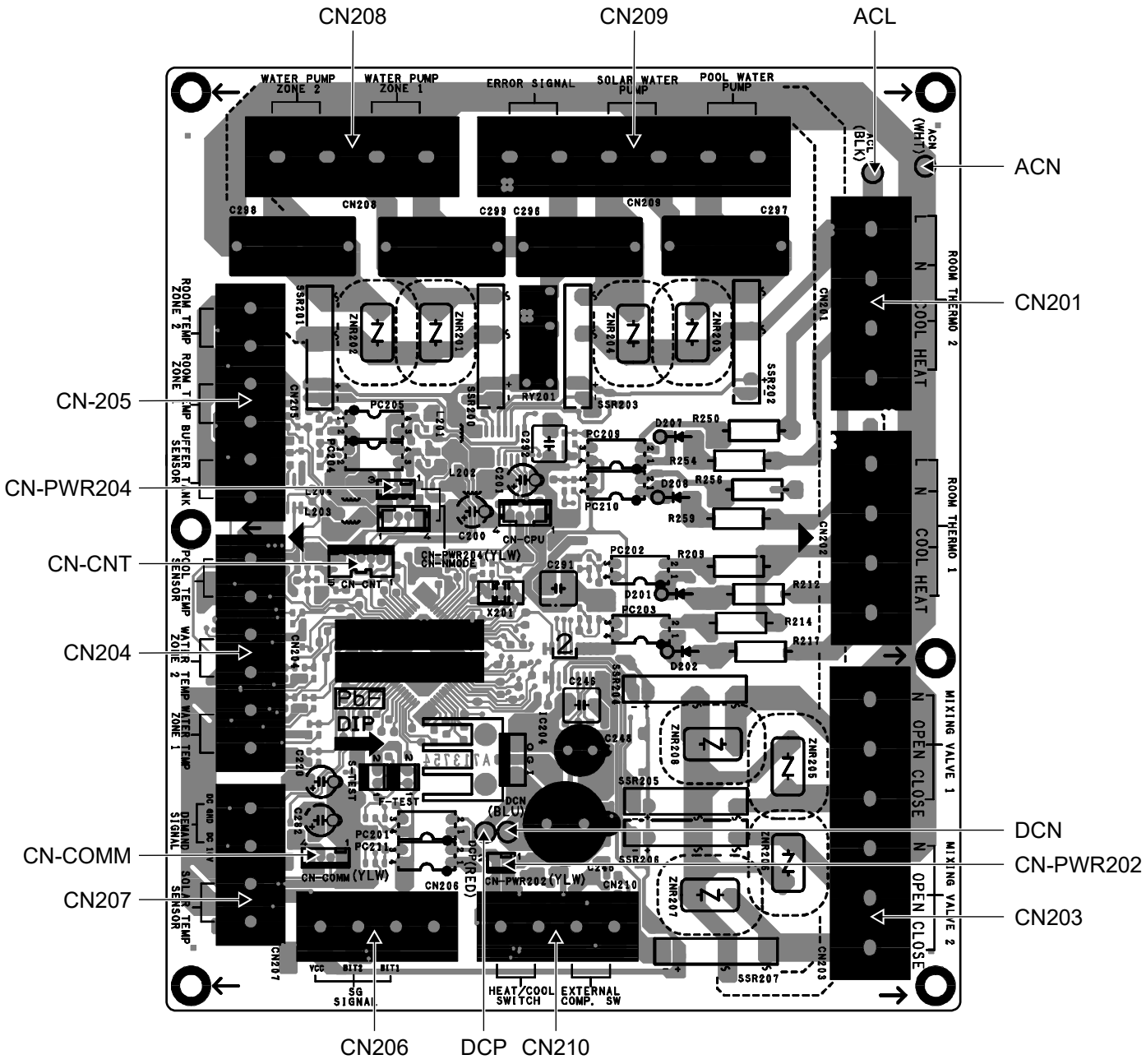
11. Printed Circuit Board

11.1 Indoor Unit

11.1.1 Main Printed Circuit Board



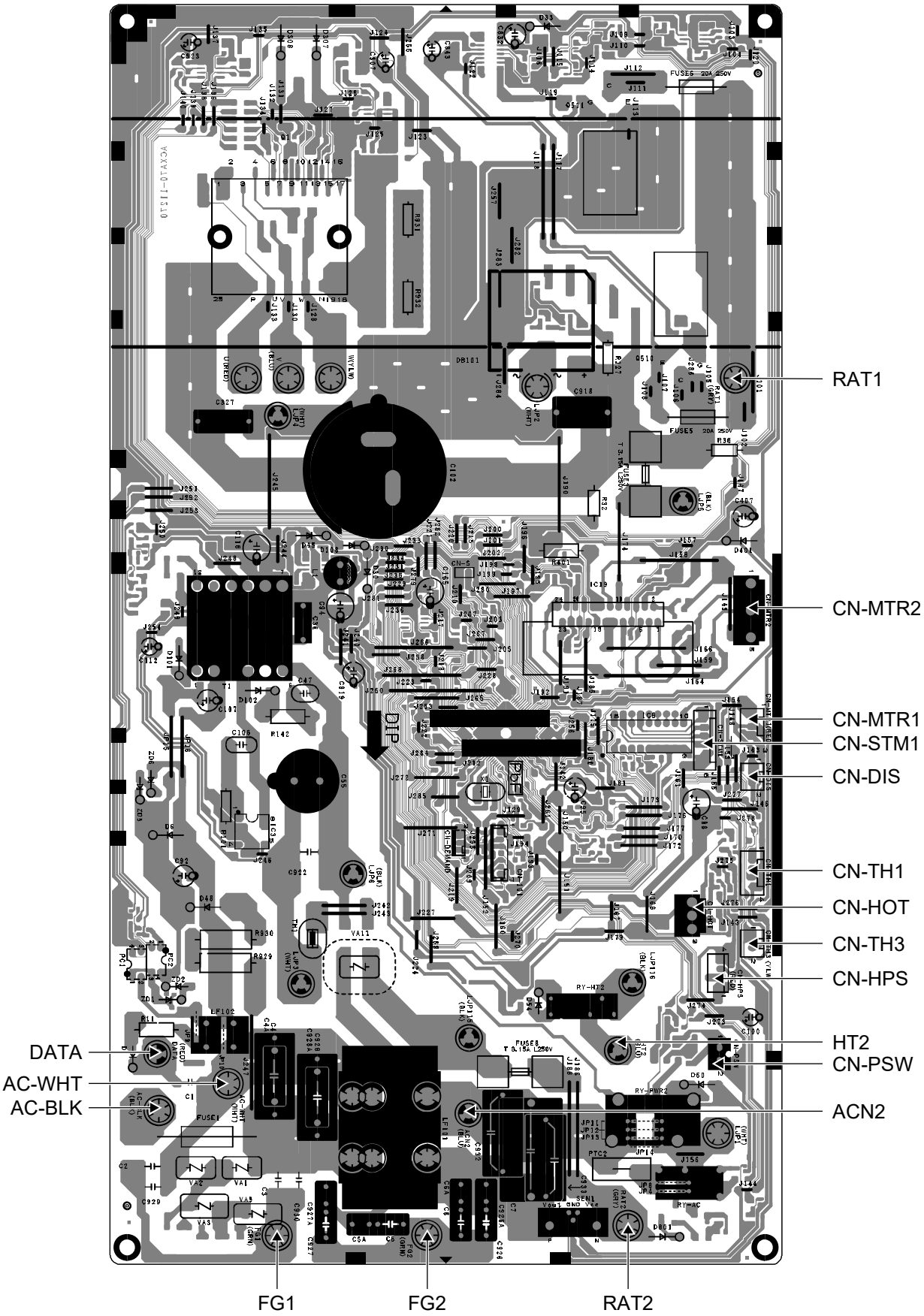
11.1.2 Sub Printed Circuit Board (Optional)



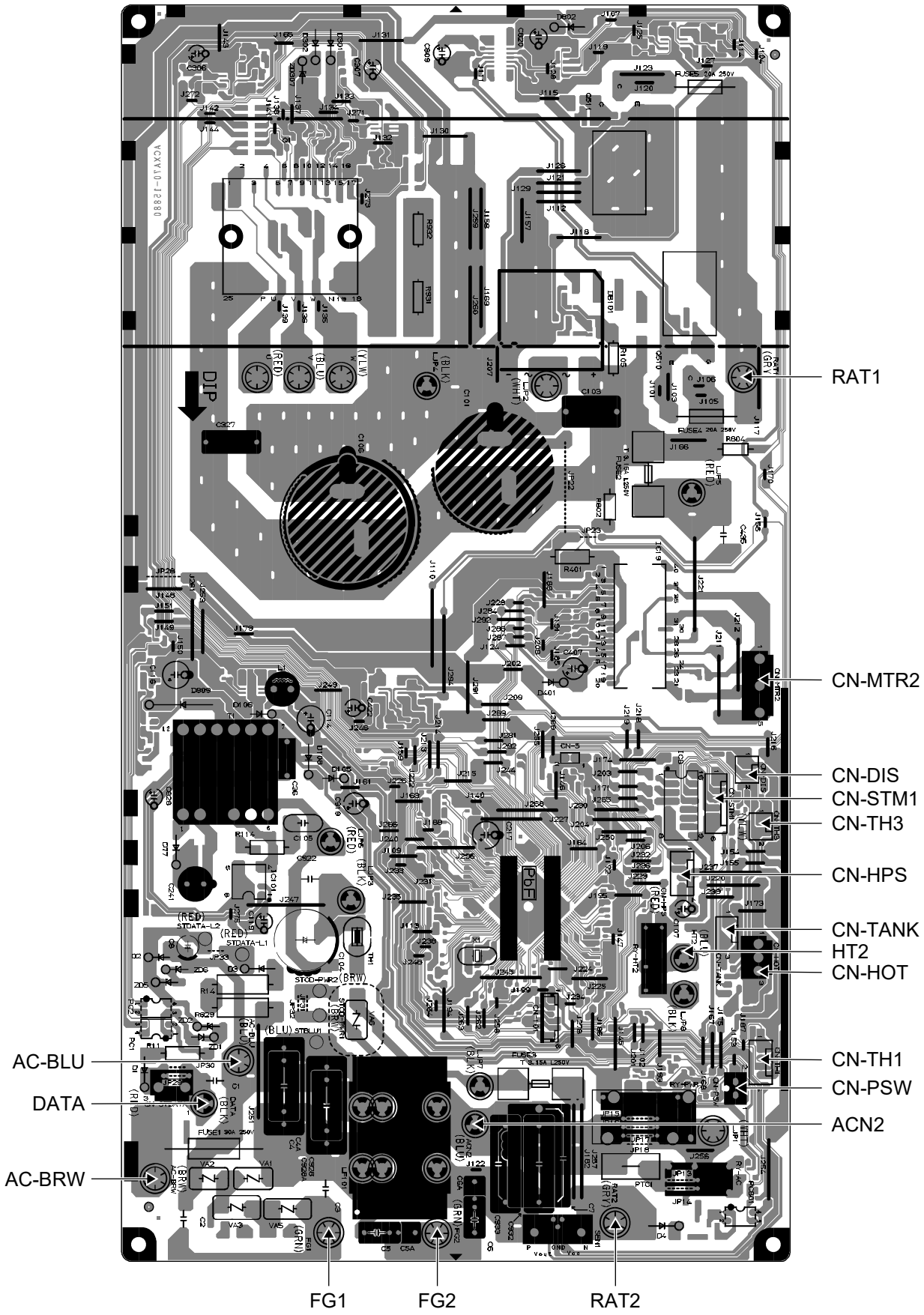
11.2 Outdoor Unit

11.2.1 Main Printed Circuit Board

11.2.1.1 WH-UDZ03KE5



11.2.1.2 WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



12. Installation Instruction

12.1 Indoor Floor Area Requirement

- **If the total refrigerant charge in the system is <1.84 kg**, no additional minimum floor area is required.
- **If the total refrigerant charge in the system is ≥1.84 kg**, additional minimum floor area requirements is complied as described below:

Symbol	Description	Unit
m_c	Total refrigerant charge in system	kg
m_{max}	Maximum refrigerant charge allowed	kg
m_{excess}	m_c - m_{max}	kg
H	Installation height	m
VA_{min}	Minimum ventilation opening area	cm^2

Total refrigerant charge in system, m_c (kg)
= Pre-charged refrigerant amount in unit (kg)
+ Additional refrigerant amount after installation (kg)

A) Determine **Maximum refrigerant charge allowed, m_{max}**

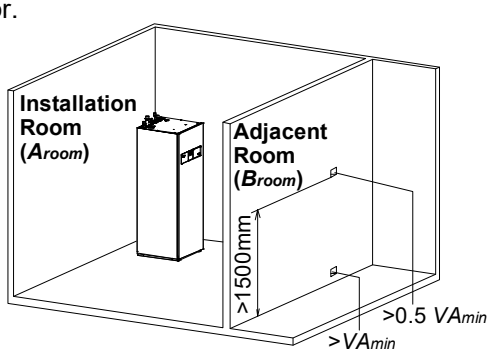
1. Calculate Installation Room Area, A_{room}.
2. Based on Table I, select m_{max} which corresponds to the calculated A_{room} value.
3. If $m_{max} \geq m_c$, the unit can be installed in the installation room with the specified installation height (***H=600mm***) in Table I and without additional room area or any additional ventilation.
4. Else, proceed to B) and C).

B) Determine **Total Floor Area of A_{room} and B_{room} compliance to $A_{min\ total}$**

1. Calculate the B_{room} area adjacent to the A_{room}.
2. Determine the $A_{min\ total}$ based on the Total Refrigerant Charge, m_c from Table II.
3. The total floor area of both A_{room} and B_{room} must exceed $A_{min\ total}$.

C) Determine **Minimum Venting Opening Area**, **VA_{min}** for natural ventilation

1. From Table III, calculate m_{excess}.
2. Then determine VA_{min} corresponding to the calculated m_{excess} for natural ventilation between A_{room} and B_{room}.
3. The unit can be installed at specific room only when the following conditions are fulfilled:
   - Two permanent openings (cannot be closed), one at bottom, another at top, for ventilation purposes are made between A_{room} and B_{room}.
   - **Bottom opening:**
     - Must comply to the minimum area requirement of VA_{min}.
     - Opening must be located ≤300mm from the floor.
     - At least 50% of required opening area must be ≤200mm from the floor.
     - The bottom of the opening shall not be higher than the point of release when the unit is installed and must be situated ≤100mm above the floor.
     - Must be as close as possible to the floor and lower than H.
   - **Top opening:**
     - The total size of the Top opening must be more than 50% of VA_{min}.
     - Opening must be located ≥1500mm above the floor.



   - The height of the openings must more than 20mm.
   - A direct ventilation opening to outside is **NOT** encouraged for ventilation opening (the user can block the opening when it is cold).
   - The value of H is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.

Table I - Maximum refrigerant charge allowed in a room

A_{room} (m²)	Maximum refrigerant charge in a room (m_{max}) (kg)
	***H*=0.6m**
1	0.138
2	0.276
3	0.414
4	0.553
5	0.691
6	0.829
7	0.907
8	0.970
9	1.028
10	1.084
11	1.137
12	1.187
13	1.236
14	1.283
15	1.328
16	1.371
17	1.413
18	1.454
19	1.494
20	1.533
21	1.571
22	1.608
23	1.644
24	1.679
25	1.714
26	1.748
27	1.781
28	1.814
29	1.846
30	1.877
31	1.909
32	1.939
33	1.969
34	1.999
35	2.028
36	2.057
37	2.085
38	2.113
39	2.141
40	2.168
41	2.195
42	2.221
43	2.248
44	2.274
45	2.299

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate A_{room} values, the value that corresponds to the lower A_{room} value from the table is considered.
  Example:
  For A_{room} = 10.5 m², the value that corresponds to "A_{room} = 10 m²" is considered.

Table II - Minimum floor area

m_c (kg)	Minimum floor area ($A_{min\ total}$ (m²))
	***H*=0.6m**
1.84	28.81
1.86	29.44
1.88	30.08
1.90	30.72
1.92	31.37
1.94	32.03
1.96	32.70
1.98	33.37
2.00	34.04
2.02	34.73
2.04	35.42
2.06	36.12
2.08	36.82
2.10	37.53
2.12	38.25
2.14	38.98
2.16	39.71
2.18	40.45
2.20	41.19
2.22	41.94
2.24	42.70
2.26	43.47
2.28	44.24
2.30	45.02

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate m_c values, the value that corresponds to the higher m_c value from the table is considered.
  Example:
  If m_c = 1.85 kg, the value that corresponds to "m_c = 1.86 kg" is considered.
- Systems with total refrigerant charge lower than 1.84 kg are not subjected to any room area requirements.
- Charges above 2.30 kg are not allowed in the unit.

Table III - Minimum venting opening area for natural ventilation

m_c (kg)	m_{max} (kg)	m_{excess} (kg) = m_c - m_{max}	Minimum venting opening area (VA_{min}) (cm²)
			***H*=0.6m**
2.3	0.1	2.20	890
2.3	0.3	2.00	809
2.3	0.5	1.80	728
2.3	0.7	1.60	647
2.3	0.9	1.40	583
2.3	1.1	1.20	552
2.3	1.3	1.00	500
2.3	1.5	0.80	430
2.3	1.7	0.60	343
2.3	1.9	0.40	242
2.3	2.1	0.20	127
2.3	2.3	0.00	0

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate m_{excess} values, the value that corresponds to the higher m_{excess} value from the table is considered.
  Example:
  m_{excess} = 1.45 kg, the value that corresponds to "m_{excess} = 1.6 kg" is considered.

Attached accessories

No.	Accessory part	Qty.
1	Adjustable Feet 	4
2	Reducing Adapter 	1
3	Drain Elbow 	1
4	Packing 	1

Optional Accessories

No.	Accessories part	Qty.
5	Remote Controller Case	1
6	Network Adaptor (CZ-TAW*) and Extension Cable (CZ-TAW1-CBL)	1
7	Optional PCB (CZ-NS5P)	1
8	Optional Back-up Heater (CZ-NB3)	1

Field Supply Accessories (Optional)

No.	Part		Model	Specifications	Maker
i	2-way valve kit *Cooling model	Electromotoric Actuator	SFA21/18	AC230V, 12 VA	Siemens
		2-port Valve	VVI46/25		Siemens
ii	Room thermostat	Wired	PAW-A2W-RTWIRED	AC230V	-
		Wireless	PAW-A2W-RTWIRELESS		
iii	Mixing valve	-	167032	AC230V, 6VA	Caleffi
iv	Pump	-	Yonos 25/6	AC 230V, 0.6 A max	Wilo
v	Buffer tank sensor	-	PAW-A2W-TSBU	-	-
vi	Outdoor sensor	-	PAW-A2W-TSOD	-	-
vii	Zone water sensor	-	PAW-A2W-TSHC	-	-
viii	Zone room sensor	-	PAW-A2W-TSRT	-	-
ix	Solar sensor	-	PAW-A2W-TSSO	-	-

■ It is recommended to purchase the field supply accessories listed in above table.

12.2 Indoor Unit

12.2.1 Select the Best Location

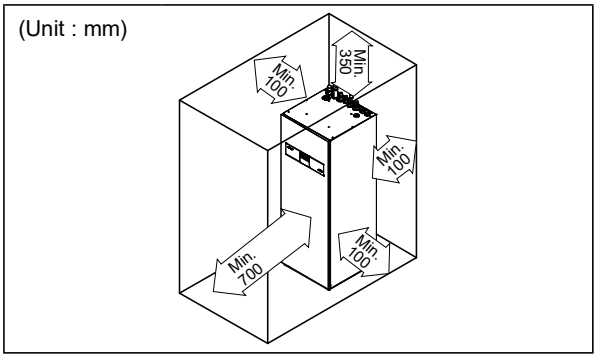
Before choosing the installation site, obtain user approval.

- Install the Tank Unit in indoors with frost free weather proof location only.
- Must install on a flat horizontal and solid hard surface.
- There should not be any heat source or steam near the Tank Unit.
- A place where air circulation in the room is good.
- A place where drainage can be easily done (e.g. Utility room).
- A place where Tank Unit's operation noise will not cause discomfort to the user.
- A place where Tank Unit is far from door way.
- A place where accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated below from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- Secure the Tank Unit to prevent it being knocked over accidentally or during earthquakes.

Please avoid installations which expose the Tank Unit to any of the following conditions:

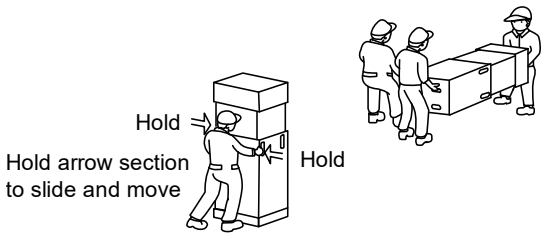
- Extraordinary environment conditions; installation in frost or exposure to unfavorable weather conditions.
- Voltage input exceeding the specified voltage.

12.2.1.1 Required Space for Installation



12.2.1.2 Transport and Handling

- Be careful during transporting the unit so that it is not damaged by impact.
- Only remove the packaging material once it has reached it is desired installation location.
- It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.
- The Tank Unit can be transported either in vertical or horizontal.
  - If it transported in horizontal, make sure Front of packaging material (printed with "FRONT") must facing upwards.
  - If it transported in vertical, use the hand holes on sides, slide and move to the desired location.
- Fix the Adjustable Feet [1], if the Tank unit installed on a uneven surface.



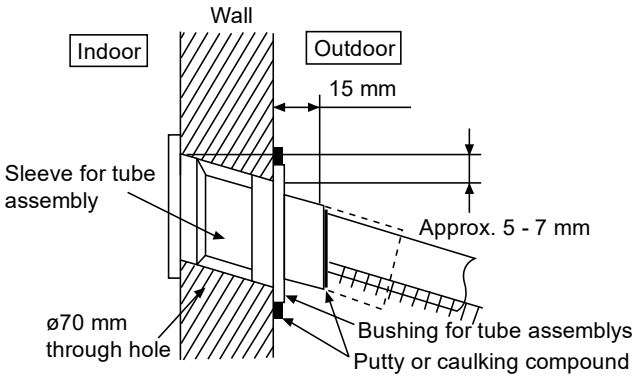
12.2.2 To Drill a Hole in the Wall and Install a Sleeve of Piping

1. Make a Ø70 mm through hole.
2. Insert the piping sleeve to the hole.
3. Fix the bushing to the sleeve.
4. Cut the sleeve until it extrudes about 15 mm from the wall.

⚠ CAUTION

When the wall is hollow, please be sure to use the sleeve for tube assembly to prevent dangers caused by mice biting the connection cable.

5. Finish by sealing the sleeve with putty or caulking compound at the final stage.

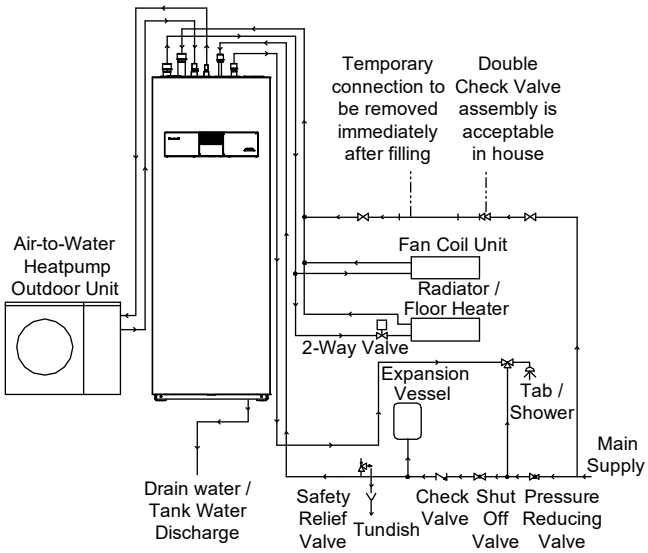


12.2.3 Piping Installation

12.2.3.1 Water Quality Requirement

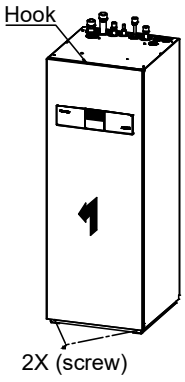
Must use water that complies with European Drinking Water Directive (EU)2020/2184. The lifespan of the Tank Unit will be shorter if groundwater (include spring water and well water) is used.
The Tank Unit shall not be used with the tap water containing contaminants such as salt, acid, and other impurities which may corrode the tank and its component.
Use sanitized water that is free of Legionella and other bacteria and microorganisms. If the water contains Legionella bacteria, it may harm the health of the user.

12.2.3.2 Typical Piping Installation



12.2.3.3 Access to Internal Components

⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

Open and Close Front Plate ⑯

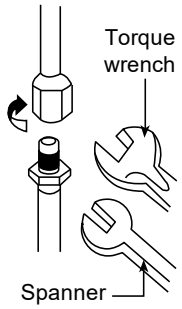
1. Remove the 2 mounting screws of Front Plate ⑯.
2. Slide it upwards to unhook the Front Plate ⑯ hook.
3. Reverse above steps 1~2 for close it.

12.2.3.4 Refrigerant Piping Installation

This Tank Unit is designed for combination with Panasonic Air-to-Water Heat Pump Outdoor Unit. If Outdoor Unit from other manufacturer are being used in combination with Panasonic Tank Unit, optimum operation and reliability of the system is not guaranteed. Thus warranty cannot be given in such case.

1. Connect Tank Unit to Air-to-Water Heatpump Outdoor Unit with correct piping size. Use Reducing Adapter [2] for Outdoor Unit WH-UDZ03KE5* Refrigerant Gas ⓔ piping connection.

Model		Piping size (Torque)		Use Reducing Adapter [2]
Tank Unit	Outdoor Unit	Gas	Liquid	
WH-ADC0309K0E5A	WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]	Yes
	WH-UDZ05KE5*, WH-UDZ07KE5*, WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]	No



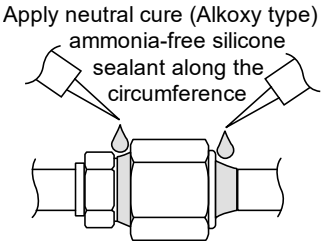
⚠ CAUTION
Do not overtighten, overtightening may cause gas leakage.
Do not pull and push refrigerant piping excessively, deformed pipe may cause refrigerant leak.

2. Please make flare after inserting flare nut (located at joint portion of tube assembly) onto the copper pipe. (In case of using long piping)
3. Do not use pipe wrench to open refrigerant piping. Flare nut may be broken and cause leakage. Use proper spanner or ring wrench.
4. Connect the piping:
   - Align the centre of piping and sufficiently tighten the flare nut with fingers.
   - Further tighten the flare nut with torque wrench in specified torque as stated in the table.

Additional Precautions For R32 Models when connecting by flaring at indoor side

❗	Ensure to do the re-flaring of pipes before connecting to units to avoid leaking.
❗	Connections made between components of refrigerant system shall be accessible for ease of maintenance.

Seal sufficiently the flare nut (both gas and liquid sides) with neutral cure (Alkoxy type) & ammonia-free silicone sealant and insulation material to avoid the gas leak caused by freezing.



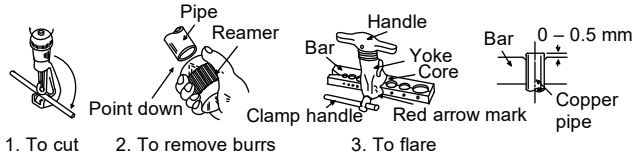
Neutral cure (Alkoxy type) & ammonia-free silicone sealant is only to be applied after pressure testing and cleaning up by following instructions of sealant, only to the outside of the connection. The aim is to prevent moisture from entering the connection joint and possible occurrence of freezing. Curing sealant will take some time. Make sure sealant will not peel off when wrapping the insulation.

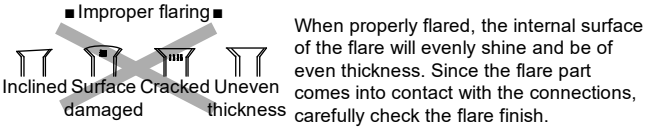
12.2.3.4.1 Checking for Gas Leakage

- Check for leakage of gas after air purging.
- See the in the installation manual for the outdoor.

12.2.3.5 Cutting and Flaring the Piping

1. Please cut using pipe cutter and then remove the burrs.
2. Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3. Please make flare after inserting the flare nut onto the copper pipes.



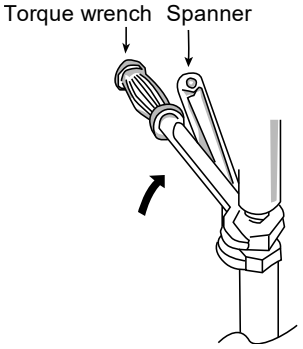


When properly flared, the internal surface of the flare will evenly shine and be of even thickness. Since the flare part comes into contact with the connections, carefully check the flare finish.

12.2.3.6 Water Piping Installation

- Please engage a licensed water circuit installer to install this water circuit.
- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not use worn out tube or detachable hose set.
- Do not apply excessive force to pipes that may damage the pipes.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Further tighten the nuts with torque wrench in specified torque as stated in the table.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall.
- If non-brass metallic piping is used for installation, make sure to insulate the pipes to prevent galvanic corrosion.
- Do not connect galvanised pipes, this will cause galvanic corrosion.
- Use correct nut for all Tank Unit tube connections and clean all tubes with tap water before installation. See Tube Position Diagram for detail.

Tube Connector	Nut Size	Torque
ⓐ & ⓑ	R 1"	88.2 N•m
ⓒ & ⓓ	RP ¾"	58.8 N•m



⚠ CAUTION
Do not overtighten, overtightening may cause water leakage.

- Make sure to insulate the water circuit pipes to prevent reduction of heating capacity.
- After installation, check the water leakage condition in connection area during test run.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Protection From Frost:
  If the Tank Unit is being exposed to frost while power supply failure or pump operating failure, drain the system. When water is idle inside the system, freezing up is very likely to happen which could damage the system. Make sure the power supply is turned off before draining.

- Corrosion Resistance:
  Duplex stainless steel is naturally corrosion resistant to mains water supply. No specific maintenance is required to maintain this resistance. However, please note that Tank Unit is not guaranteed for use with a private water supply.
- It is recommended to use a tray (field supply) to collect water from the Tank Unit if water leakage occur.

Recommended piping installation sequence:
(a) → (c) → (e) → (f) → (b) → (d)

(A) Space Heating/Cooling Pipework

- Connect Tank Unit Tube Connector ⓐ to outlet connector of Zone 1 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓑ to inlet connector of Zone 1 Panel/Floor heater.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Refer below table for the rated flow rate of each particular Outdoor Unit.

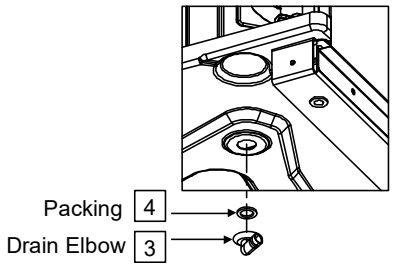
Model		Rated Flow Rate (l/min)	
Tank Unit	Outdoor Unit	Cool	Heat
WH-ADC0309K0E5A	WH-UDZ03KE5*	9.2	9.2
	WH-UDZ05KE5*	14.3	14.3
	WH-UDZ07KE5*	19.2	20.1
	WH-UDZ09KE5*	23.5	25.8

(B) Domestic Hot Water Tank Pipework

- A Safety Relief Valve (field supply) with below specification is required to be installed along the line of the tube connector (c) of Tank Unit.
  Refer Typical Piping Installation section to locate this valve.
  Recommended Safety Relief Valve specifications:
  - Pressure setting: 0.80 MPa (8 bars)
- Piping must always be installed in a continuously downward direction. It must not be longer than 2m, with no more than 2 elbows, and must not allow condensation to build up or freezing to occur.
- The pipe from this drainage outlet fitting must not be shut off. The discharge must be freed.
- The end of this pipework must be in such a way so that the outlet is visible and can not cause any damage. Keep away from electrical components.
- It is recommended to fit a tundish into this pipework. Tundish should be visible and positioned away from frost environment and electrical components.
- It's strongly recommended to install an expansion vessel (field supply) in the Domestic Hot Water Tank circuit. Refer Typical Piping Installation section to locate the expansion vessel.
  - Recommended pre-charge pressure of the expansion vessel (field supply) = 0.35MPa (3.5 bars)
- In high water pressure or water supply is above 500kPa, please install the Pressure Reducing Valve for water supply. If the pressure higher than that, it might damage the Tank Unit.
- A Pressure Reducing Valve (field supply) with below specification is strongly advised to be installed along the line of the tube connector ⓒ of Tank Unit. Refer Typical Piping Installation section to locate both of these valves.
  Recommended Pressure Reducing Valve specifications:
  - Set pressure: 0.35 MPa (3.5 bars)
- Must connect a faucet to Tank Unit Tube Connector ⓓ and main water supply, in order to supply water with appropriate temperature for shower or tap usage. Failure to do so might cause scalding.
- Failure to connect the tube appropriately might causing the Tank Unit malfunction.

(C) Drain Elbow and Hose Installation

- Fix the Drain Elbow 3 and Packing 4 to the bottom of Drain Water Hole ⓖ.



- Use inner diameter 17 mm drain hose in the market, fix to Drain Elbow 3 .
- This hose must to be installed in a continuously downward direction and in a frost-free environment. Improper drain piping may cause water leakage hence damage the furnitures.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Guide the drain hose to outdoor as illustrated.

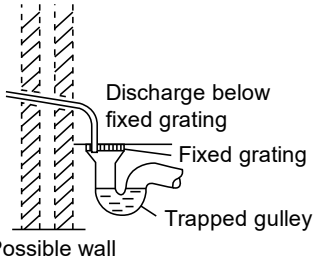
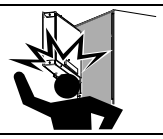


Illustration of guide drain hose to outdoor

- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulphuric gas etc.
- If necessary, use hose clamp to further tighten the hose at drain hose connector to prevent leakage.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.

12.2.4 Connect the Cable to the Tank Unit

⚠ WARNING
This section is for authorized and licensed electrician only. Work behind the Control Board Cover ⑤ secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

⚠ CAUTION	
Please take extra precaution when open the control board cover ⑤ and control board ⑥ for unit installation and servicing. Failure to do so may cause injury.	

12.2.4.1 Fixing of Power Supply Cable and Connecting Cable

1. Connecting cable between Tank Unit and Outdoor Unit shall be approved polychloroprene sheathed flexible cord, type designation 60245 IEC 57 or heavier cord. See below table for cable size requirement.

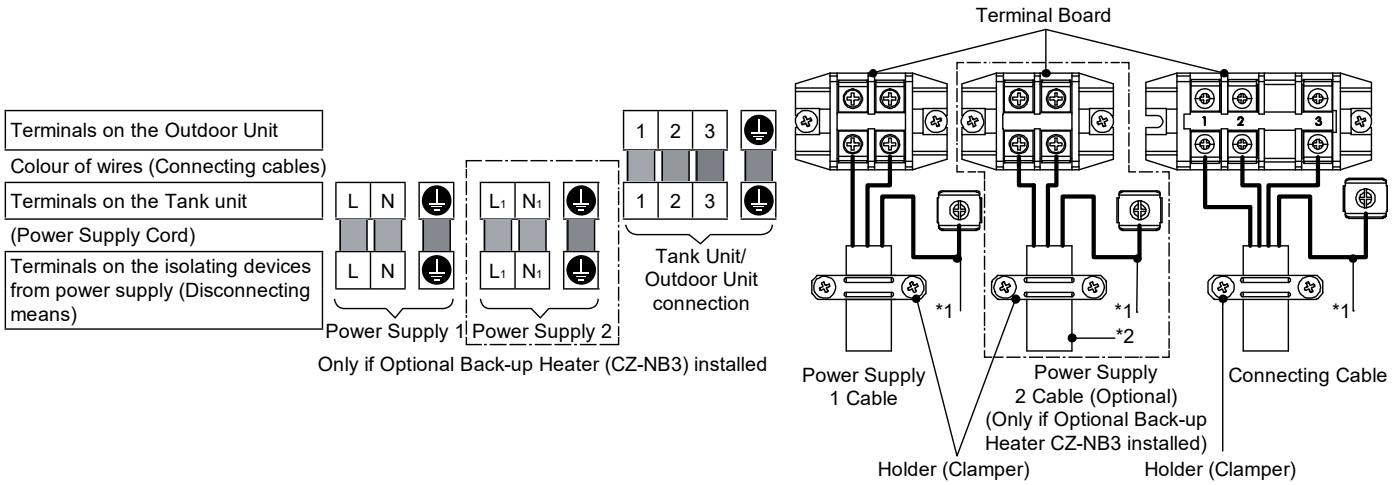
Model		Connecting Cable Size
Tank Unit	Outdoor Unit	
WH-ADC0309K0E5A	WH-UDZ03KE5*, WH-UDZ05KE5*	4 × min 1.5 mm^2
	WH-UDZ07KE5*, WH-UDZ09KE5*	4 × min 2.5 mm^2

   - Ensure the colour of wires of Outdoor Unit and the terminal no. are the same to the Tank Unit respectively.
   - Earth wire shall be longer than the other wires as shown in the figure for the electrical safety in case of the slipping out of the cord from the Holder (Clamper).

2. An isolating device must be connected to the power supply cable.
   - Isolating device (disconnecting means) should have minimum 3.0 mm contact gap.
   - Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord (Only if Optional Back-up Heater (CZ-NB3) installed) and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means). See below table for cable size requirement.

Model		Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
Tank Unit	Outdoor Unit				
WH-ADC0309K0E5A	WH-UDZ03KE5*, WH-UDZ05KE5*	1	3 × min 1.5 mm^2	15/16A	30mA, 2P, type A
		2	3 × min 1.5 mm^2	16A	30mA, 2P, type AC
	WH-UDZ07KE5*, WH-UDZ09KE5*	1	3 × min 2.5 mm^2	25A	30mA, 2P, type A
		2	3 × min 1.5 mm^2	16A	30mA, 2P, type AC

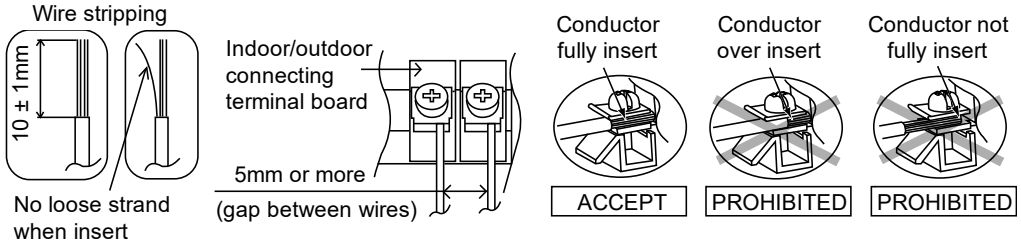
3. To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through a bushing (located at the bottom of Control Board ⑥) before terminal board. The bushing must be used and must not be removed.



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25}

*1 - Earth wire must be longer than other cables for safety reasons
*2 - Power supply cord 2 is optional and connected only if Optional Back-up Heater (CZ-NB3) installed.

12.2.4.2 Wire Stripping and Connecting Requirement



12.2.4.3 Connecting Requirement

For Tank Unit WH-ADC0309K0E5A with WH-UDZ03KE5*, WH-UDZ05KE5*, WH-UDZ07KE5*, WH-UDZ09KE5*

- The equipment's Power Supply 1 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-3 and can be connected to current supply network.

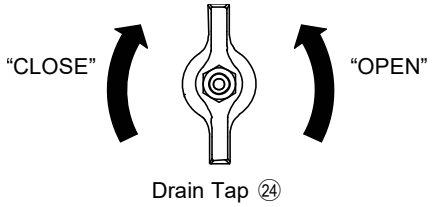
12.2.5 Charging and Discharging the Water

- Make sure all the piping installations are properly done before carry out below steps.

12.2.5.1 Charge the Water

For Domestic Hot Water Tank

1. Set the Drain Tap ㉔ to “CLOSE”.

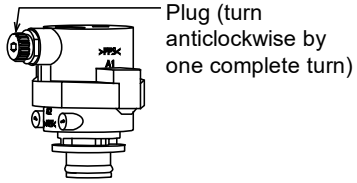


Drain Tap ㉔

2. Set all Tap / Shower “OPEN”.
3. Start filling water to the Domestic Hot Water Tank via Tube Connector ⓒ. After 20~40min, water should flow out from Tap / Shower.
   Else, please contact your local authorized dealer.
4. Check and make sure no water leaking at the tube connecting points.
5. Set the Drain Tap ㉔ to “OPEN” for 10 seconds to release air from this pipeline. Then set it “CLOSE”.
6. Turn the Safety Relief Valve (field supply) knob slightly and hold for 10 seconds to release air from this pipeline. Then recover the knob to original position.
7. Ensure Step 5 & 6 is carried out each time after charging water to Domestic Hot Water Tank.
8. To prevent back pressure from happening to the Safety Relief Valve (field supply), do turn the Safety Relief Valve (field supply) knob.

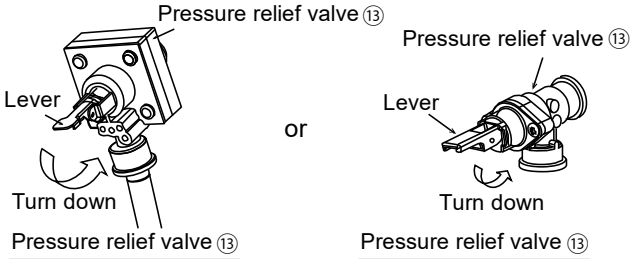
For Space Heating / Cooling

1. Turn the plug on the Air Purge Valve ⑫ outlet anticlockwise by one complete turn from fully closed position.



Air purge valve ⑫

2. Set the Pressure Relief Valve ⑬ lever “DOWN”.



3. Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Space Heating / Cooling circuit via Tube Connector ⓐ. Stop filling water if the free water flow through Pressure Relief Valve ⑬ discharge hose.
4. Turn ON the Tank Unit and make sure Water Pump ④ is running.
5. Check and make sure no water leaking at the tube connecting points.
6. The water may drip from this discharge hose. Therefore must guide the hose without close or block the outlet of the hose.

12.2.5.2 Discharge the Water

For Domestic Hot Water Tank

1. Turn OFF power supply.
2. Set the Drain Tap ㉔ to “OPEN”.
3. Open Tap / Shower to allow air inlet.
4. Turn the Safety Relief Valve (field supply) knob slightly and hold it until all air is released from this pipeline. Then recover the knob to original position after ensured the pipeline is emptied.
5. After discharge, set Drain Tap ㉔ to “CLOSE”.

12.2.6 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings.

12.2.6.1 Check Water Pressure

*(0.1 MPa = 1 bar)
Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure from Remote Controller). If necessary add water into Tank Unit (via Tube Connector ⓐ).

12.2.6.2 Check Pressure Relief Valve ⑬

- Check for correct operation of Pressure Relief Valve ⑬ by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keep on draining out from the Tank Unit, switch off the system, and then contact your local authorized dealer.

12.2.6.3 Expansion Vessel ⑪ Pre Pressure Checking

For Space Heating / Cooling

- Expansion Vessel ⑪ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit's piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m.

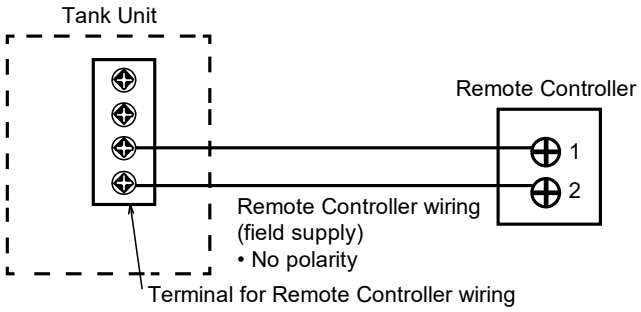
12.2.7 Installation of Remote Controller as Room Thermostat

- Remote Controller ① mounted to the Tank Unit can be moved to the room and serve as Room Thermostat.

12.2.7.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1. By the window, etc. exposed to direct sunlight or direct air.
  2. In the shadow or backside of objects deviated from the room airflow.
  3. Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4. Location near heat source.
  5. Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

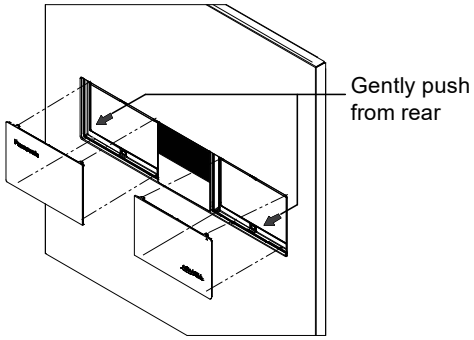
12.2.7.2 Remote Controller Wiring



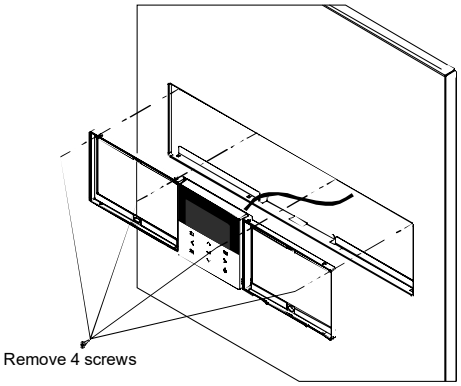
- Remote Controller cable shall be (2 × min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals of Tank Unit (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.

12.2.7.3 Remove the Remote Controller from Tank Unit

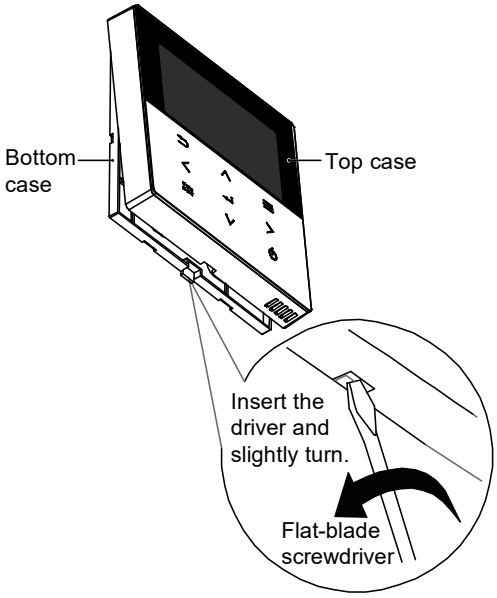
1. Remove both Left Decoration Panel ② and Right Decoration Panel ③ from Front Plate ⑯ with gently push the panels from back.



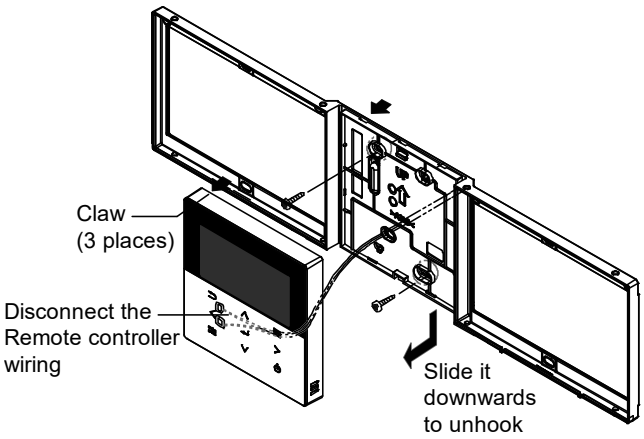
2. Remove the 4 screws and take out the holder with Remote Controller ①.



3. Remove the top case from the bottom case.

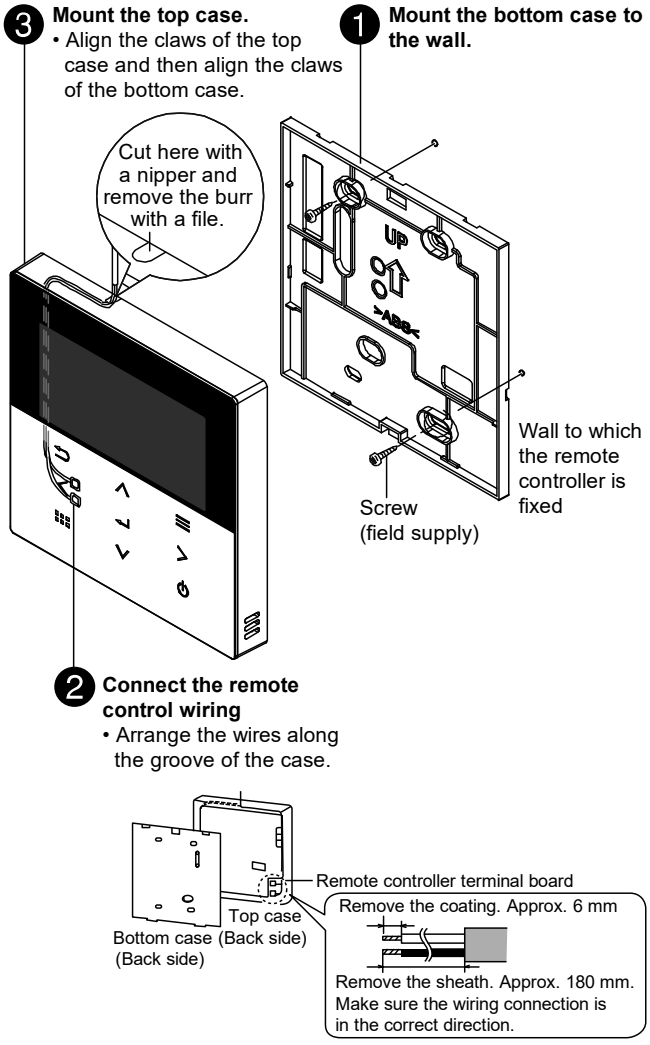


4. Remove the wiring between Remote controller ① and Tank Unit terminal.

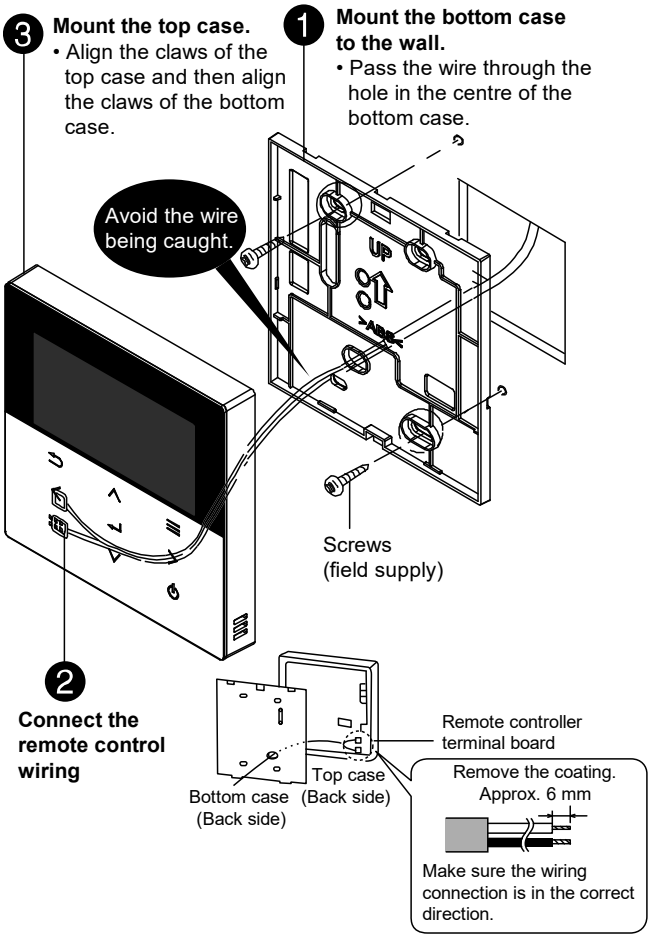


12.2.7.4 Mounting the Remote Controller

For exposed type
Preparation: Make 2 holes for screws using a driver.



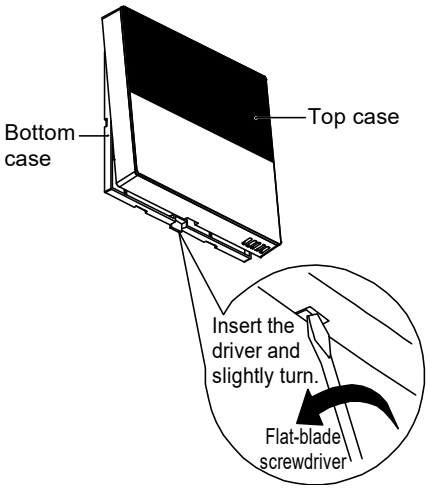
For embedded type
Preparation: Make 2 holes for screws using a driver.



12.2.7.5 Replace the Remote Controller Cover

- Replace the existing Remote Controller with Remote Controller Case [5] to close the hole left after remove the Remote Controller.

1. Refer Section “Remove The Remote Controller From Tank Unit” for remove Remote Controller.
2. Remove the top case from the bottom case of Remote Controller Case [5] .



3. Reverse the steps 1 to 4 of section “Remove The Remote Controller From Tank Unit” to fix Remote Controller Case [5] on Tank Unit.

12.2.8 Test Run

1. Before test run, make sure below items have been checked:-
   a) Pipework are properly done.
   b) Electric cable connecting work are properly done.
   c) Tank Unit is filled up with water and trapped air is released.
   d) Please turn on the power supply after filling the tank until full.
2. Switch ON the power supply of the Tank Unit. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run.
  Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

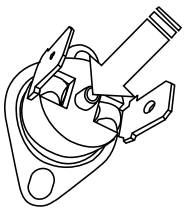
3. For normal operation, Water Pressure reading should be in between 0.05 MPa and 0.3 MPa (0.5 bar and 3 bar). If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4. After test run, please clean the Magnetic Water Filter Set ⑨. Reinstall it after finish cleaning.

12.2.8.1 Check Water Flow of Water Circuit

Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]

12.2.8.2 Reset Overload Protector (Prefixed on Optional Back-up Heater (CZ-NB3))

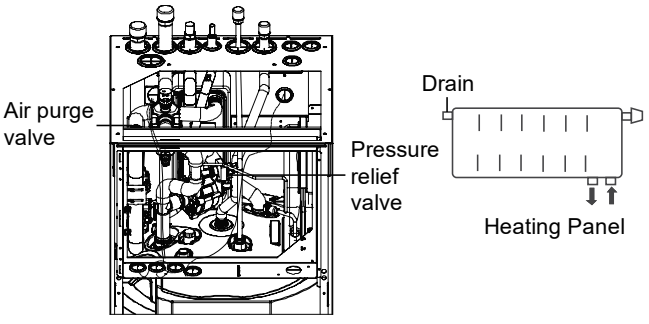
Overload Protector serves the safety purpose to prevent the water over heating. When the Overload Protector a trip at high water temperature, take below steps to reset it.
1. Take out the cover.
2. Use a test pen to push the centre button gently in order to reset the Overload Protector.
3. Fix the cover to the original fixing condition.



- Must ensure that Overload Protector is fixed/installed on the Optional Back-up heater (CZ-NB3).
- Overload Protector Specifications:
  - Operating Temperature : 80 ±3°C (OFF)
  - Rated Voltage, Current : AC 250V, 16A; DC 12V, 50mA
  - Type: Single pole single throw type thermostat with disk bimetal.
- For installation works of Optional Back-up Heater (CZ-NB3), refer to the Optional Back-up Heater (CZ-NB3) Installation Instruction.

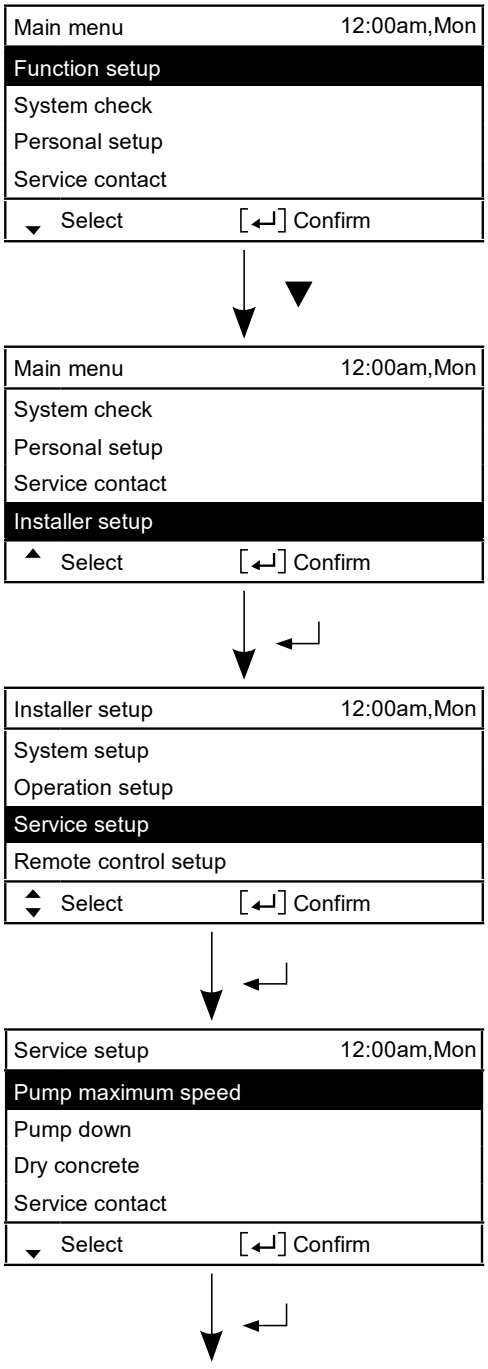
12.2.8.3 Before Test Run

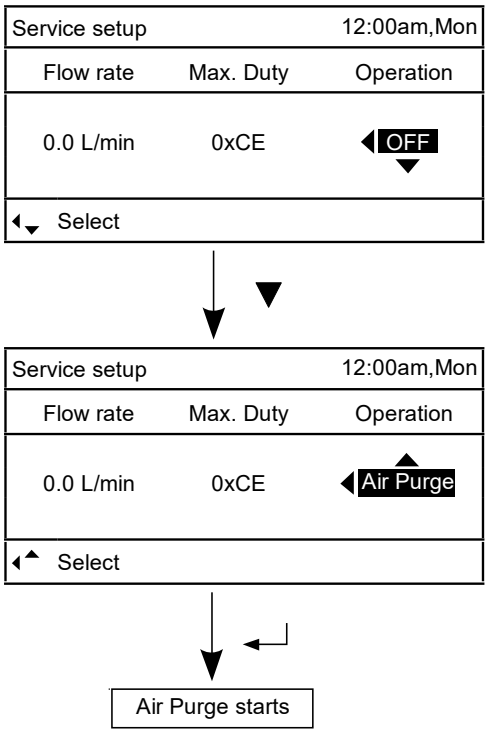
- Before test-running the water circulation circuit, loosen the plug on the unit’s air purge valve counter-clockwise. Make sure to thoroughly release the air from the drain plugs on each terminal.



12.2.8.4 During Test Run (Air Purging)

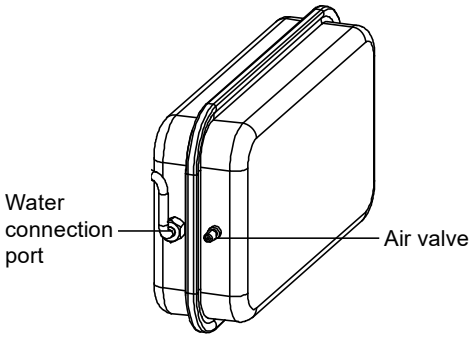
- Make sure to carry out thorough air purging during commissioning to prevent pump from dry running.
- Access Remote Controller service setup for initiate air purging function.





Note:

- When air purge operation during initial commissioning, ensure constant water pressure applied by supplying water to the circulation circuit.
- Do not leave pump running dry for a long time, it may cause pump malfunction.
- Ensure to check pre-charge pressure of the expansion vessel during installation, as air is gradually escape after a certain period. If the static pressure in the expansion vessel falls below 0.5 bar, cavitation and pump lock may happen.
- It is strongly recommended to check the expansion vessel pre-charged pressure once a year. (The pressure may be reduced by half in two years)
- After the air purge operation, carry out normal operation. When you can no longer hear the sound of air flowing from the water circulation circuit, turn the air purge valve plug clockwise to tighten it.



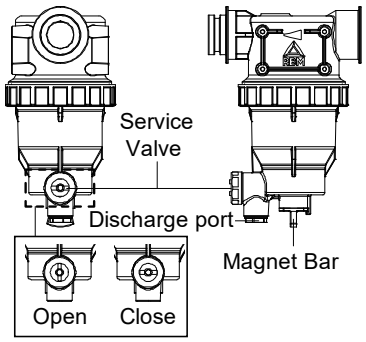
Expansion Vessel

12.2.9 Maintenance

- In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

12.2.9.1 Maintenance for Magnetic Water Filter Set ⑨

1. Turn OFF power supply.
2. Place a container below Magnetic Water Filter Set ⑨.
3. Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑨.
4. By using Allen key (8mm), remove the Cap of Discharge Port.
5. By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
6. Reinstall the Cap of Discharge Port and Magnet Bar.
7. Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 12.2.5 for details.)
8. Turn ON power supply.



12.2.9.2 Maintenance for Safety Relief Valve (Field Supply)

- It is strongly recommended to operate the valve by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

12.2.9.3 Proper Pump Down Procedure

⚠ WARNING

Strictly follow the steps below for proper pump down procedure. Explosion may occur if the steps are not followed as per sequence.

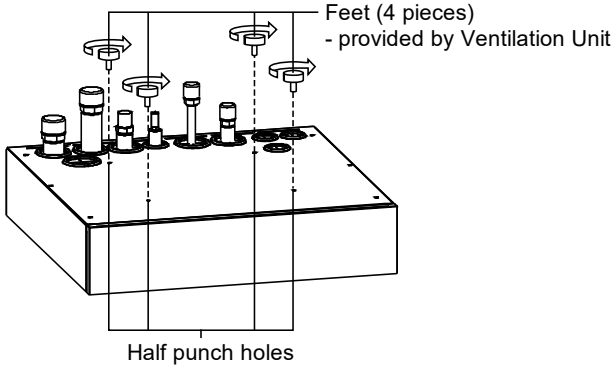
1. When the Tank Unit is not in operation (standby), enter the Service setup menu in the Remote Controller and select Pump down operation to turn it ON. (See APPENDIX for detail)
2. After 10~15 minutes, (after 1 or 2 minutes in case very low ambient temperature (< 10°C)), fully close 2 way valve on Outdoor Unit.
3. After 3 minutes, fully close 3 way valve on Outdoor Unit.
4. Press the “OFF/ON” switch on the Remote Controller ① to stop pump down operation.
5. Remove the refrigerant piping.

12.2.9.4 Installation of Ventilation Unit on Top of Tank Unit (Optional)

- For installation works of Ventilation Unit on top of Tank Unit, refer to the Ventilation Unit Installation Manual.

⚠ CAUTION

Before install Ventilation Unit, fix the Feet that provided by Ventilation Unit to the half punch holes on Top Panel of Tank Unit, Otherwise, heavy Ventilation Unit may fall and cause injury.



12.3 Outdoor Unit

Attached accessories

No.	Accessories part	Qty.
1	Drain elbow 	1
2	Rubber cap 	7 (For WH-UDZ03KE5*) 3 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
3	Discharge grille 	1 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
4	Metal plate 	2 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
5	Screw 	8 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)

Optional accessories

No.	Accessories part	Qty.
6	Base Pan Heater CZ-NE2P	1

- It is strongly recommended to install a Base Pan Heater (optional) if the outdoor unit is install in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.
- Applicable Piping Kit (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*) CZ-52F5,7,10BP
- Applicable Piping Kit (For WH-UDZ03KE5*) CZ-4F5,7,10BP

12.3.1 Select the Best Location

- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- For WH-UDZ03KE5*, avoid installation in areas where the ambient temperature may drop below -20°C.
- For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*, avoid installations in areas where the ambient temperature may drop below -25°C.
- Keep the spaces indicated by arrows from wall, ceiling, fence or other obstacles.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- If outdoor unit installed near sea, region with high content of sulphur or oily location (e.g. machinary oil, etc), it lifespan maybe shorten.
- If piping length is over 10 m, additional refrigerant should be added as shown in the table.

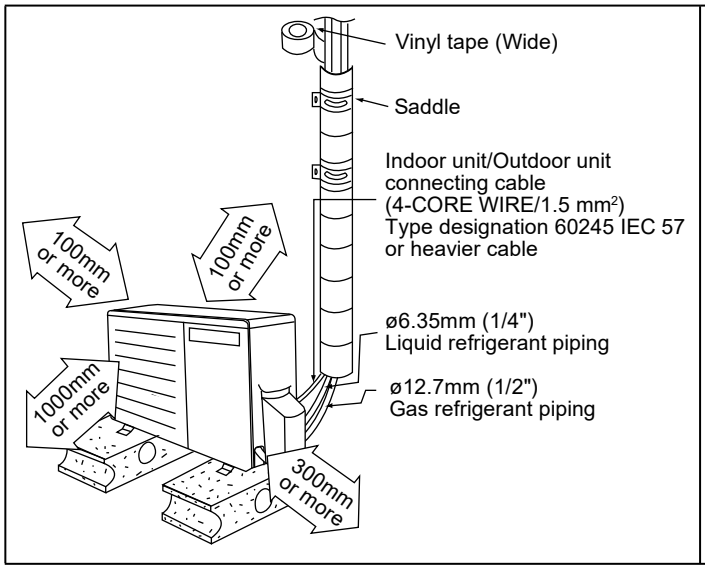
Model	Piping size		Pre-charged Refrigerant (kg)	Rated Length (m)		Max. Elevation (m)	Min. Piping Length (m)	Max. Piping Length (m)	Additional Refrigerant (g/m)
	Gas	Liquid		For Heat Pump Indoor Unit	For Hydromodule + Tank				
WH-UDZ03KE5*	ø12.7mm (1/2")	ø6.35mm (1/4")	0.90	7	7	20	3	25	20
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8")	ø6.35mm (1/4")	1.30	7	7	30	3	50	25

Example: WH-UDZ03KE5*
If piping length is 15m, the quantity of additional refrigerant should be 100g. [(15-10)m x 20 g/m = 100g]

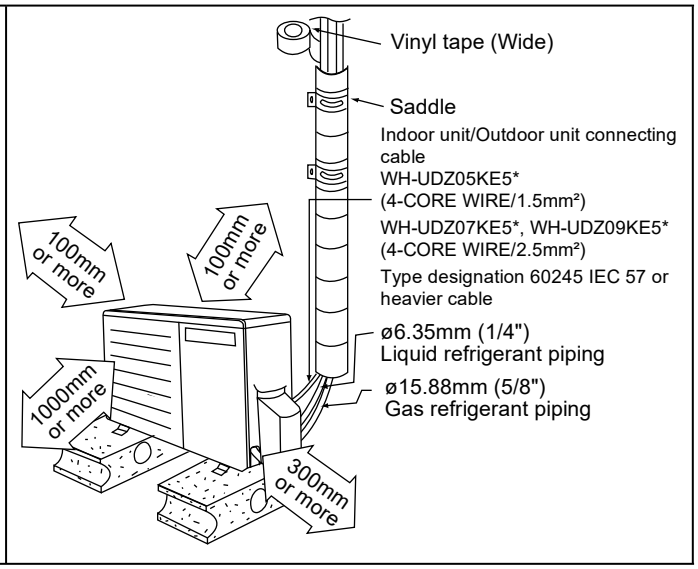
12.3.2 Install the Outdoor Unit

12.3.2.1 Installation Diagram

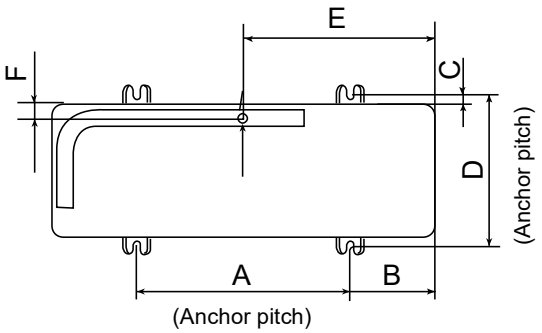
- It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.
- This illustration is for explanation purposes only.



For WH-UDZ03KE5*



For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*



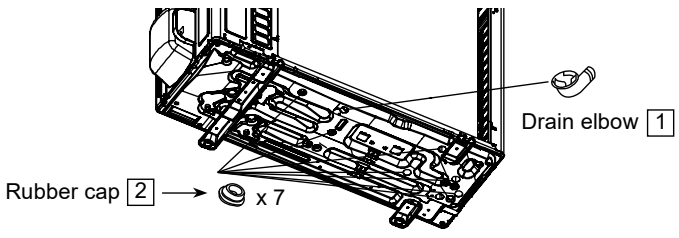
Model	A	B	C	D	E	F
WH-UDZ03KE5*	540	160	20	330	430	46
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	613	130	24	360.5	543	32

(Unit : mm)

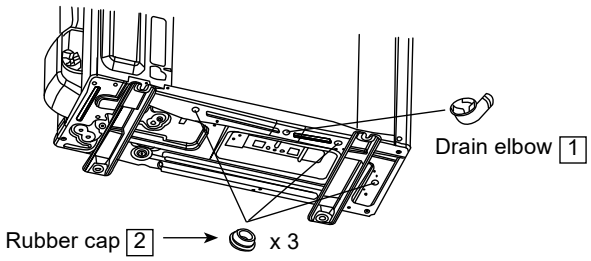
- After selecting the best location, start installation according to Installation Diagram.
  1 Fix the unit on concrete or rigid frame firmly and horizontally by bolt nut (ø10 mm).
  2 When installing at roof, please consider strong wind and earthquake. Please fasten the installation stand firmly with bolt or nails.

12.3.2.2 Disposal of Outdoor Unit Drain Water

- When a Drain elbow [1] is used, please ensure to follow below:
  - the unit should be placed on a stand which is taller than 50 mm.
  - cover the ø20mm holes with Rubber cap [2] (refer to illustration below).
  - use a tray (field supply) when necessary to dispose the outdoor unit drain water.
- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow [1] and Rubber cap [2], for the drain water freezes and the fan will not rotate.



WH-UDZ03KE5*



WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*

12.3.3 Install the Discharge Grille

- This section is for WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5* only.
- Make sure to install the outdoor unit on concrete or rigid frame by bold nut before install the discharge grille [3].
- Hook the metal plate [4] on the cabinet front plate at 2 locations and tighten with screw [5]. (Figure a)
- Hook the discharge grille [3] to the cabinet front plate at 4 locations. (Figure b)
- Tighten the discharge grille to the cabinet front plate at 6 locations with screw [5]. (Figure c)

⚠ CAUTION
Make sure to install the discharge grille to the outdoor unit before powering ON to protect againts a rotating fan

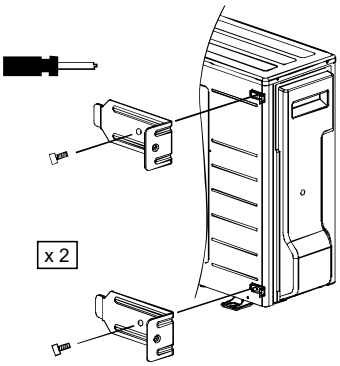


Figure a

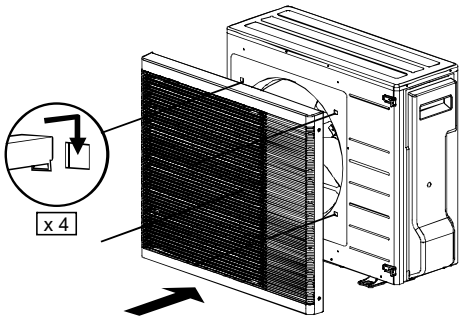


Figure b

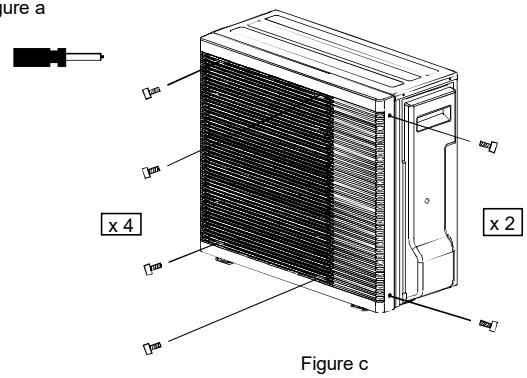
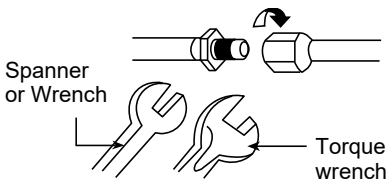


Figure c

12.3.4 Connecting the Piping

12.3.4.1 Connecting the Piping to Outdoor Unit

Decide piping length and then cut by using pipe cutter. Remove burrs from cut edge. Make flare after inserting the flare nut (locate at valve) onto the copper pipe. Align center of piping to valves and then tighten with torque wrench to the specified torque as stated in the table.

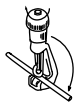


Be sure to use two spanners to tighten.
(If the nuts are overtightened, it may cause the flares to break or leak.)

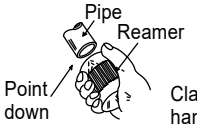
Model	Piping size (Torque)	
	Gas	Liquid
WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]

12.3.4.1.1 Cutting and Flaring the Piping

1 Please cut using pipe cutter and then remove the burrs.
2 Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3 Please make flare after inserting the flare nut onto the copper pipes.



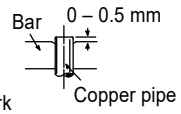
1. To cut

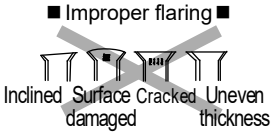


2. To remove burrs



3. To flare



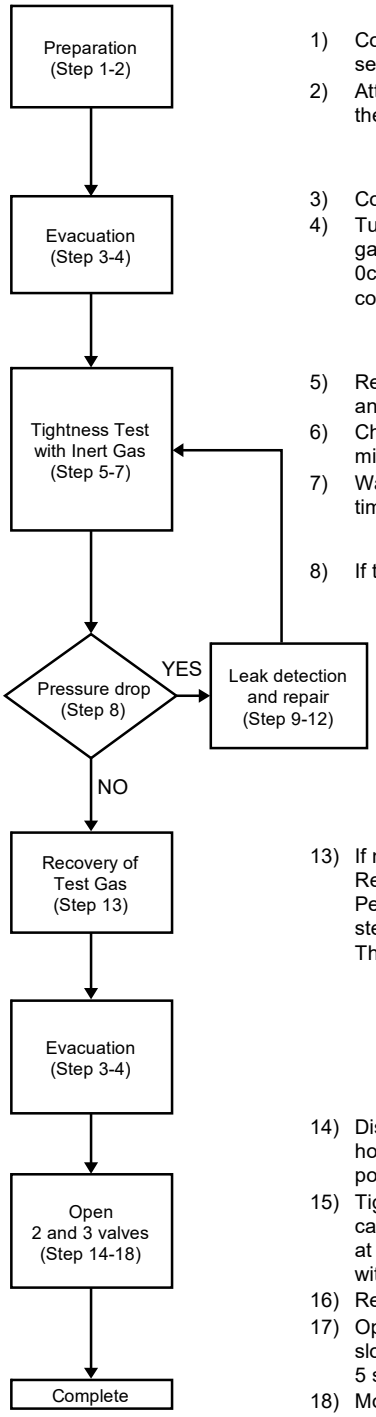


When properly flared, the internal surface of the flare will evenly shine and be of even thickness. Since the flare part comes into contact with the connections, carefully check the flare finish.

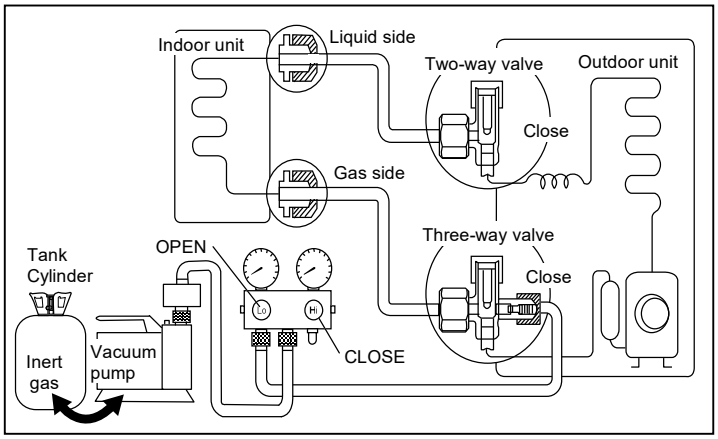
12.3.5 Air Tightness Test on the Refrigerating System

⊘	Do not purge the air with refrigerants but use a vacuum pump to vacuum the installation.
❗	There is no extra refrigerant in the outdoor unit for air purging.

- Before system is charged with refrigerant and before the refrigerating system is put into operation, below site test procedure and acceptance criteria shall be vertified by the certified technicians, and/or the installer.
- Be sure to check whole system for gas leakage.



1) Connect a charging hose with a push pin to the Low side of a charging set and the service port of the 3-way valve.
2) Attach the gauge manifold set correctly and tightly. Make sure that both valves of the manifold gauge (low pressure and high pressure) is in close position.
3) Connect the center hose of the manifold gauge to a vacuum pump.
4) Turn on the power switch of the vacuum pump, then turn open the low side manifold gauge valve and make sure that the needle in the gauge moves from 0cmHg (0 MPa) to –76 cmHg (–0.1 MPa) or vacuum until 500 microns is achieved. This process continues for approximately ten minutes. Then close the low side manifold gauge valve.
5) Remove the vacuum pump from the centre hose and connect the center hose to cylinder of any applicable inert gas as test gas.
6) Charge test gas into the system and wait until the pressure within the system to reach min. 1.04MPa (10.4barg).
7) Wait and monitor the pressure reading on the gauges. Check if there is any pressure drop. Waiting time depends on the size of the system.
8) If there is any pressure drop, perform step 9-12. If there is no pressure drop, perform step 13.
9) Use Gas Leak Detector to check for leaks. Must use the detection equipment with a sensitivity of 5 grams per year of test gas or better.
10) Move the probe along the Air-to-Water Heatpump system to check for leaks, and mark for repair.
11) Any leak detected and marked shall be repaired.
12) After repair, repeat evacuation steps 3-4 and tightness test steps 5-7. Check the pressure drop as in step 8.
13) If no leak, Recover the test gas. Perform evacuation of steps 3-4. Then proceed to step 14.



14) Disconnect the charging hose from the service port of the 3-way valve.
15) Tighten the service port caps of the 3-way valve at a torque of 18 N•m with a torque wrench.
16) Remove the valve caps of both of the 2-way valve and 3-way valve.
17) Open both of the valves, using a hexagonal wrench (4mm). It is recommended to allow refrigerant slowly flow into the refrigerant system to prevent refrigerant freezing. Slightly open 2-way valve for 5 seconds then close the valve. Repeat this action for 3 cycles then fully open the valve.
18) Mount back the valve caps onto the 2-way valve and the 3-way valve to complete this process.

Notes:

Recommended use of any of the following leak detector,

I) Universal Sniffer leak detector
II) Electronic halogen leak detector
III) Ultrasonic Leak Detector

12.3.6 Connect the Cable to the Outdoor Unit

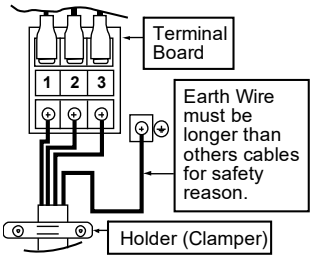
(FOR DETAIL REFER TO WIRING DIAGRAM AT UNIT)

1 Remove the control board cover from the unit by loosening the screw.

2 Connecting cable between indoor unit and outdoor unit shall be approved polychloroprene sheathed flexible cable (see below table), type designation 60245 IEC 57 or heavier cable.

Model	Flexible cable specification
WH-UDZ03KE5*, WH-UDZ05KE5*	4 x (1.5 mm^2)
WH-UDZ07KE5*, WH-UDZ09KE5*	4 x (2.5 mm^2)

Terminals on the indoor unit	1	2	3	⏚
Colour of wires				
Terminals on the outdoor unit	1	2	3	⏚

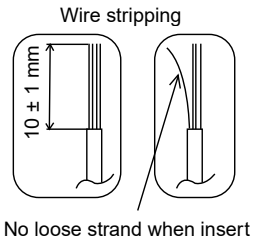


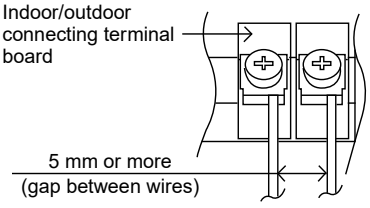
⚠ WARNING
⏚ This equipment must be properly earthed.

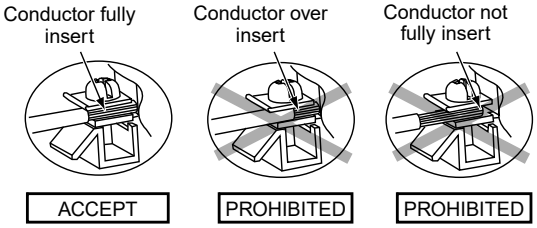
3 Secure the cable onto the control board with the holder (clamper).

4 Attach the control board cover back to the original position with screw.

12.3.6.1 Wire Stripping and Connecting Requirement







12.3.7 Pipe Insulation

1 Please carry out insulation at pipe connection portion as mentioned in Indoor/Outdoor Unit Installation Diagram. Please wrap the insulated piping end to prevent water from going inside the piping.

2 If drain hose or connecting piping is in the room (where dew may form), please increase the insulation by using POLY-E FOAM with thickness 6 mm or above.

⚠ CAUTION
If cleaning of the outdoor unit is necessary during installation or servicing, do not clean the outdoor unit with any hydrocarbon based solvent.

12.4 Appendix

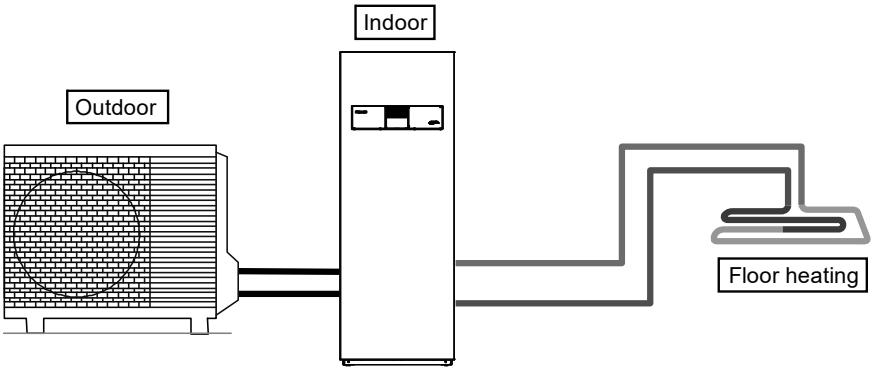
12.4.1 Variation of System

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.
(NOTE) : For this model, both external room thermistor of Zone 1 and external room thermostat of Zone 1 must always be connected to main indoor PCB only regardless of Optional PCB (CZ-NS5P) connection.

12.4.1.1 Introduce Application Related to Temperature Setting

12.4.1.1.1 Temperature Setting Variation for Heating

1. Remote Controller



Setting of remote controller

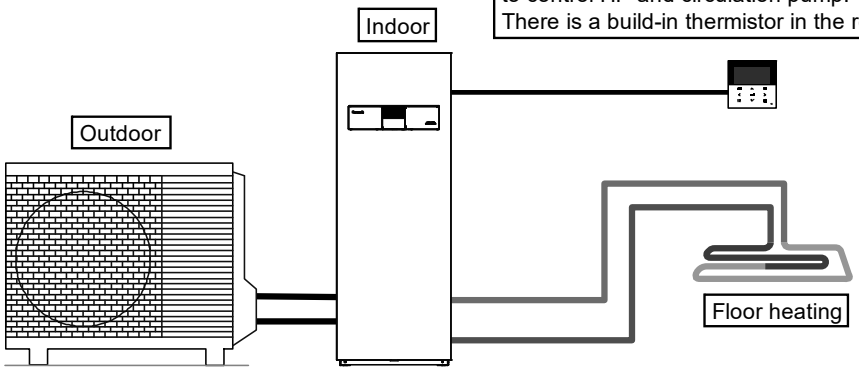
Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Water temperature

Connect floor heating or radiator directly to the Tank Unit.
Remote controller is installed on Tank Unit.
This is the basic form of the most simple system.

2. Room Thermostat

Tank Unit receive Room Thermo signal (ON/OFF) from Remote controller to control HP and circulation pump.
There is a build-in thermistor in the remote controller.



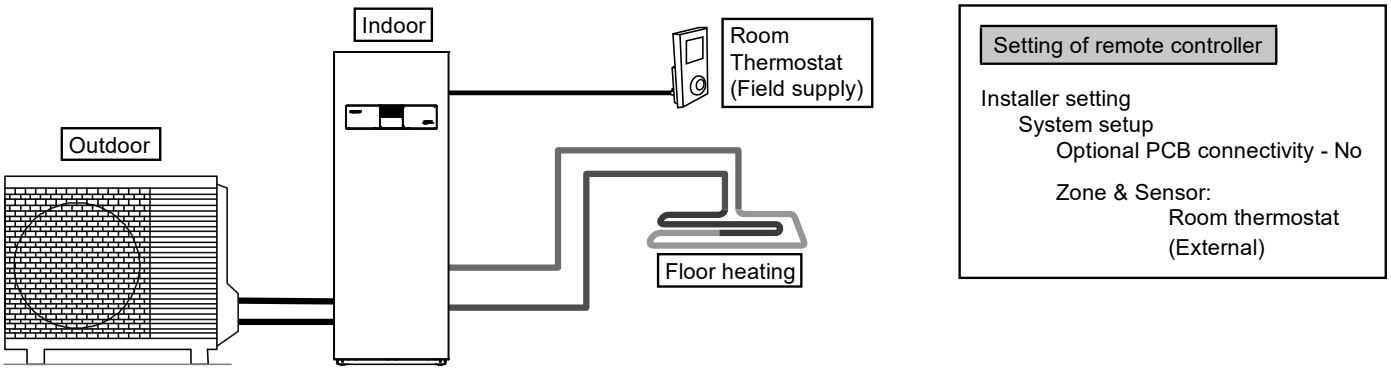
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
Internal

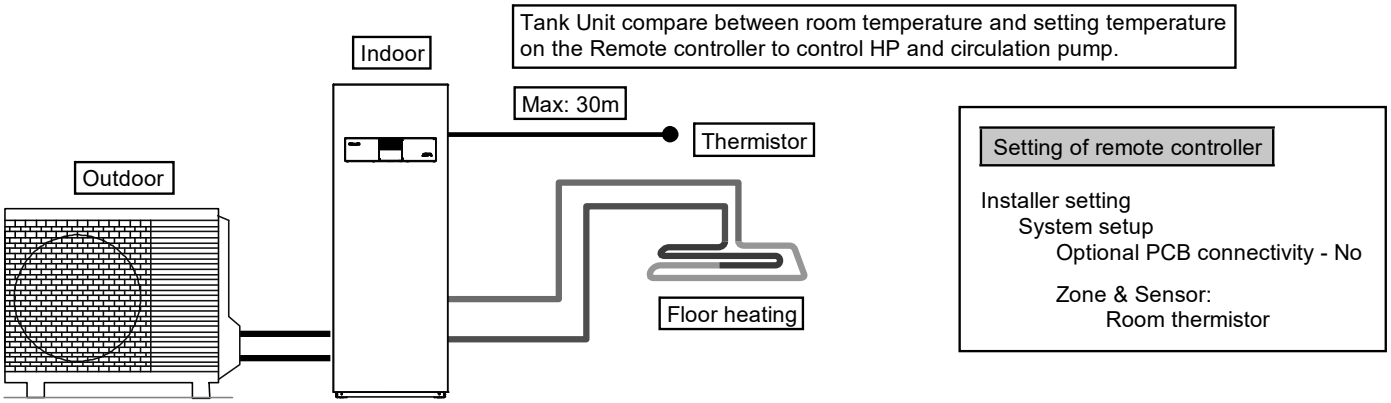
Connect floor heating or radiator directly to the Tank Unit.
Remove remote controller from Tank Unit and install it in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat



Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



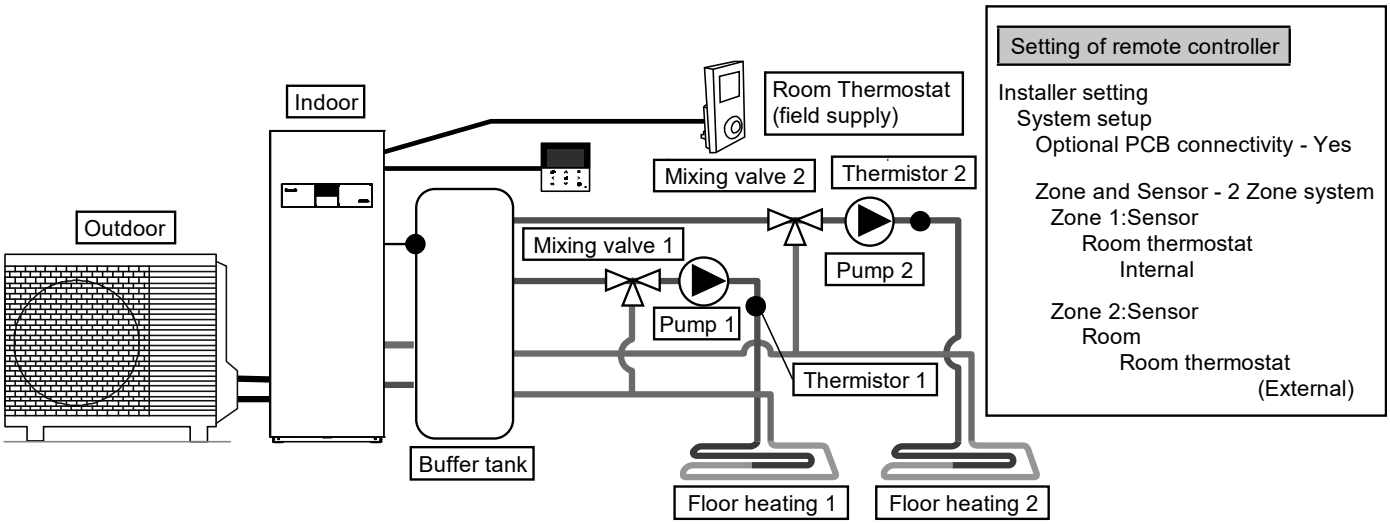
Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, compensation curve is shifted according to the thermo ON/OFF situation.

- (Example) If room temperature increasing speed is;
  very slow → shift up the compensation curve
  very fast → shift down the compensation curve

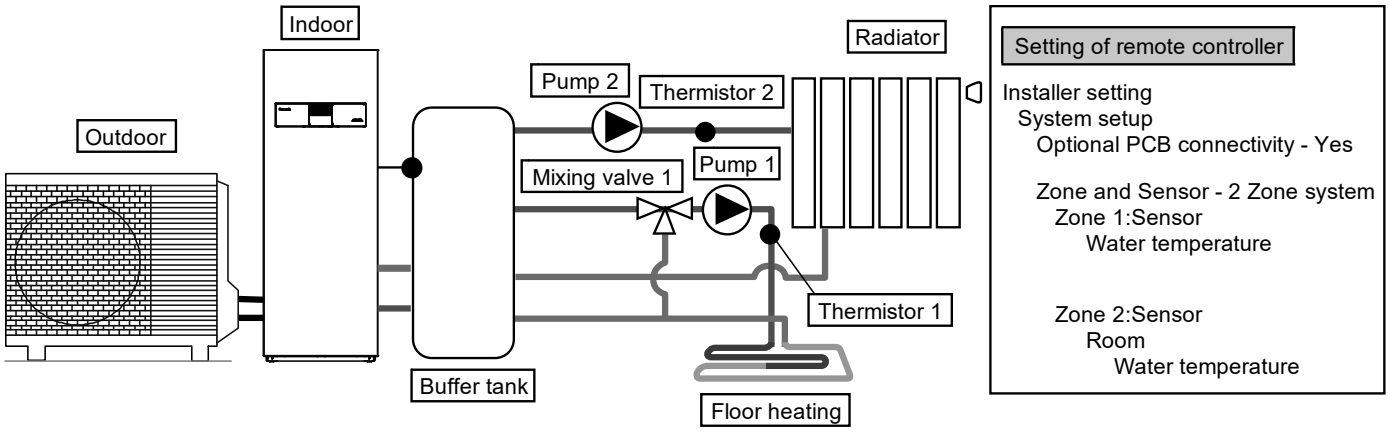
12.4.1.1.2 Examples of Installations

Floor heating 1 + Floor heating 2



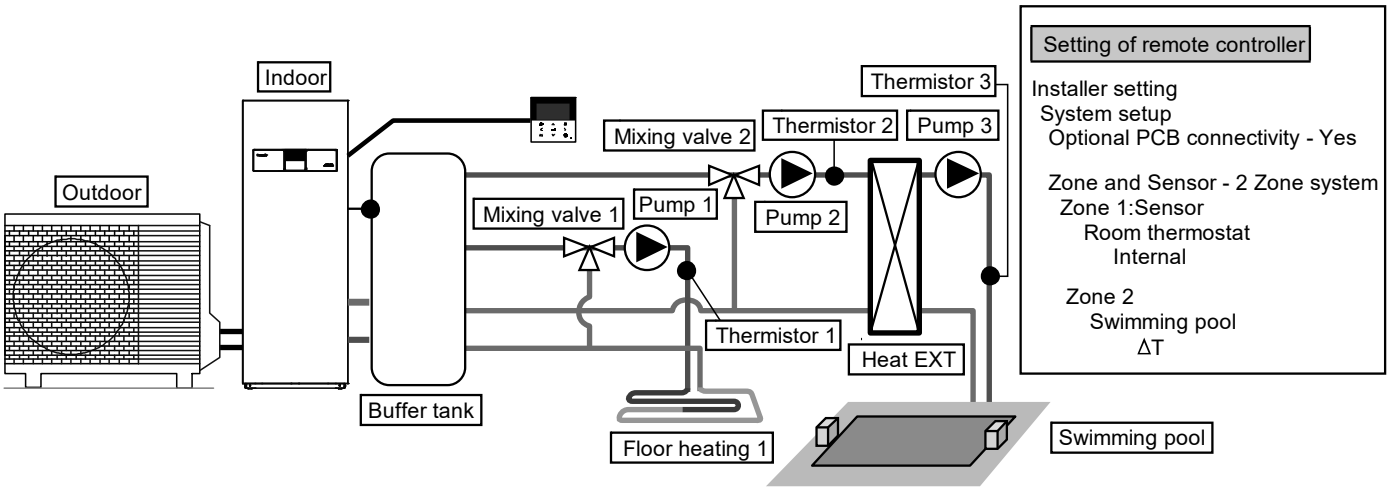
Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Remove remote controller from Tank Unit, install it in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Floor heating + Radiator



Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
Remote controller is installed on Tank Unit.
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the Optional PCB (CZ-NS5P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

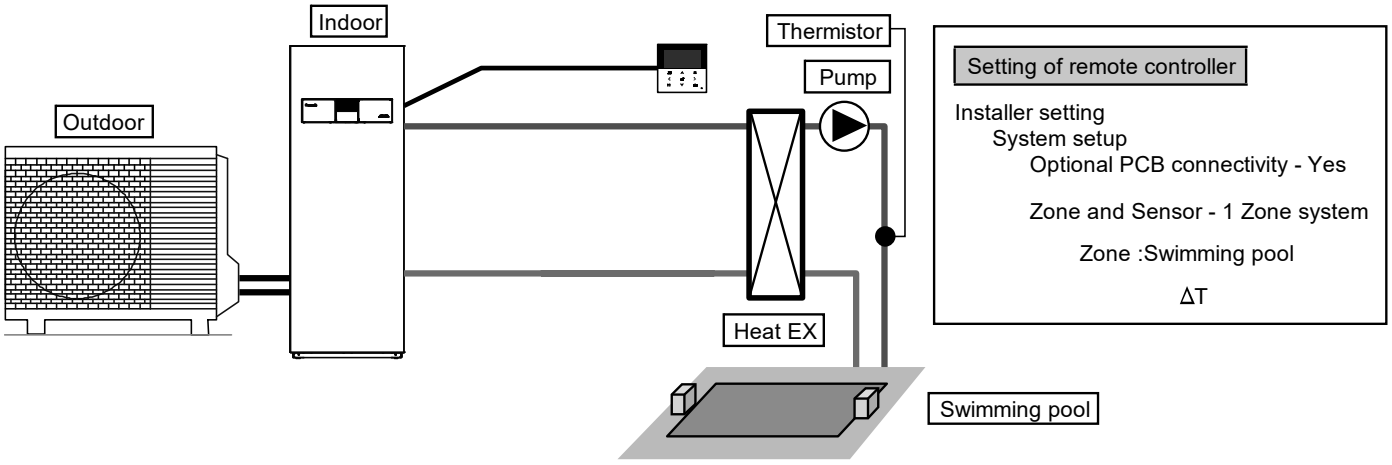
Floor heating + Swimming pool



Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Remove remote controller from Tank Unit and install in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the Optional PCB (CZ-NS5P).

❇ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

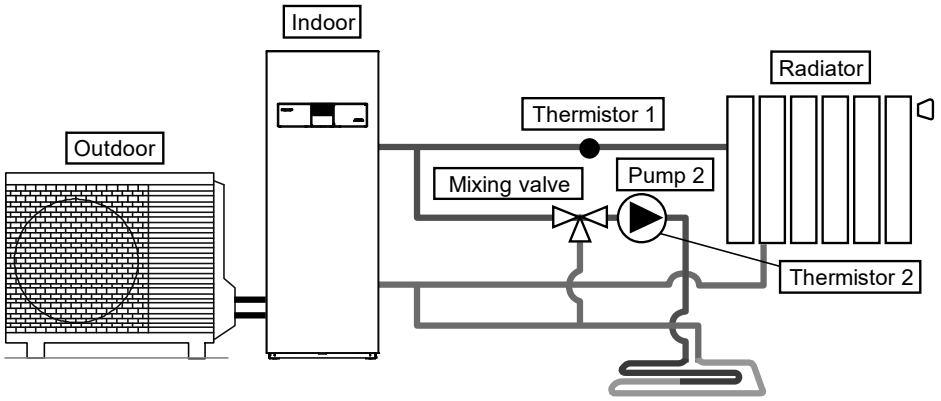
Swimming pool only



This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Tank Unit without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Remove remote controller from Tank Unit and install in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the Optional PCB (CZ-NS5P).

In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



Setting of remote controller

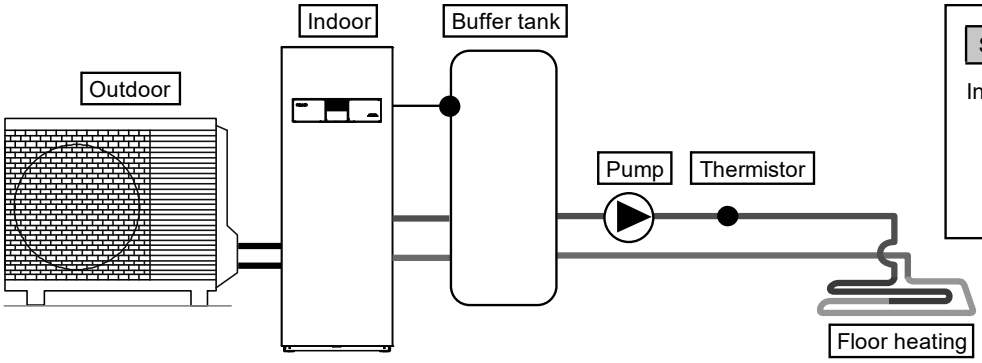
Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1:Sensor
Water temperature
Zone 2:Sensor
Room
Water temperature
Operation setup
Heat
ΔT for heating ON – 1°C
Cool
ΔT for cooling ON – 1°C

This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Tank Unit served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the Optional PCB (CZ-NS5P).

(NOTE)

- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
- Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

Buffer tank connection

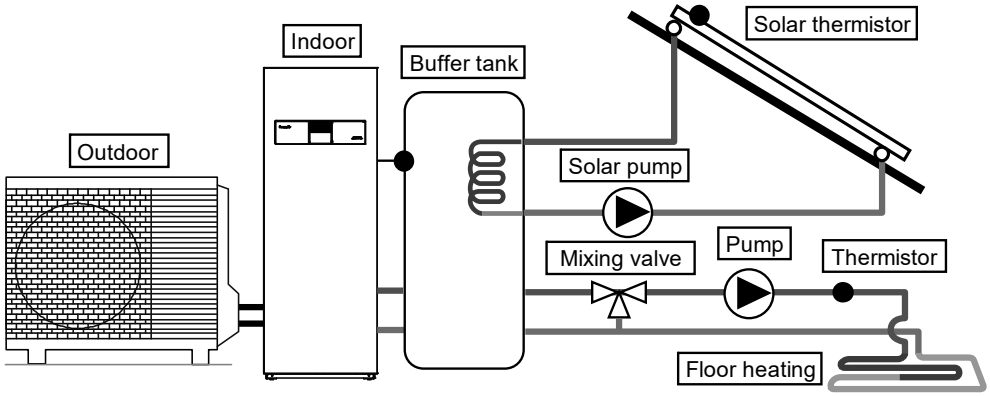


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No
Buffer Tank connection - Yes
ΔT for buffer tank

This is an application that connects the buffer tank to the Tank Unit.
Buffer tank’s temperature is detected by buffer tank thermistor (specified by Panasonic).
Without connection of Optional PCB, external pump can be used for circulation in the floor heating circuit.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Buffer tank + Solar

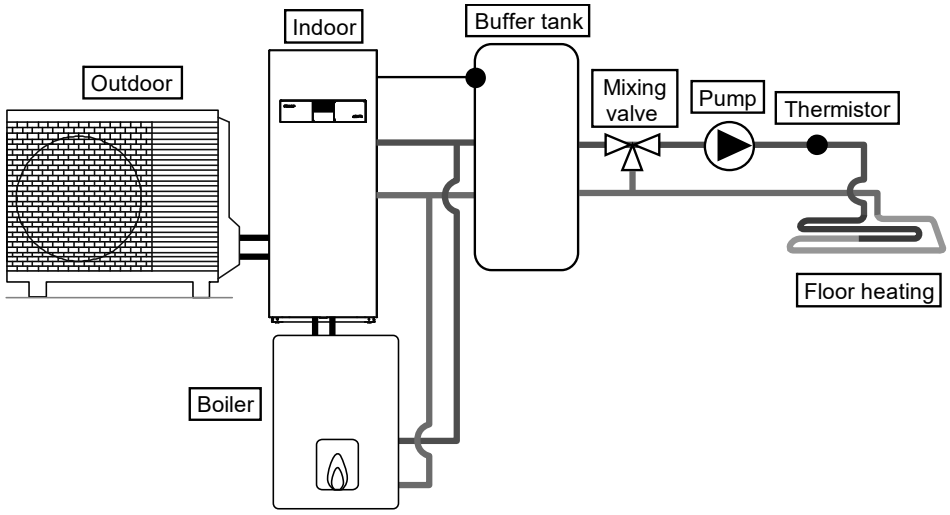


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank
Solar connection - Yes
Buffer tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the buffer tank to the Tank Unit before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Boiler connection



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Bivalent - Yes
Turn ON: outdoor temp.
Control pattern

This is an application that connects the boiler to the Tank Unit, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
Besides that, an application that connects to the DHW tank's circuit to heat up tank 's hot water is also possible.
Boiler output can be control by either SG ready input from optional PCB or Auto control by 3 modes selection pattern. (Operation setting of boiler shall be responsible by installer.)
This system requires Optional PCB (CZ-NS5P) for SG ready input control.
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when select Advanced Parallel setting.)
Note : Buffer tank thermistor must be connected to main indoor PCB only.

⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

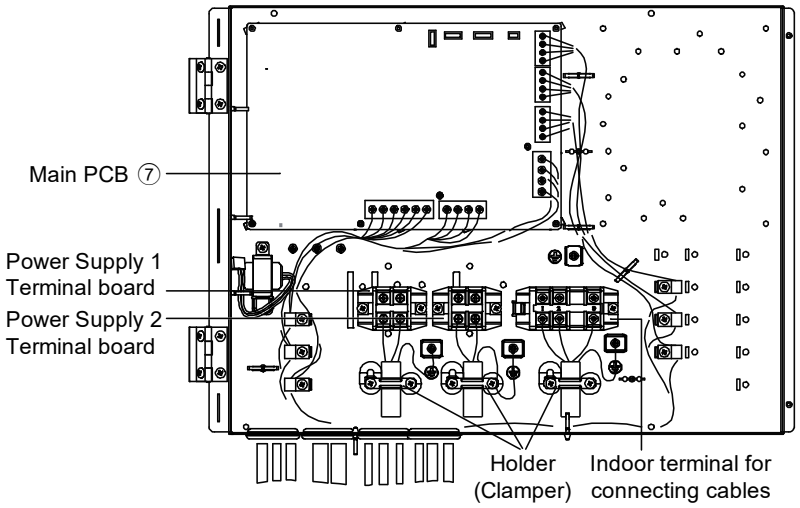
⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the Tank Unit does NOT exceed 55°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

12.4.2 How to Fix Cable

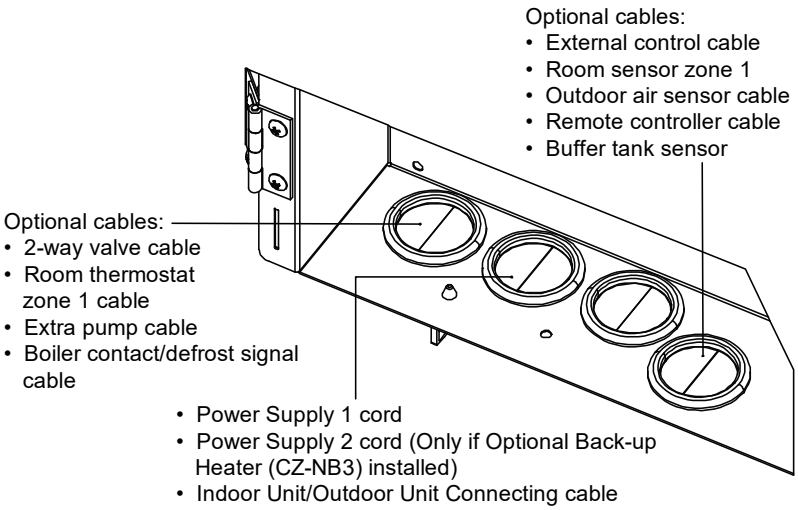
12.4.2.1 Connecting with External Device (Optional)

- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB ⑦

1. Two-way valve shall be spring and electronic type, refer to “Field Supply Accessories” table for details. Valve cable shall be (3 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
   *note: - Two-way Valve shall be CE marking compliance component.
   - Maximum load for the valve is 9.8VA.
2. Room thermostat zone 1 cable must be (4 or 3 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
3. Extra pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
4. Boiler contact/defrost signal cable shall be (2 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
5. External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 × min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
   *note: - Switch used shall be CE compliance component.
   - Maximum operating current shall be less than $3A_{rms}$.
6. Room sensor zone 1, outdoor air sensor and buffer tank sensor cable shall be (2 × min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.

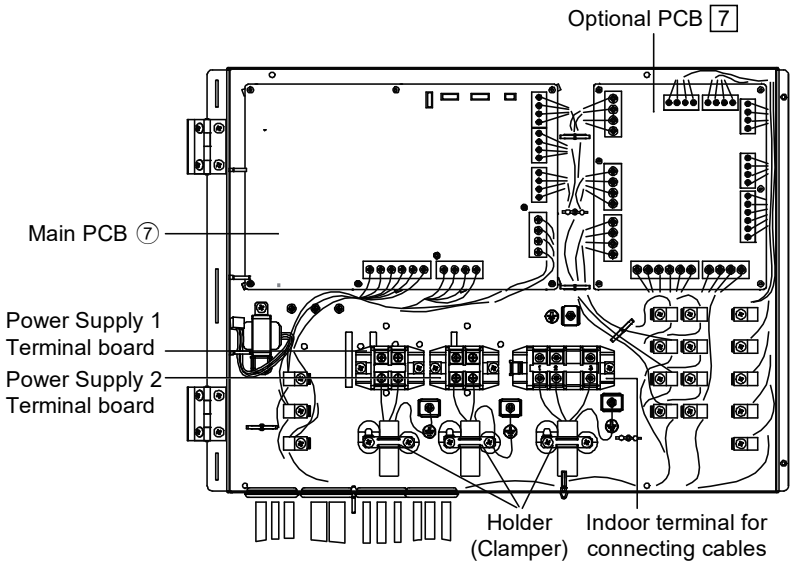


How to guide the optional cables and power supply cord
(view without internal wiring)

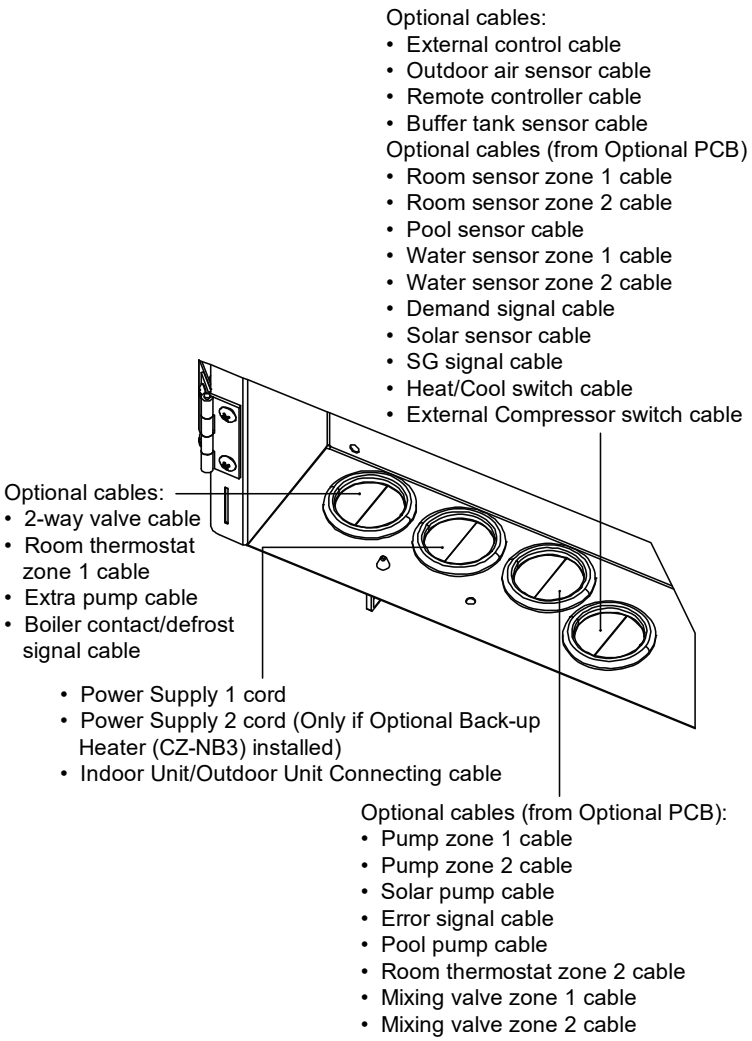


- For connection to Optional PCB [7]

1. By connecting Optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in Optional PCB. Temperature of each zone can be controlled independently by remote controller.
2. Pump zone 1 and zone 2 cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
3. Solar pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
4. Pool pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
5. Room thermostat zone 2 cable shall be (4 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
6. Mixing valve zone 1 and zone 2 cable shall be (3 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
7. Room sensor zone 1 and zone 2 cable shall be (2 × min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
8. Pool water sensor and solar sensor cable shall be (2 × min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
9. Water sensor zone 1 and zone 2 cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
10. Demand signal cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
11. SG signal cable shall be (3 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
12. Heat/Cool switch cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
13. External compressor switch cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide the optional cables and power supply cord
(view without internal wiring)

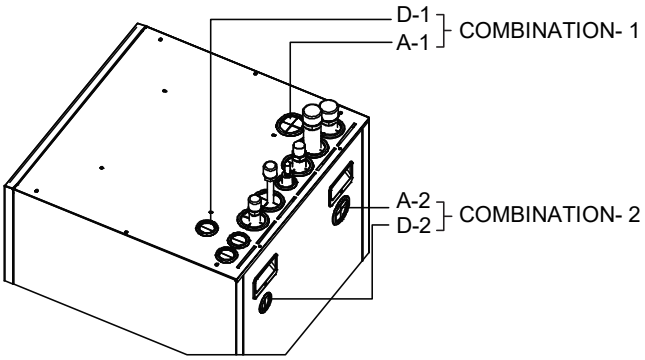


Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

12.4.2.2 Guide Optional Cables and Power Supply Cords to Bushings

⚠ CAUTION
Wire guiding shall be free from hot surfaces. Else, cable insulator damage and electrical shock may happen.
Wire ways shall be smooth and free from sharp edges. Else, cable insulator damage and electrical shock may happen.

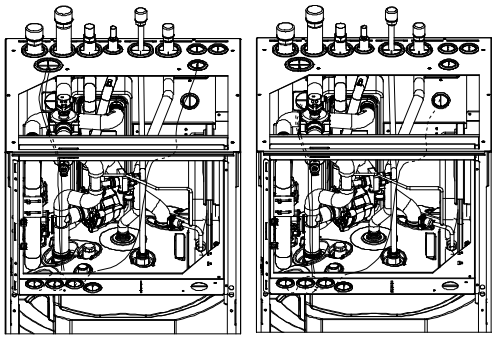
- Use either “COMBINATION-1” or “COMBINATION-2” for guiding Optional Cables and Power Supply Cords to Bushings.



- A-1 and A-2 Bushings are for:
  - Power Supply 1 cord
  - Power Supply 2 cord
  - Indoor Unit/Outdoor Unit Connecting cable
  - Pump zone 1 cable
  - Pump zone 2 cable
  - Solar pump cable
  - Room thermostat zone 1 cable
  - Room thermostat zone 2 cable
  - Mixing valve zone 1 cable
  - Mixing valve zone 2 cable
  - 2-way valve cable
  - Extra pump cable
  - Boiler contact cable

- D-1 and D-2 Bushings are for:
  - External control cable
  - Outdoor air sensor cable
  - Remote controller cable
  - Room sensor zone 1 cable
  - Room sensor zone 2 cable
  - Buffer tank sensor cable
  - Pool sensor cable
  - Water sensor zone 1 cable
  - Water sensor zone 2 cable
  - Demand signal cable
  - Solar sensor cable
  - SG signal cable
  - Heat/Cool switch cable
  - External Compressor switch cable

- Ensure all sensor cables are not touching with Front Plate ⑱
- Guide the wiring inside the unit like below figure. Once all wiring work done, tie the cable / cord with the banding strap (field supply), to prevent them touching with hot surfaces such as Optional Back-up Heater (CZ-NB3), bare copper pipes and etc.



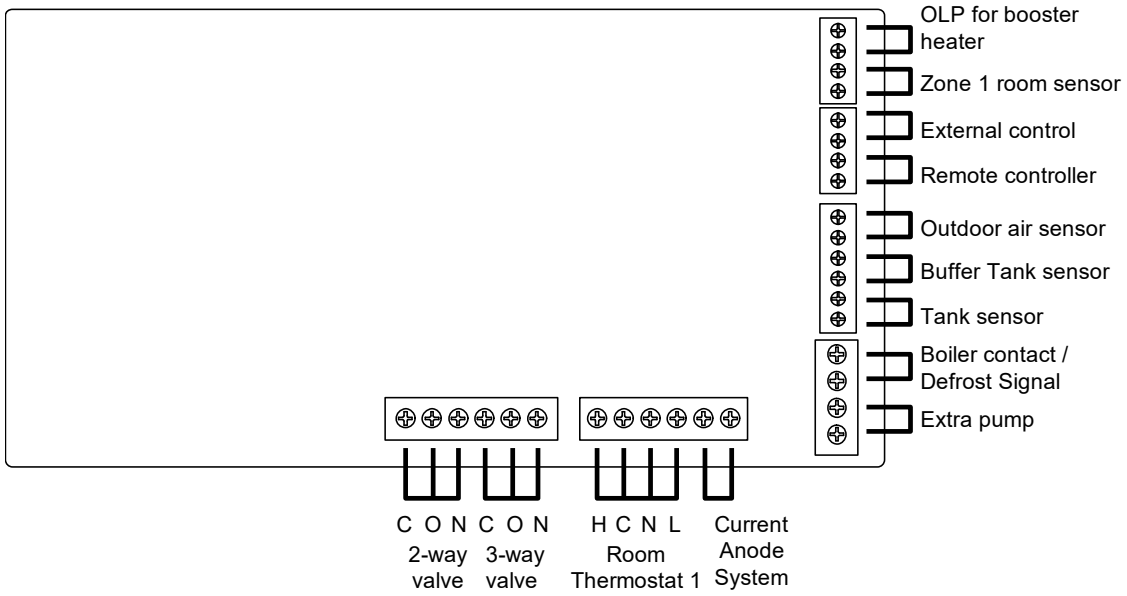
Wiring for “COMBINATION-1” | Wiring for “COMBINATION-2”

12.4.2.3 Connecting Cables Length

When connecting cables between Tank Unit and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Mixing valve	50
Room thermostat	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Room sensor	30
Outdoor air sensor	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

12.4.2.4 Connection of the Main PCB



■ Signal inputs

Optional Thermostat	L N=AC230V, Heat, Cool=Thermostat heat, Cool terminal
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

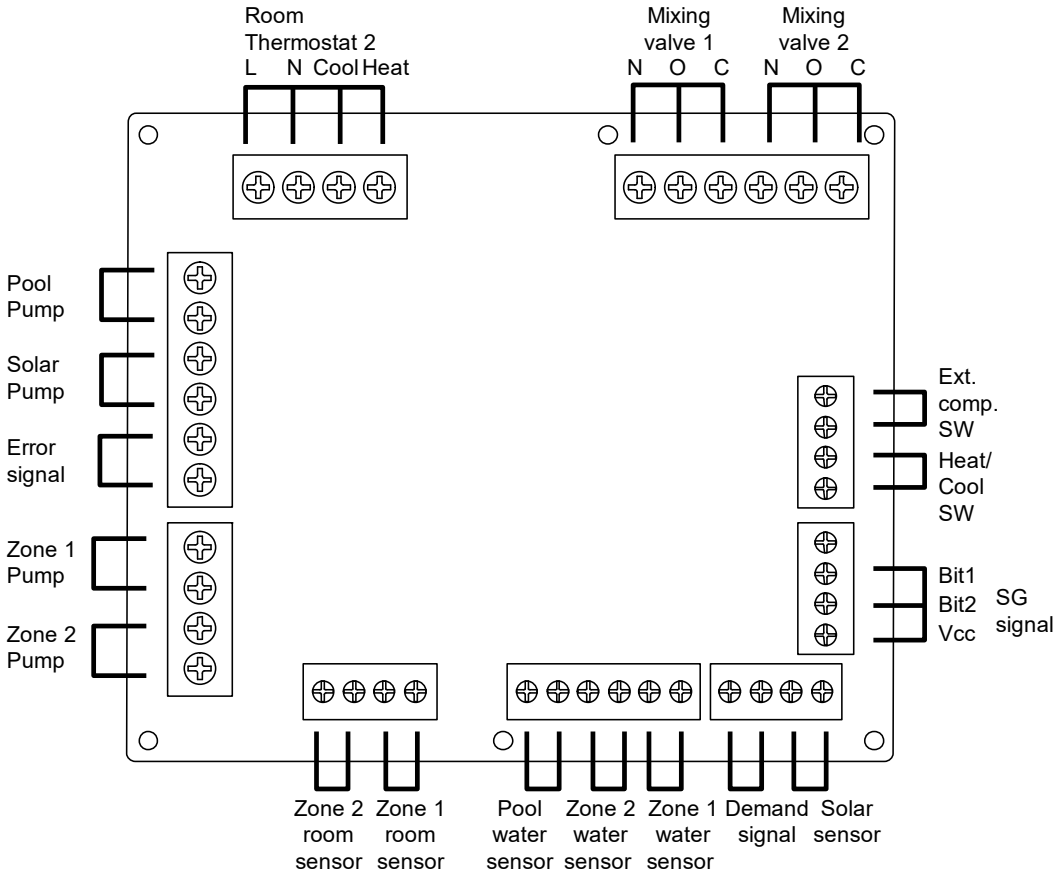
■ Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)	AC230V, 12 VA
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)	AC230V, 12 VA
Extra pump	AC230V (Used when Tank Unit pump capacity is insufficient)	AC 230V, 0.6 A max
Boiler contact / Defrost signal	Dry contact (System setup necessary)	

■ Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT ❇It does not work when using the Optional PCB
Outdoor air sensor	PAW-A2W-TSOD (Total cable length shall be 30m or less)
Buffer tank sensor	PAW-A2W-TSBU

12.4.2.5 Connection of Optional PCB (CZ-NS5P)



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp. OFF, Short=Comp. ON (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

■ Outputs

Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s	AC230V, 6VA
Pool pump	AC230V	AC 230V, 0.6 A max
Solar pump	AC230V	AC 230V, 0.6 A max
Zone pump	AC230V	AC 230V, 0.6 A max

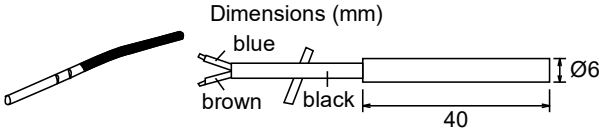
■ Thermistor inputs

Zone room sensor	PAW-A2W-TSRT
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

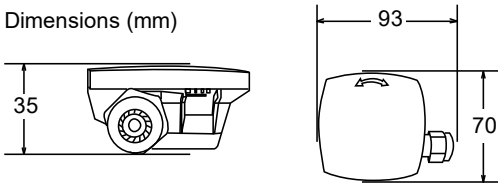
12.4.2.6 Recommended External Device Specification

- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.

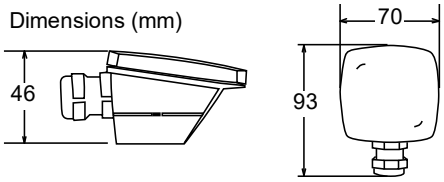
1. Buffer tank sensor: PAW-A2W-TSBU
   Use for measurement of the buffer tank temperature.
   Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



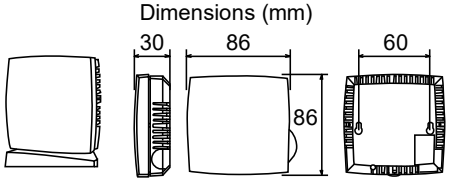
2. Zone water sensor: PAW-A2W-TSHC
   Use to detect the water temperature of the control zone.
   Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



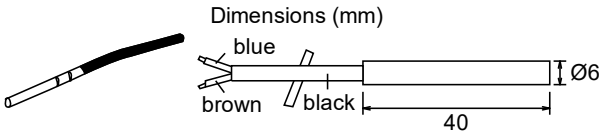
3. Outdoor sensor: PAW-A2W-TSOD
   If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
   In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



4. Room sensor: PAW-A2W-TSRT
   Install the room temperature sensor to the room which requires room temperature control.



5. Solar sensor: PAW-A2W-TSSO
   Use for measurement of the solar panel temperature.
   Insert the sensor into the sensor pocket and paste it on the solar panel surface.

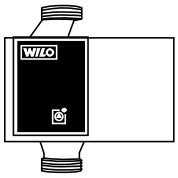


6. Please refer to the table below for sensor characteristic of the sensors mentioned above.

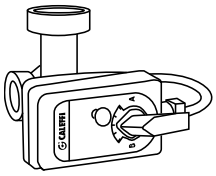
Temperature (°C)	Resistance (kΩ)
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo



- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

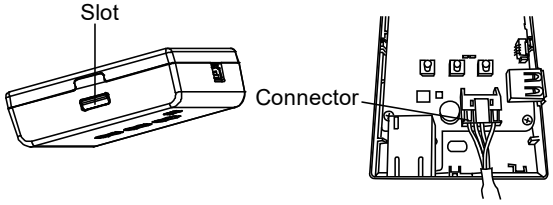


⚠ WARNING

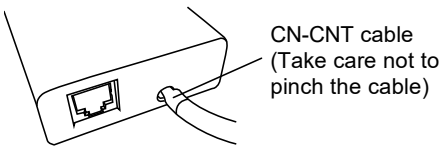
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

12.4.2.7 Network Adaptor [6] Installation (Optional)

1. **Remove the Control Board Cover ⑤, then connect the cable included with this adaptor to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Tank Unit so that there is no pinching.
   - If an Optional PCB has been installed in the Tank Unit, connect to the CN-CNT connector of the Optional PCB.
2. **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**

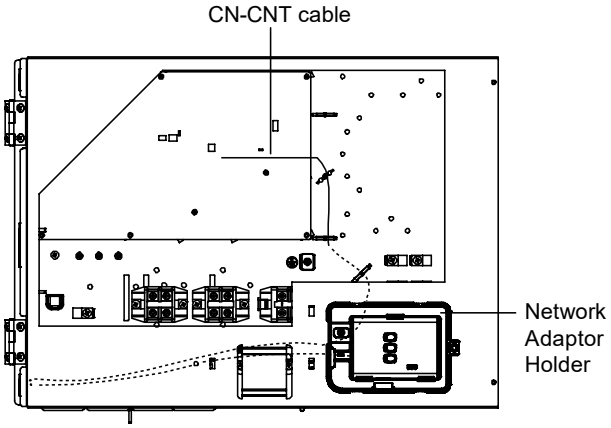


3. **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**

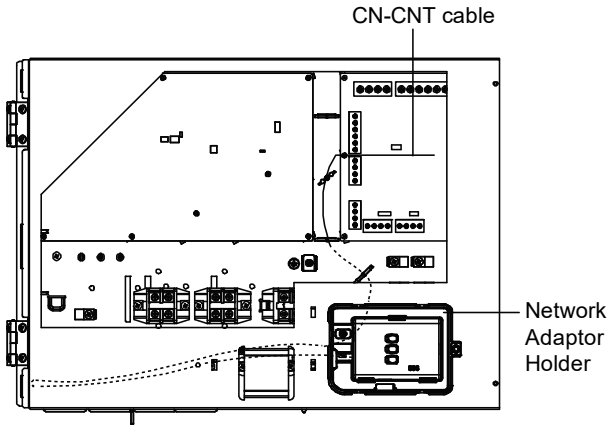


4. **Fix the Network Adaptor [6] to Network Adaptor Holder.**
   Guide the cable as shown in the diagram so that external forces cannot act on the connector in the adaptor.

   Connection examples:



Without Optional PCB

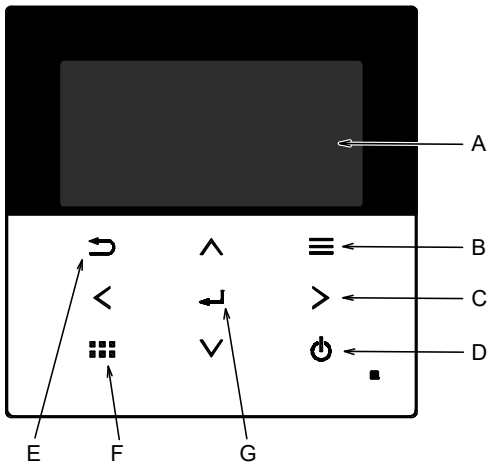


With Optional PCB

12.4.3 System Installation

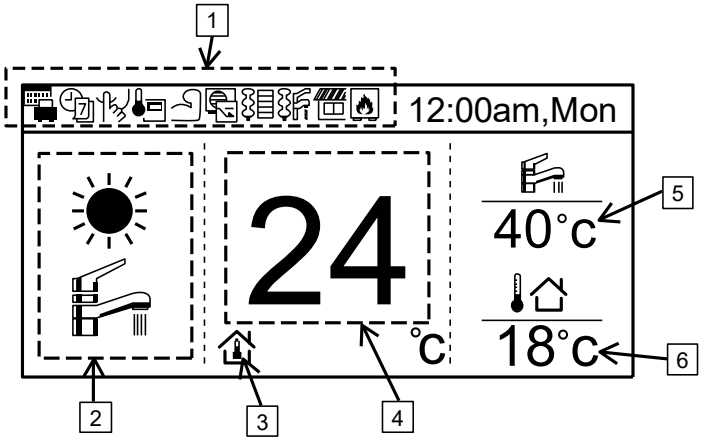
12.4.3.1 Remote Controller Outline

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.



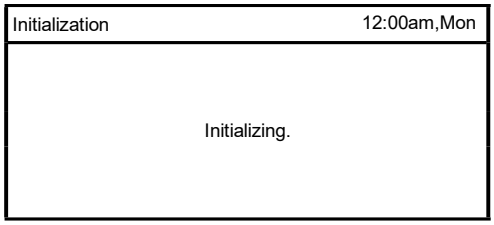
LCD Display
(Actual - Dark background with white icons)

Name	Function
A: Main screen	Display information
B: Menu	Open/Close main menu
C: Triangle (Move)	Select or change item
D: Operate	Start/Stop operation
E: Back	Back to previous item
F: Quick Menu	Open/Close Quick menu
G: OK	Confirm

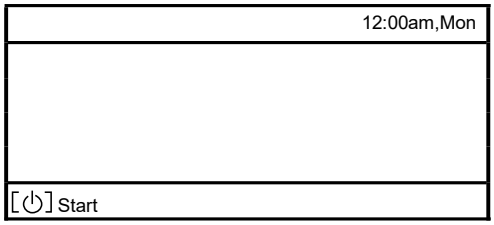


	Name	Function		
1:	Function icon	Display set function/status		
		Holiday mode	Demand control	
		Weekly timer	Room heater *Only displayed when Back-up heater function is enabled.	
		Quiet mode	Tank heater *Only displayed when Back-up heater function is enabled.	
		Remote controller room thermostat	Solar	
		Powerful mode	Boiler	
2:	Mode	Display set mode/current status of mode		
		Heating	Cooling	
		Auto	Hot water supply	Auto heating / Auto cooling
		Heat pump operating		
3:	Temp setting	Set room temp	Compensation curve	Set direct water temp / Set pool temp
4:	Display Heat temp	Display current heating temperature (it is set temperature when enclosed by line)		
5:	Display tank temp	Display current tank temperature (it is set temperature when enclosed by line) The zone 2 will shown in LCD display if is connected with Optional PCB.		
6:	Outdoor temp	Display outdoor temp		

First time of power ON (Start of installation)



When power is ON, firstly initialization screen appears (10 sec)



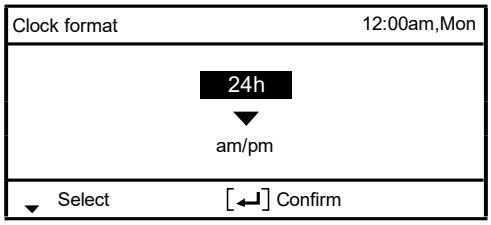
When initialization screen ends, it turns to normal screen.



When any button is pressed, language setting screen appears.
(NOTE) If initial setting is not performed, it does not go into menu.

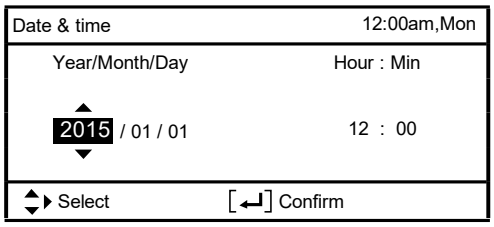
When there are two remote controllers installed from the beginning, the first remote controller to set and confirm language will be recognised as main remote controller.

Set language & confirm



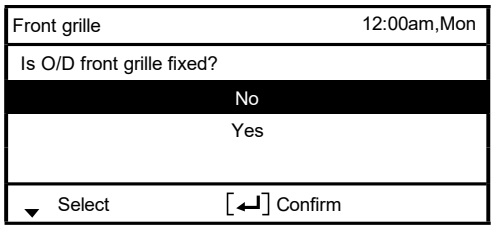
When language is set, setting screen of time display appears (24h/am/pm)

Set time display & confirm

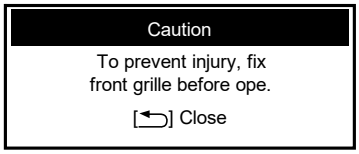


YY/MM/DD/Time setup screen appears

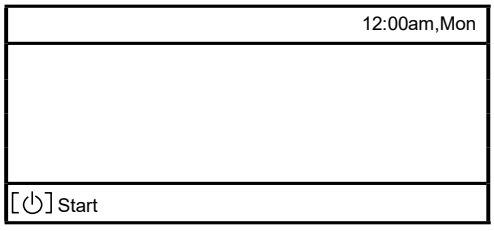
Set YY/MM/DD/Time & confirm



If set No & confirm, a caution message will be displayed to ensure outdoor front grille is installed before proceed to operate the unit.

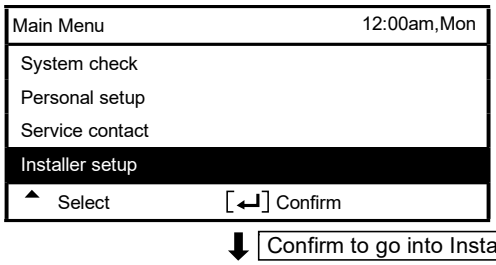


Set Yes & confirm if outdoor front grille has been installed

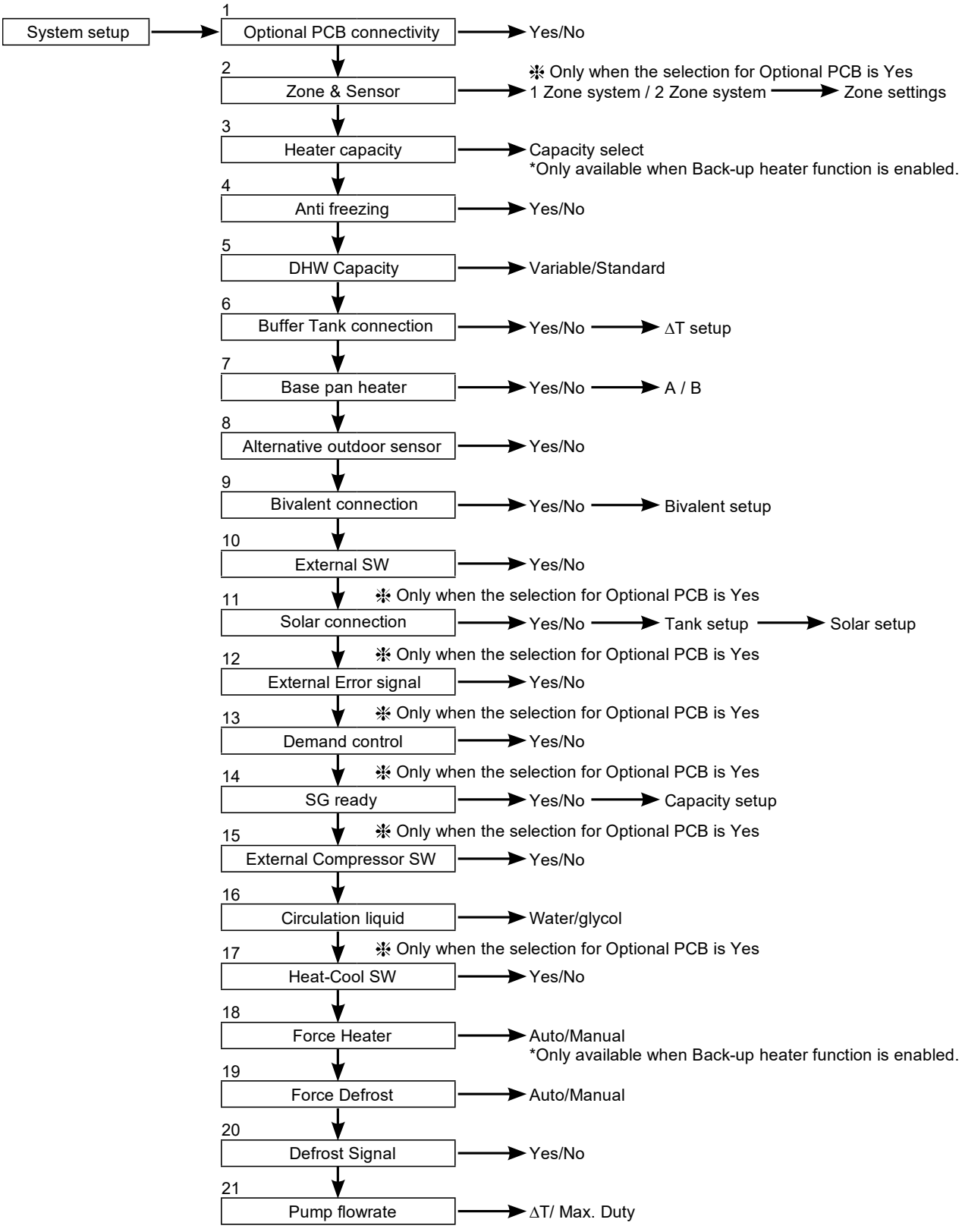


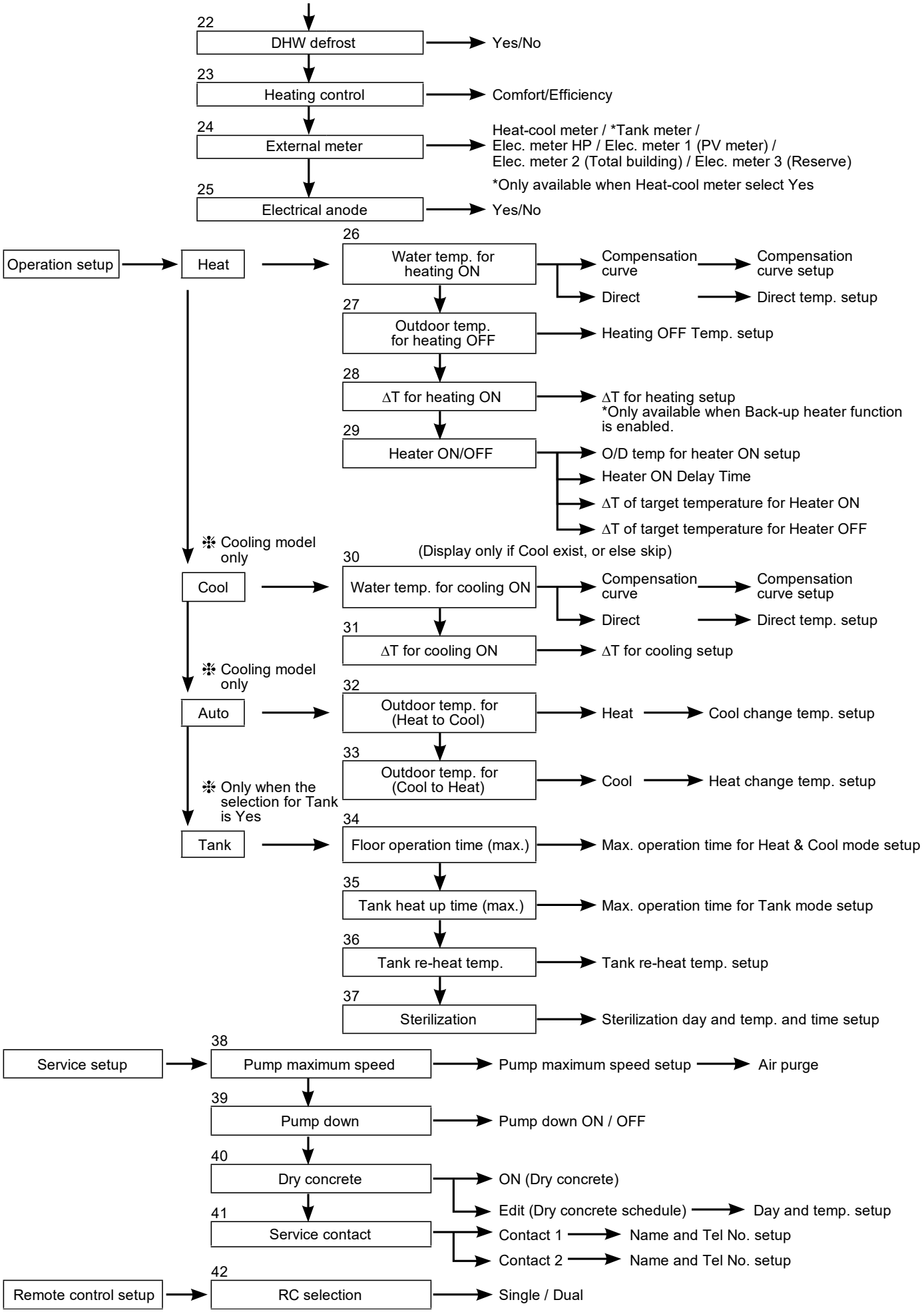
Back to initial screen

Press menu, select Installer setup



12.4.3.2 Installer Setup



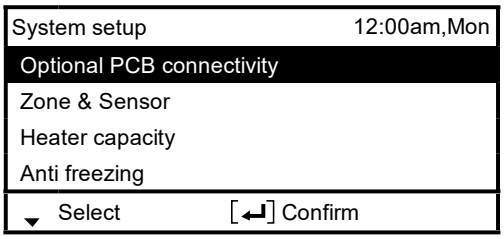

22
DHW defrost
Yes/No
23
Heating control
Comfort/Efficiency
24
External meter
Heat-cool meter / *Tank meter /
Elec. meter HP / Elec. meter 1 (PV meter) /
Elec. meter 2 (Total building) / Elec. meter 3 (Reserve)
*Only available when Heat-cool meter select Yes
25
Electrical anode
Yes/No
Operation setup
Heat
26
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
27
Outdoor temp. for heating OFF
Heating OFF Temp. setup
28
ΔT for heating ON
ΔT for heating setup
*Only available when Back-up heater function is enabled.
29
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
❇ Cooling model only
(Display only if Cool exist, or else skip)
Cool
30
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
31
ΔT for cooling ON
ΔT for cooling setup
❇ Cooling model only
Auto
32
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
33
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
❇ Only when the selection for Tank is Yes
Tank
34
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
35
Tank heat up time (max.)
Max. operation time for Tank mode setup
36
Tank re-heat temp.
Tank re-heat temp. setup
37
Sterilization
Sterilization day and temp. and time setup
Service setup
38
Pump maximum speed
Pump maximum speed setup
Air purge
39
Pump down
Pump down ON / OFF
40
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
41
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup
Remote control setup
42
RC selection
Single / Dual

12.4.3.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install Optional PCB.
Please select Yes after installing Optional PCB.

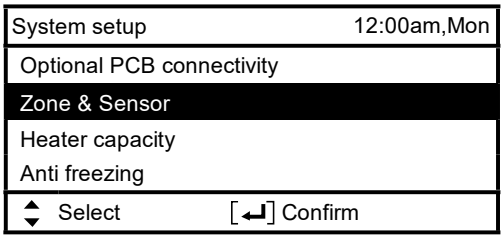
- 2-zone control
- Pool
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

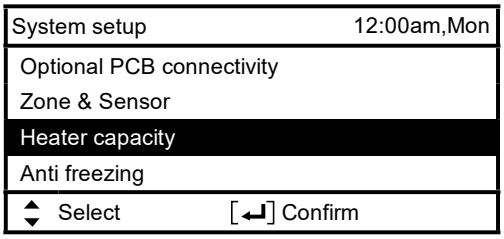
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

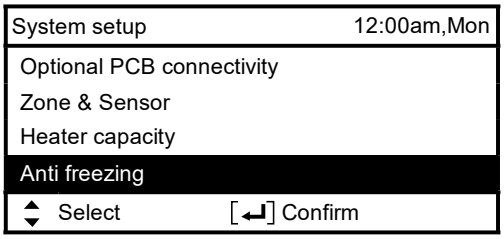
(NOTE) There are models which cannot select Heater capacity.
*Only available when Back-up heater function is enabled.



4. Anti freezing | Initial setting: Yes

Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

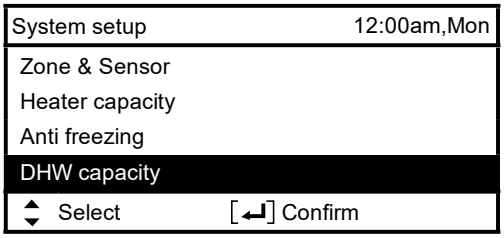
(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



5. DHW Capacity | Initial setting: Variable

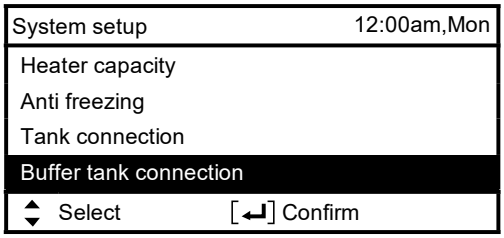
Select “Variable”
- The heat pump runs with efficient boiling, which is an energy-saving mode. However, if hot water usage is high and the tank water temperature is low, the heat pump will switch to fast heat-up mode, which uses higher heating capacity to heat up the tank. When “Efficiency” is selected in the “23. Heating control” setting, the tank heat-up process will be delayed until a much lower tank water temperature is reached, which improves the efficiency of the heating process. To ensure sufficient hot water, select either “Standard” in this setting or “Comfort” in the “23. Heating control” setting.

Select “Standard”
- The heat pump operates at its rated heating capacity during tank heating.



6. Buffer Tank connection | Initial setting: No

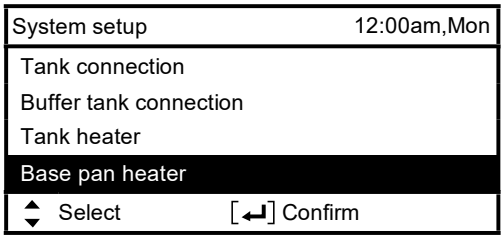
Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
If the buffer tank capacity is not so large, please set larger value for ΔT.



7. Base pan heater | Initial setting: No

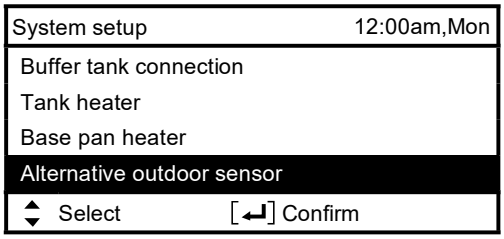
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



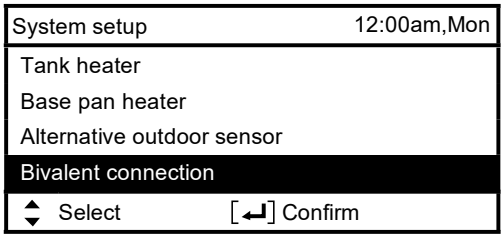
8. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



9. Bivalent connection | Initial setting: No

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.



After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)

1) SG ready (Only available to set when optional PCB set to YES)
   - SG Ready input from optional PCB terminal control ON/OFF of boiler and heat pump as below condition

SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

* This bivalent SG ready input is sharing same terminal as [14. SG ready] connection. Only one of these two setting can be set at the same time.
  When one is set, another setting will reset to not set.

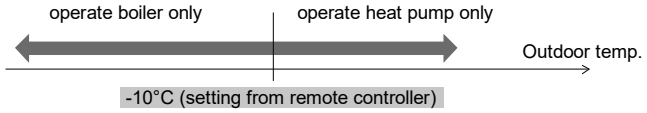
2) Auto

There are 3 different modes in the boiler auto pattern operation. Movement of each modes are shown below.

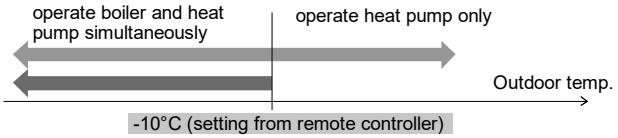
② Alternative (switch to boiler operation when drops below setting temperature)
③ Parallel (allow boiler operation when drops below setting temperature)
④ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is “ON”, “boiler contact” is “ON”, “_”(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

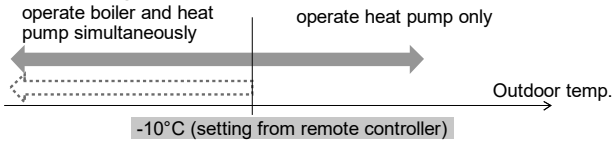


Parallel mode



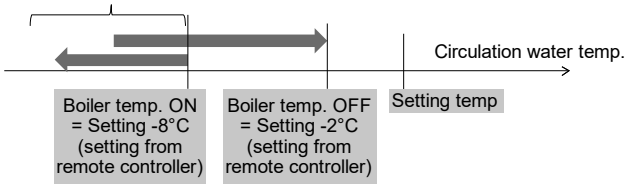
Advanced Parallel mode

For heating

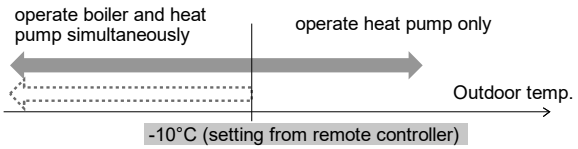


Although heat pump operates but water temperature does not reach this temperature for more than 30 mins (setting from remote controller)

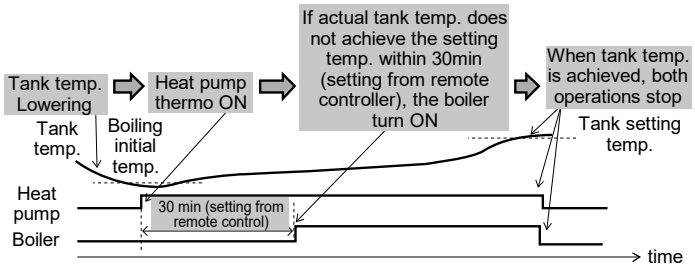
AND



For DHW tank



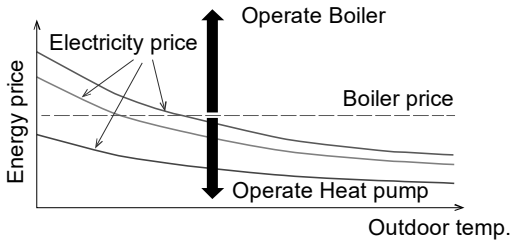
AND



In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of “Heating/Tank” mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

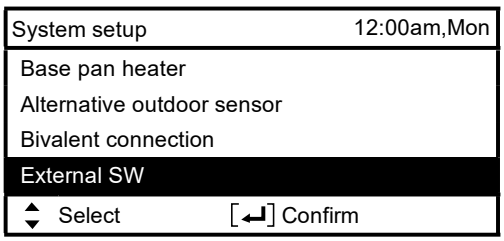
3) Smart

There are Energy price (both electricity and boiler) and Schedule to be set on remote controller.
Operation setting of Energy price and Schedule shall be responsible by installer.
Based on these settings, system will calculate the final price for both electricity and boiler.
When final price of Electricity is lower than Boiler’s, heat pump will operate.
When final price of Electricity is higher than Boiler’s, boiler will operate.



10. External SW | Initial setting: No

Able to turn ON/OFF the operation by external switch.

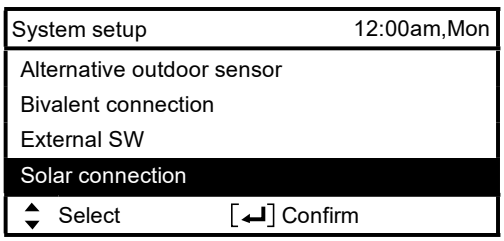


11. Solar connection | Initial setting: No

Set when solar water heater is installed.

Setting include items below.

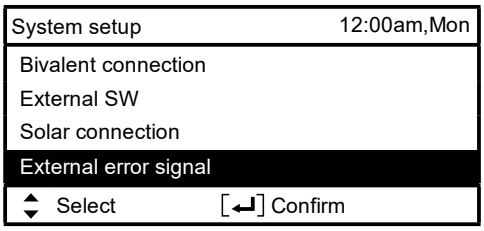
① Set either buffer tank or DHW tank for connection with solar water heater.
② Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to operate the solar pump.
③ Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to stop the solar pump.
④ Anti-freezing operation start temperature (please change setting based on usage of glycol.)
⑤ Solar pump stop operation when it exceeds high limit temperature (when tank temperature exceed designated temperature (70~90°C))



12. External Error Signal | Initial setting: No

Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

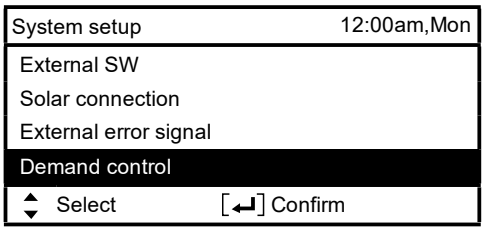
(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off "close" from the display, error signal will still remain ON.



13. Demand control | Initial setting: No

Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.



Analog input [v]	Rate [%]
0.0	not activate
0.1 ~ 0.6	
0.7	10
0.8	not activate
0.9 ~ 1.1	10
1.2	15
1.3	10
1.4 ~ 1.6	15
1.7	20
1.8	15
1.9 ~ 2.1	20
2.2	25
2.3	20
2.4 ~ 2.6	25
2.7	30
2.8	25
2.9 ~ 3.1	30
3.2	35
3.3	30
3.4 ~ 3.6	35
3.7	40
3.8	35

Analog input [v]	Rate [%]
3.9 ~ 4.1	40
4.2	45
4.3	40
4.4 ~ 4.6	45
4.7	50
4.8	45
4.9 ~ 5.1	50
5.2	55
5.3	50
5.4 ~ 5.6	55
5.7	60
5.8	55
5.9 ~ 6.1	60
6.2	65
6.3	60
6.4 ~ 6.6	65
6.7	70
6.8	65
6.9 ~ 7.1	70
7.2	75
7.3	70

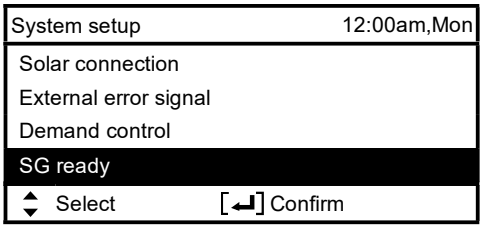
Analog input [v]	Rate [%]
7.4 ~ 7.6	75
7.7	80
7.8	75
7.9 ~ 8.1	80
8.2	85
8.3	80
8.4 ~ 8.6	85
8.7	90
8.8	85
8.9 ~ 9.1	90
9.2	95
9.3	90
9.4 ~ 9.6	95
9.7	100
9.8	95
9.9 ~	100

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

14. SG ready | Initial setting: No

Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2



Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Set by SG ready setting of remote controller

(When SG ready set to YES, Bivalent control pattern will set to Auto.)
(NOTE) Does not display if there is no Optional PCB.

SG Ready HPU STOP Consumption Mode Control Setting

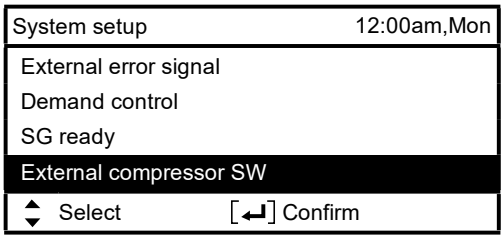
Remote control setting		Main PCB DIP SW (SW 2)	MODE
SG ready	External compressor SW	DIP SW 3	
No	No	OFF	No SG ready and External compressor SW function
		ON	No SG ready and External compressor SW function
No	Yes	OFF	Normal External compressor SW control
		ON	Normal External compressor SW control
*Yes	No	OFF	Normal SG ready control
		*ON	SG ready control limit operation to 4.2kW (SG ready = 10 or 11, heater OFF while heatpump ON, Heat pump OFF if >4.2kW and Heater ON 1 element)
*Yes	Yes	OFF	Normal SG ready and External compressor SW control
		*ON	SG ready control limit operation to 4.2kW (SG ready = 10 or 11, heater OFF while heatpump ON, Heat pump OFF if >4.2kW and Heater ON 1 element)

15. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

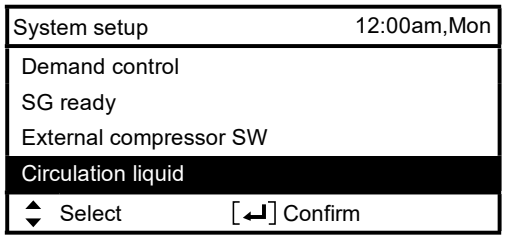


16. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

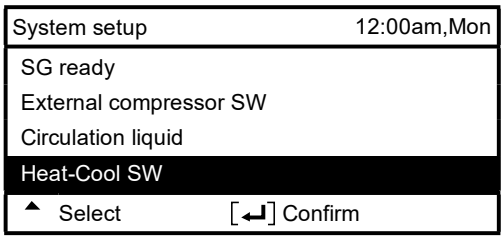


17. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

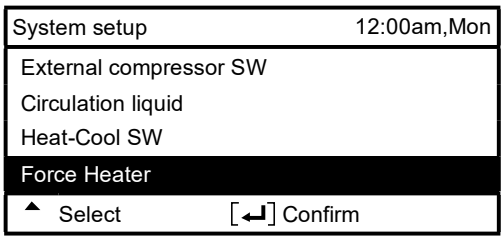


18. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

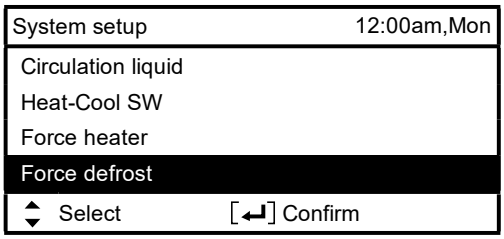
Heater source will ON during force heater mode.
*Only available when Back-up heater function is enabled.



19. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

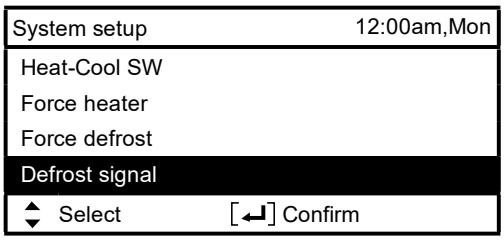
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition. (Even auto is selected, user still can turn on force defrost through quick menu)



20. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

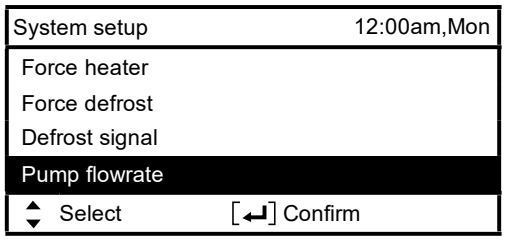
When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).



21. Pump flowrate | Initial setting: ΔT

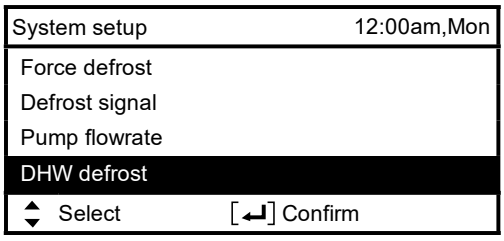
If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



22. DHW defrost | Initial setting: Yes

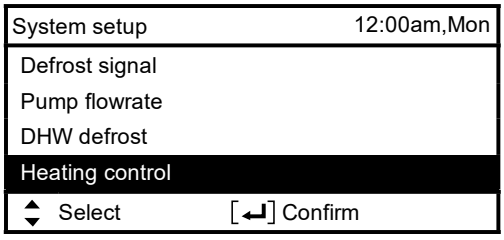
When DHW defrost set to YES, hot water of domestic hot water tank will be used during defrost cycle.
When DHW defrost set to NO, hot water of floor heating circuit will be used during defrost cycle.



23. Heating control | Initial setting: Comfort

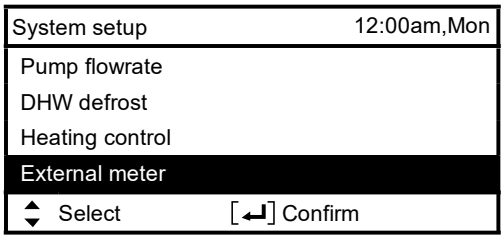
Compressor frequency control can be selected from two modes:
Comfort and Efficiency.
Select “Comfort”
- The compressor operates at maximum frequency at the upper zone limit and reaches the set temperature faster.

Select “Efficiency”
- The compressor frequency is gradually increased to reduce power consumption. When “Variable” is selected in the “5. DHW capacity” setting, the tank heat-up process will be delayed until a much lower tank water temperature is reached, which improves the efficiency of the heating process. To ensure sufficient hot water, select either “Comfort” in this setting or “Standard” in the “5. DHW capacity” setting. It is a function for Heating + Tank operation.



24. External meter

Initial setting : [Heat-cool meter : No]
[Tank meter : No] *only available when Heat-cool meter select Yes
[Elec. meter HP : No]
[Elec. meter 1 (PV meter) : No]
[Elec. meter 2 (Total building) : No]
[Elec. meter 3 (Reserve) : No]



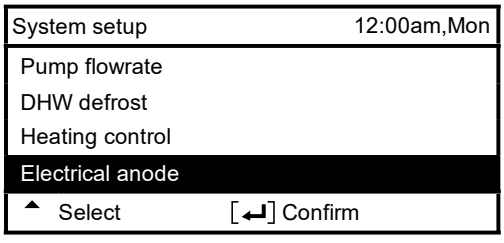
There are two systems for generation meter connection : one generation meter system (Heat-cool meter) or two generation meter system (Heat-cool meter and Tank meter)
Both systems can provide all generation data of heating, cooling and DHW directly from external meter.
If Heat-cool meter is set to Yes, it will read from external meter for heat pump’s energy generation data during heating, cooling and DHW operation*1.
If Heat-cool meter is set to No, it will base on unit’s calculation for heat pump’s energy generation data during heating, cooling and DHW operation.
If Tank meter is set to Yes, it will read from external meter for heat pump’s energy generation data during DHW operation*1.
If Elec. meter HP is set to Yes, it will read from external meter for heat pump’s energy consumption data.
If Elec. meter HP is set to No, it will base on unit’s calculation for heat pump’s energy consumption data.
If Elec. meter 1 (PV meter) is set to Yes, it will read from external meter for energy generation data of solar system and display it on Cloud system.
If Elec. meter 2 (Building) is set to Yes, it will read from external meter for energy consumption data of the building and display it on Cloud system.
If Elec. meter 3 (Reserve) is set to Yes, it will read from external meter for energy consumption data obtained from reserved electricity meter and display it on Cloud system.

*1 Set Heat-cool meter to Yes and set Tank meter to No when 1 generation meter system is installed.
Set Heat-cool meter to Yes and set Tank meter to Yes when 2 generation meter system is installed.

Remark : Elec. meter HP refers to Electricity meter that measures Heat Pump unit’s consumption.
Elec. meter 1 / 2 / 3 refers to Electricity meter no. 1 / no. 2 / no. 3

25. Electrical anode | Initial setting: No

When Electrical anode set to YES, anode will be turned on.
When Electrical anode set to NO, anode will not be turned on.



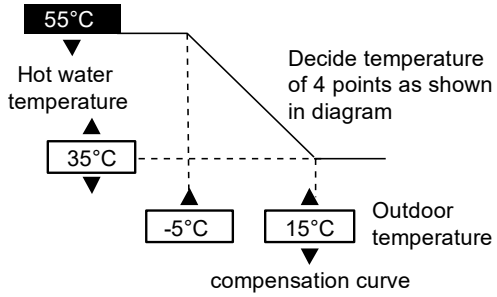
12.4.3.4 Operation Setup

Heat

26. Water temp. for heating ON | Initial setting: compensation curve

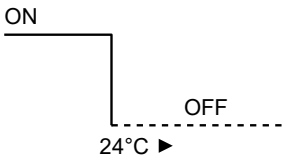
Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



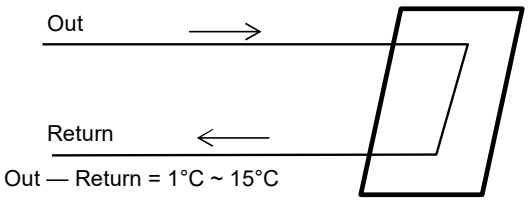
27. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C

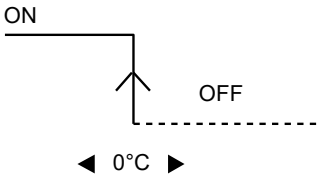


28. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C



29. Heater ON/OFF *Only available when Back-up heater function is enabled.

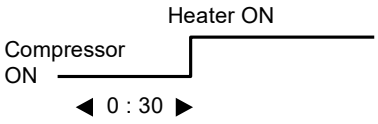


a. Outdoor temp. for heater ON | Initial setting: 0°C

Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.

b. Heater ON delay time | Initial setting: 30 minutes

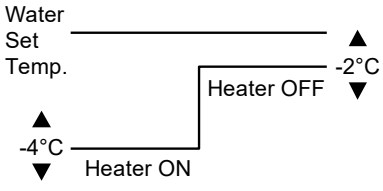


Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

c. Heater ON: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp | Initial setting: -2°C



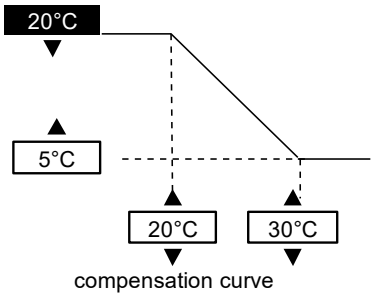
Set water temperature for heater to turn off at heat mode.
Setting range is -8°C ~ 0°C

Cool

30. Water temp. for cooling ON | Initial setting: compensation curve

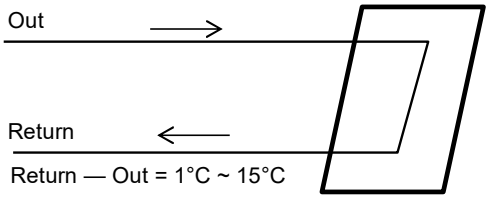
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



31. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

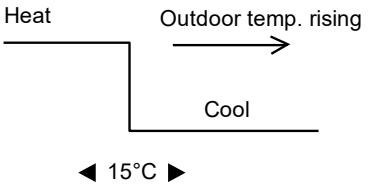


Auto

32. Outdoor temp. for (Heat to Cool) | Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 5°C ~ 25°C

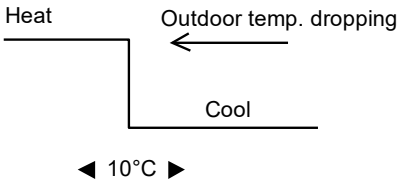
Timing of judgement is every 1 hour



33. Outdoor temp. for (Cool to Heat) | Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 25°C

Timing of judgement is every 1 hour

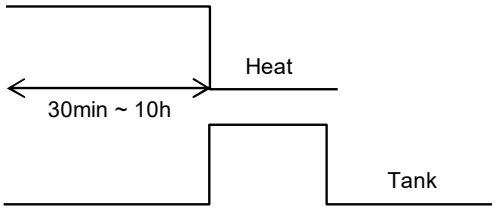


Tank

34. Floor operation time (max.) Initial setting: 8h

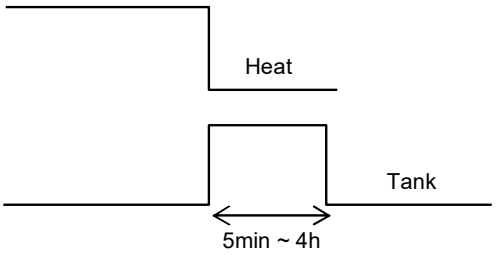
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



35. Tank heat up time (max.) Initial setting: 60min

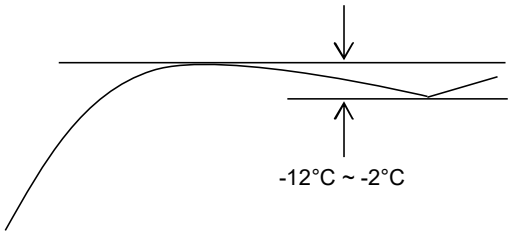
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



36. Tank re-heat temp. Initial setting: -8°C

Set temp to perform reboil of tank water.
(When boiled by heat pump only, (51°C – Tank re-heat temp) shall become max temp.)

Setting range is -12°C ~ -2°C

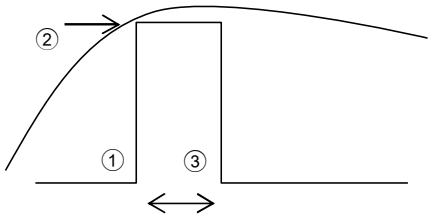


37. Sterilization Initial setting: 65°C 10min

Set timer to perform sterilization.

① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~75°C ❇ If use back-up heater, it is 65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.

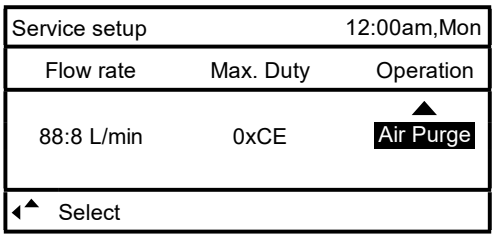


12.4.3.5 Service Setup

38. Pump maximum speed | Initial setting: Depend on model

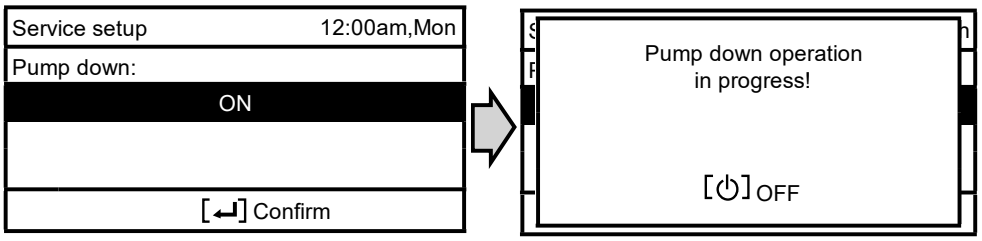
Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.



39. Pump down

Operate pump down operation

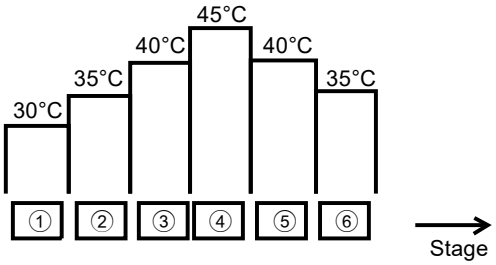


40. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

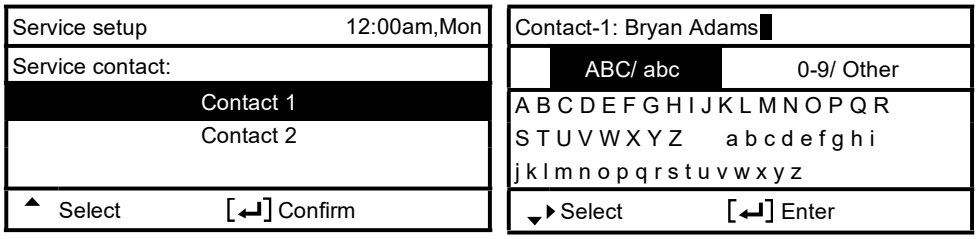
When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



41. Service contact

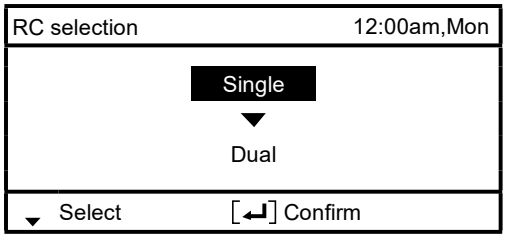
Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)



12.4.3.6 Remote Control Setup

42. RC selection | Initial setting: Single

Set to “Single” when only one remote controller is installed.
Set to “Dual” when two remote controllers are installed.



12.5 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

12.5.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

12.5.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it.
In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Back-up heater (Use/Do not use Back-up heater) (Optional)
Default is without
(NOTE) It is different from to use/not to use Back-up heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer. When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

12.5.3 Check Water Pressure from Remote Controller

1. Press ☰ SW and scroll to “System check”.
2. Press ↵ and scroll to “System information”.
3. Press ↵ and search for “Water pressure”.

